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EXECUTIVE SUMMARY

Liquid Air Combined Cycle (LACC) is a hybrid liquid air energy storage (LAES) system combining energy storage with a combustion turbine to enable large-scale, long-duration energy storage (LDES) while reducing fuel intensity compared to the current state-of-the-art. The LACC technical approach employs proven equipment (cryogenic refrigeration, storage, tanks, pumps, gas turbines, exhaust heat recovery equipment, and turbines) to limit technical risk to a novel organic Rankine cycle (ORC), which was evaluated during this project and found to be feasible. Moreover, LACC storage is safe and relatively compact, to facilitate siting close to loads and within metropolitan regions. The air storage medium is freely available and eliminates supply chain constraints.

LACC uses cryogenic air as a storage medium and a gas turbine as the source of heat to drive the discharge process. LACC is distinguished from other LAES technologies by several factors. The charge and discharge processes are decoupled so that cryogenic liquid air is the only storage medium. Other systems also store the higher temperature thermal energy from the liquefaction process in an additional medium. Subsequently, LACC focuses on maximization of the discharge energy and power. LACC also permits the use of commercially available cryogenic refrigeration and storage technologies to increase competition.

This project identified product requirements to support market entry and commercialization of the LACC in modular units of approximately 117 MW, each drawing liquid air from customary cryogenic storage tanks capable of storing 75 GWh of dispatchable energy, more than pumped storage hydro or compressed air energy storage technologies.

An economic analysis identified the specific liquid air consumption (quantity of liquid air per unit of discharge energy) as a critical parameter. Minimizing the air consumption reduces the specific capital cost (\$/kW) for charging and discharging equipment by reducing the size of piping and turbomachinery. Likewise, the specific cost of energy capacity (\$/kWh) is reduced by increasing the energy deliverable from a given size tank.

The cycle was analyzed to identify the optimal equipment selection and operating conditions, which in turn were combined with quotes and cost estimates to calculate the cost of energy from an LACC system.

A substantial effort was focused on the ORC, which draws low-temperature heat from the gas turbine exhaust and condenses at low temperature using the cryogenic liquid air as a heat sink. Alternative turbomachinery arrangements were evaluated for feasibility and cost.

A technology maturation plan lays out a low-risk approach to development of the novel ORC components and demonstration of LACC technology at pilot scale.

TABLE OF CONTENTS

1. Introduction.....	1
1.1. The need for energy storage	2
1.2. The need for long-duration energy storage	3
1.3. Hybrid LDES.....	4
1.4. The Liquid Air Combined Cycle.....	4
1.4.1. Basic principles and benefits of LACC.....	5
1.4.2. Competition	6
1.4.3. Commercial applications	6
1.5. Project objective	7
2. Requirements	9
2.1. Energy storage terminology	9
2.1.1. Chernyakhovskiy’s duration framework.....	10
2.1.2. SMUD’s durations.....	11
2.1.3. Multi-day and seasonal storage	11
2.2. Competitive positioning	13
2.3. Product definition	14
2.4. Overall plant economics	15
2.4.1. Key definitions	15
2.4.2. Levelized cost.....	17
2.4.3. Capacity factor	18
2.4.4. Determination of capacity factor for LACC systems.....	18
2.4.5. Stored energy rate.....	19
2.4.6. Marginal cost considerations.....	19
2.4.7. Maximum capacity factor.....	20
2.5. Design considerations: charging and storage sub-systems.....	21
2.6. Design considerations: discharge sub-system	22
2.6.1. Gas turbine	22
2.6.2. Flue gas condensation	23
2.6.3. Fuel selection.....	23
2.6.4. ORC working fluid.....	23
2.6.5. Rating condition	24
2.6.6. Coupling of inlet air cooling	25
2.7. Reference process design	25

2.7.1.	Performance of discharge system.....	27
2.7.2.	Cost of discharge system.....	27
2.8.	Summary.....	28
3.	Cryogenic charging and storage system	29
3.1.	Decoupling of charge and discharge	29
3.2.	Charge cycle trade studies	30
3.2.1.	Claude cycle	31
3.2.2.	Linde-Hampson cycle.....	32
3.2.3.	Charge system cost estimate.....	33
4.	ORC component sizing	35
4.1.	Turbines	35
4.1.1.	Axial turbine sizing	35
4.1.2.	Radial turbine sizing.....	43
4.2.	Heat exchangers.....	46
4.2.1.	ORC-to-air preheater	46
4.2.2.	ORC condenser / air gasifier	49
4.2.3.	ORC recuperator.....	50
5.	Commercial-scale LACC discharge cycle design.....	52
5.1.	Process model enhancements	52
5.2.	Cost modeling.....	54
5.3.	Discharge cycle trade studies	55
5.4.	Discharge cycle design optimization.....	66
5.5.	Final design selection	68
5.5.1.	Performance.....	70
5.5.2.	Cost estimate	70
6.	LACC economics.....	72
6.1.	Final design economic assumptions	72
6.2.	Effect of plant configuration	73
6.3.	Cost sensitivity	76
6.3.1.	Cost reduction opportunities	77
6.4.	Comparison to alternatives	78
6.4.1.	LDES	78
6.4.2.	Gas turbines	79
7.	Technology maturation plan	80

7.1. Technology readiness level	80
7.2. Post project plans.....	83
7.2.1. Engineering optimization	83
7.2.2. Demonstration-scale system.....	84
7.2.3. ORC component validation	84
7.2.4. Commercialization	85
8. Demonstration-scale LACC design	86
8.1. Performance.....	86
8.2. Plant arrangement.....	87
8.3. Cost.....	87
9. Summary and conclusions	88
10. References	90
Appendix: Publications	92

LIST OF FIGURES

Figure 1. Commercial-scale LACC process	5
Figure 2. Reference process design	25
Figure 3. Adiabatic LAES process. Figure taken without permission from Vecchi et al. [23]. ...	29
Figure 4. Claude cycle diagram	32
Figure 5. Linde-Hampson cycle diagram.....	32
Figure 6. Charge system start-up costs	34
Figure 7. Rotor layout for a 9-stage 10 krpm axial turbine	36
Figure 8. Isentropic efficiency	39
Figure 9. Minimum hub diameter and maximum blade tip diameter	39
Figure 10. Bearing span	40
Figure 11. Comparison of 6- and 10-stage 5 krpm turbine layouts	40
Figure 12. Critical speed ratio.....	41
Figure 13. Shaft shear stress to allowable shear stress ratio	41
Figure 14. Bearing surface speed to maximum surface speed.....	42
Figure 15. Comparison of shaft shear stress ratio, bearing speed ratio, rotational speed, and efficiency	43
Figure 16. Conceptual design of the high-speed shaft.....	45
Figure 17. Conceptual design of the low-speed shaft.....	45
Figure 18. Schematic of gasketed plate heat exchanger for the air preheater.....	47
Figure 19. Plate fin heat exchanger schematic for the air preheater. Stream 1 is air and stream 2 is ethane.....	48
Figure 20. Schematic for gasketed plate heat exchanger for the ORC condenser / air gasifier....	50
Figure 21. Schematic of gasketed plate heat exchanger for the ORC recuperator.	51
Figure 22. Reference process design	52
Figure 23. Graphical interface for discharge cycle process model.....	53
Figure 24. Comparison between the liquid air mass flow rate and three dependent metrics. a) Discharge capital costs, b) discharged energy, c) stored energy rate.	57
Figure 25. Comparison of the amount of energy discharged, SER, and discharge capital costs as the liquid air flow rate was swept between 260-290 tonne/hr. a) Discharge capital costs v. discharged energy, b) discharge capital costs vs. stored energy rate, c) discharged energy vs. stored energy rate.	58
Figure 26. Comparison between the ORC mass flow rate and three dependent metrics. a) Discharge capital costs, b) discharged energy, c) stored energy rate.	59
Figure 27. Comparison of the amount of energy discharged, SER and discharge capital costs as the liquid air flow rate was swept between 165-170 tonne/hr. a) Discharge capital	

costs vs. discharged energy, b) discharge capital costs vs. stored energy rate, c) discharged energy vs. stored energy rate.	60
Figure 28. Comparison between the liquid air turbine inlet temperature and three dependent metrics. a) Discharge capital costs, b) discharged energy, c) stored energy rate.	61
Figure 29. Comparison of the amount of energy discharged SER and discharge capital costs as the liquid air turbine inlet temperature was swept between 400-600°C. a) Discharge capital costs vs. discharged energy, b) discharge capital costs vs. stored energy rate, c) discharged energy vs. stored energy rate.	62
Figure 30. Comparison between the ORC turbine inlet temperature and three dependent metrics. a) Discharge capital costs, b) discharged energy, c) stored energy rate.	63
Figure 31. Comparison of the amount of energy discharged, SER and discharge capital costs as the ORC turbine inlet temperature was swept between 200-350°C. a) Discharge capital costs vs. discharged energy, b) discharge capital costs vs. stored energy rate, c) discharged energy vs. stored energy rate.	64
Figure 32. Comparison between the liquid air turbine efficiency and three dependent metrics. a) Discharge capital costs, b) discharged energy, c) stored energy rate.	65
Figure 33. Comparison of the amount of energy discharged, SER and discharge capital costs as the liquid air efficiency was swept between 80-95%. a) Discharge capital costs vs. discharged energy, b) discharge capital costs vs. stored energy rate, c) discharged energy vs. stored energy rate.	66
Figure 34. Factorial analysis of the cycle modeling. a) Selection of points with low TIT for both the liquid air and ORC subsystems. b) Selection of points with high TIT for the liquid air subsystem and low TIT for the ORC subsystem. c) Selection of points with high TIT for both subsystems.	67
Figure 35. Final design	68
Figure 36. Process conditions – final design	69
Figure 37. Total overnight cost for a single tank system.....	74
Figure 38. Specific cost of storage for a single tank system.....	74
Figure 39. Capacity factor for different plant configurations	75
Figure 40. Annual discharge energy for various plant configurations.....	75
Figure 41. Cost of energy for various plant configurations	76
Figure 42. COE for gas turbine power plants fueled with hydrogen or equipped with carbon capture.....	79
Figure 43. Process flow diagram of engineering optimized LACC.....	84
Figure 44. ORC component validation loop	85
Figure 45. Demonstration-scale performance model.....	86

LIST OF TABLES

Table 1. Chernyakhovskiy’s energy storage framework [16]	11
Table 2. Firm dispatchable capacity as a percentage of peak MW demand for Northern and Southern regions [13].	13
Table 3. Summary of the design requirements	15
Table 4. Key definitions for determining cost of energy (COE)	15
Table 5. Gas turbine alternatives	22
Table 6. Refrigerant alternatives.....	24
Table 7. Discharge system performance	27
Table 8. Power and efficiency study for charge system	31
Table 9. Air liquefaction cost summary.....	31
Table 10. Charge system cost breakdown.....	33
Table 11. Rotor layout scaled dimensions.	36
Table 12. Component inertia properties.....	37
Table 13. Conceptual design of two-shaft, four stage radial turbine.	44
Table 14. Conditions for the air preheater	46
Table 15. Conditions for the ORC condenser / air gasifier.....	49
Table 16. Conditions for the ORC recuperator	50
Table 17. Specific costs of major equipment.....	54
Table 18. Independent variables for the discharge trade studies	55
Table 19. Discharge train performance – final design.....	70
Table 20. Discharge train equipment cost estimate – final design	70
Table 21. List of assumptions for techno-economic analysis	72
Table 22. Minimum cost and configurations for different assumptions.	77
Table 23. Costs of LDES technologies in 2030.....	78
Table 24. Technology readiness levels for LACC.....	80
Table 25. Demonstration scale performance	87

1. INTRODUCTION

This is the final report for project DE-FE0032002, Liquid Air Combined Cycle™ (LACC) for Power and Storage, which was performed for the U.S. Department of Energy's Office of Fossil Energy's Advanced Energy Storage Program, which aims to advance energy storage technologies and integrate them with fossil assets to reduce barriers to wide-spread deployment. Integrating energy storage with electric generating units can provide the short and long-term reliability needed as the use of variable renewable power generation.

The LACC technology, patented by Pintail Power (US Patent No. 11,053,818 B2 issued July 6, 2021), is a unique approach to liquid air energy storage (LAES) that focuses on reducing the capital and operating costs of large-scale energy storage by combining proven equipment and processes adapted from the power and process industries. An important design principle of LACC is systems engineering, with clear interfaces between sub-systems to minimize interaction, reduce technical and financial risk, and permit the use of best-in-class equipment provided by top-tier vendors. This approach is analogous to that used to demonstrate and commercialize the now-dominant Natural Gas Combined Cycle (NGCC) power plant. LACC energy storage is based on these principles:

- Air is plentiful and free, non-toxic, and safely stored at low pressure as a cryogenic liquid.
- Liquid air production uses proven cryogenic refrigeration processes and equipment.
- Liquid air's high energy density enables tens of GWh to be stored within a compact footprint.
- Low-cost or excess variable renewable energy (VRE) makes discharge efficiency more important than charge efficiency.
- Discharge efficiency depends on the temperature difference between source and sink.
- Integration of stored energy reduces the overall carbon intensity of power generation.

Accordingly, LACC aims to maximize the specific energy produced during discharge (kWh/kg of air) by using high-temperature exhaust from combustion turbines (CTs) as the heat source and cascading heat engines that reject to low-temperature cryogenic liquid air as the heat sink. LACC integration with fossil energy contrasts sharply with other LAES technologies that focus instead on minimizing the specific energy consumed during charging.

The project team of Southwest Research Institute (SwRI) and Pintail Power (Pintail) analyzed and optimized the LACC, aiming to develop conceptual design and specification for commercial-scale systems. LACC integrates cryogenic energy storage with gas turbine exhaust heat to provide large-scale, long-duration energy storage with fuel savings using largely proven equipment. LACC avoids the geotechnical, hydrological, and land-use constraints of pumped hydro storage (PHS), or the underground caverns required for compressed air energy storage (CAES).

The cost-effectiveness of an LACC system requires that the capital costs for charging and discharging equipment be minimized while maximizing the ability to convert liquid air to electric

energy. Accordingly, the project undertook to (1) define cost and performance trades for charge and discharge cycle components, (2) perform system optimization of the charge and discharge cycles, (3) develop an optimized commercial-scale LACC specification from techno-economic trade studies, and (4) analyze technology gaps and develop a technology maturation plan.

1.1. The need for energy storage

The transition to zero-carbon energy systems is projected to proceed by the displacement of fossil-fuel based resources with renewable resources such as wind or solar power. The United States Energy Information Administration references case projects that renewable resources will account for approximately 60% of the 1,000 GW of new generating capacity deployed from 2020 to 2050, with natural gas fired generation accounting for most of the balance [1]. Over this time period, in the reference case, renewables would double as a share of overall electric generation, from 21% to 42%, with natural gas generation going from 40% to 36%, while the amount of coal and nuclear generation declines.

Public and political pressure may accelerate the adoption of renewable energy by national, state, and local governments. For example, the U.S. State of California has adopted legislation requiring that 100% of retail electricity be provided from renewable and zero-carbon resources by 2045. The California Energy Commission expects this to mean a tripling of solar generating capacity to 97.6 GW [2]. For comparison, this solar generating capacity would substantially exceed the 2020 baseline forecast peak user demand of 63.4 GW [3], with the mismatch satisfied by 48.8 GW of battery storage and 4.0 GW of Long Duration Energy Storage (LDES).

Similar trends are evident globally with the International Energy Agency's World Energy Outlook [4], projecting that renewables would increase from 17% to 43% of the electricity and heat sectors between 2020 and 2050 under the Stated Policies Scenario. More aggressive scenarios project higher renewable proportions in 2050: 71% for the Sustainable Development Scenario and 77% for the Net Zero Emissions by 2050 scenario.

Even at present-day deployment of renewable resources, electric grid operators must curtail renewable resources for economic or reliability reasons or because there is excess renewable power. The California Independent System Operator (CAISO) curtailed 1,441 GWh of utility-scale solar and wind generation in the first 11 months of 2021 [5]. The anticipated tripling of solar generation, which is most productive during the mid-day, would be expected to substantially increase curtailment in the future. In the Southwest Power Pool, wind curtailment was more than 3,500 GWh through the first 10 months of 2021 [6]. With deployment of renewable resources expected to continue, large-scale energy storage would be an alternative to curtailment.

In addition to renewable curtailment, day-ahead and real-time electric energy markets now experience periods of negative prices. During 2020 between the hours of 11:00 and 16:00, prices in the CAISO day-ahead market were near or below zero 7.1% of the time (130 hours per year) [7]. Although low energy prices are undesirable for the developers and operators of renewable resources, they are an attractive prospect for Energy Storage Systems (ESS), as well as for electric vehicles or production of hydrogen by electrolysis.

Renewable resources are also characterized by their variability on a range of time scales. Diurnal variability is most evident, with solar power available only during sunlight hours and wind power often indirectly driven by the daily sun cycle. Seasonal variations in renewable output are also apparent as solar intensity varies with the sun angle. Likewise, wind generation varies throughout the year, with the highest performance during the spring and lowest during the summer in the United States [8]. In addition, weather conditions, such as clouds or calms, or extreme events, such as wildfires, windstorms, or a polar vortex, can reduce renewable production for many consecutive days.

Electric grids incorporating large amounts of renewable resources will require complementary dispatchable resources, capable of operating over the entire range of time scales of renewable variability. The cost of LDES is one of the more challenging aspects of maintaining the reliability of a low-carbon grid with high renewable content.

1.2. The need for long-duration energy storage

The duration of an ESS is “the length of time that a storage system can sustain power output at its maximum discharge (power), typically expressed in hours” [9]. The definition of ‘long’ is a matter of perspective related to the mission of the ESS [10]. The ARPA-E DAYS program focused on LDES for storage durations >10 hours to <100 hours, intended “for needs that go beyond daily cycling but are short of seasonal energy time-shift applications” [11].

The economics of LDES can be improved by separating the power-related functions (charging and discharging) from the energy related function of storage. For example, bulk storage media, such as water, air, molten salt, or liquid electrolytes, can be stored in large reservoirs, caverns, or tanks, to flow through power conversion equipment to affect PHS, CAES, thermal energy storage, and flow battery storage. This approach relies on the economy of storing large volumes of low-cost storage media. Thermal-mechanical systems also benefit from scale-up of piping, turbomachinery, and electrical components.

The LDES perspective of the ARPA-E DAYS program was for relatively near-term deployment of LDES competitive with natural gas combined cycles in the then-current grid [12]. The LDES would be charged from renewable power costing \$25/MWh. Charging and discharging power were constrained to be equal, and only ‘pure-play’ ESS were evaluated (e.g., adiabatic CAES without fuel for air heating). An incremental levelized cost of \$50/MWh was allowed for capital and operations and maintenance (O&M) expenses. These parameters then defined a locus of acceptable capital cost for power equipment (\$/kW) and capital cost for energy storage (\$/kWh). With these constraints, the maximum capital cost of storage (including medium, container, and associated equipment and installation) was found to be about \$20/kWh, while the power equipment could not exceed \$1,000/kW.

The deployment of LDES in a decarbonized power system was investigated using an electricity system capacity expansion model with hourly resolution to assess the impact of LDES on the overall mix of resources, including renewable, nuclear, and dispatchable resources in northern and southern regions of the United States [13]. Rather than evaluating alternative ESS technologies, key ESS parameters were parametrically varied to determine the optimum resource mix. Parameters included storage duration, round trip efficiency (RTE), capital cost of charge and

discharge equipment, capital cost of the energy storage, and charge-to-discharge power ratio. The deployment of storage was constrained by the economic competitiveness of combustion turbine-based resources with either post-combustion carbon capture and storage (CCS) or hydrogen fuel produced by steam methane reformation of natural gas with carbon capture. For LDES to displace the fuel-based dispatchable resources, low-cost energy storage (\$/kWh) and high discharge efficiency (conversion of stored energy to electricity) were found to be the most important factors. Full displacement of the fuel-based dispatchable resources (natural gas with CCS and hydrogen fuel combustion turbines) placed severe economic constraints on LDES: energy storage cost <\$1/kWh, RTE>50%, and power cost <\$400/kW.

1.3. Hybrid LDES

Hybrid LDES can potentially overcome these economic constraints by coupling energy storage with combustion turbines in ESS such as CAES, Liquid Air Combined Cycle (LACC), and Liquid Salt Combined Cycle [14]. In each of these approaches, the integration of energy storage with combustion turbines is synergistic, resulting in the following:

- Low fuel heat rate (kJ/kWh-discharge), typically about half that of the simple cycle combustion turbine, due to addition of energy from storage
- Low stored energy rate (kWh-charge/kWh-discharge) by supplementing the stored energy with simple cycle power and exhaust heat
- Low stored media rate (kg-storage/kWh-discharge)

The first two synergies reduce the Marginal Cost of Energy (MCOE) compared to conventional combustion turbine systems using the same fuel, which is especially important with expensive fuels, such as renewable fuels and hydrogen. The third synergy reduces the size and cost of the storage cavern or storage tank and the capital cost of the charging system. In the case of CAES and LACC, using air as the storage medium, cost of storage depends entirely on the cost of the vessel containing the air.

1.4. The Liquid Air Combined Cycle

The LACC is a hybrid LDES technology intended for managing the variability of renewable resources on a range of scales by providing large-scale (10+ GWh) and long-duration (days to weeks) storage.

Figure 1 illustrates the commercial-scale LACC, which uses two ambient air streams at different times for cryogenic liquefaction during charging and as combustion air during discharging. The system consumes electric power to drive refrigeration during charging and consumes fuel and liquefied air during discharging. Discharge power is produced by a turbomachinery cascade that resembles CAES: a conventional combustion turbine whose exhaust heat is captured for heating pressurized air to power an air turbine. Pressurized liquid air is re-gasified using heat rejected from an organic Rankine cycle (ORC), which extracts additional energy from the combustion turbine exhaust gas. A cooling loop maintains gas turbine performance on hot days by controlling the inlet air temperature.

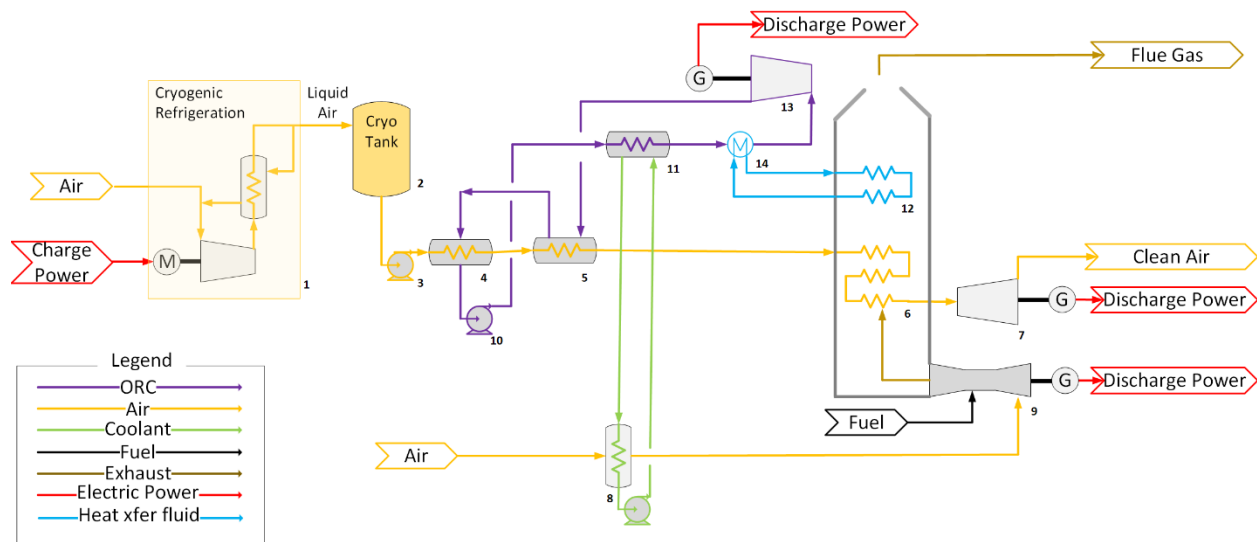


Figure 1. Commercial-scale LACC process

1.4.1. Basic principles and benefits of LACC

ESS technologies have three major components whose cost and performance influence their suitability for different scales and applications: charging process, storage medium, and discharge process. Increasing deployment of VRE resources will drive a need for multi-GWh scale ESS in the future, where the cost of the storage medium is paramount. LACC is expected to be especially valuable at large scale because:

- The charging process (cryogenic liquefaction) benefits from scale on specific capital cost [\$/ (kg/s) of liquid air production] and specific operating efficiency (kWh/kg of liquid air produced).
- The storage medium (liquid air) is free and ubiquitous, has high energy density (e.g., compared to compressed air), requires no active cooling measures, and has low storage loss rates (<0.1% per day). Large (>10 GWh capacity), stainless steel-lined concrete storage tanks, which are proven in the LNG business can be used for liquid air storage.
- The LACC coupling with fossil exhaust heat increases the specific conversion efficiency (kWh/kg of liquid air), which in turn reduces the size and capital cost of the charging process and storage medium.

At 10 GWh scale or greater, the “free” air storage medium provides compelling economic benefits over lithium battery technology (\$100/kWh aspirational target is \$1 billion for storage medium) or Pintail’s Liquid Salt Combined Cycle [15] (\$25/kWh is \$250 million for storage medium). LACC can be sited anywhere, unlike PSH and CAES, which require geologic formations. This facilitates the siting of storage close to load centers or renewable resources to reduce transmission costs and congestion and improve local reliability.

1.4.2. Competition

The future need for renewable backup spanning many consecutive days could be met by these competitive approaches.

- **Energy storage.** Of energy storage systems, only PSH has the potential of being deployed at scale that approaches LACC. Geotechnical and siting challenges limit the potential for PSH deployments. Due to its compactness, LACC would be the only large-scale LDES storage technology that could be safely deployed in a metropolitan area.
- **Hydrogen fueled gas turbines.** Large-scale electrolysis of hydrogen is projected [15] to cost between \$1.50/kg and \$5.00/kg (\$10.55/GJ to \$35.16/GJ) in 2050, depending on the scenario (degree of optimism), production conditions, and cost of electricity used for electrolysis. The storage and pipeline transport of hydrogen to the burner tip would be an additional expense. Two potential scenarios emerge for gas turbine-based systems, depending on the need for renewable backup and the resulting capacity factor of the dispatchable resources.
 - **Simple cycle gas turbines** would likely be preferred for low capacity factor, as is the case with natural gas. However, the cost of expensive hydrogen fuel would make the LACC fuel savings attractive, as they could be about \$50/MWh or more.
 - **Combined cycle gas turbines** typically are more cost-effective when capacity factors exceed 50%, but given the high cost of hydrogen, a modern hydrogen-fueled combined cycle could be a cost-effective alternative at lower capacity factors.
- **Natural gas fired combined cycle with carbon capture and storage (CCS)** would be an alternative depending on how the unit is dispatched. Startup of the CCS unit is likely to take many hours before rated capture efficiency is obtained, so cyclic duty would not be effective at greenhouse gas (GHG) reduction.

The team believes that hydrogen fueled gas turbines are the chief competition for LACC systems, which would likewise be hydrogen fueled in order to provide zero-carbon dispatchable power. The primary competitive advantage for LACC would be fuel savings, the benefit of which would depend on the cost of hydrogen delivered to the burner tip. Savings could be enhanced by capturing the heat of condensation of the water of combustion, which can be fed into the low temperature ORC to improve efficiency. The condensate could be recycled to an electrolyzer, which could be advantageous if on-site hydrogen production is found to be economic compared to pipeline transportation.

1.4.3. Commercial applications

Competitive advantages of LACC include:

- A storage medium that is both freely available without supply chain constraints, and free

- Compact, high density storage, which can also be relatively inexpensive when using customary large-scale storage tanks such as used for LNG
- The low loss rate, less than 0.1%/day, which enables seasonal storage applications

These cost and scale advantages may be particularly appealing in applications requiring storage at the scale of a PSH system (tens of GWh). LACC could be safely sited near or within a metropolitan area to provide long duration storage closer to load and more cost-effectively than PSH.

The major challenge to LACC commercialization is the hybrid nature of the technology – its requirement of a fuel source to power the gas turbine. However, the fuel requirement can be turned to provide reliable fuel supply for polar vortex events which have caused challenges for gas fired power plants. Using liquid air at -195°C pipeline gas can be liquefied at -178°C using relatively inexpensive heat exchangers. During periods of low gas prices, some liquid air could be circulated through a heat exchanger and the cold gas returned to a boil-off gas (BOG) compressor for re-liquefaction. The LNG would provide on-site fuel storage, without risk of pipeline delivery disruptions due to well-head freeze-offs, such as has occurred in ERCOT, and diversion of pipeline gas from power to heating loads as occurs in the Northeast.

LACC can provide synergistic benefits to the use of hydrogen as an alternative to carbon-based fuels. Using a heat exchanger and BOG compressor as above, the density of hydrogen can be increased by a factor of three or more to reduce the footprint of on-site hydrogen storage. The discharge can be configured to condense the water of combustion from the gas turbine exhaust, making about 18% more thermal energy available for use in the ORC bottoming cycle. The condensate could be recycled to the on-site hydrogen electrolyzer to minimize water requirements.

The team believes that the additional fuel synergies will enable smaller LACC systems to be competitive.

1.5. Project objective

The overall project objective was to develop specifications for LACC systems at commercial and demonstration scales following techno-economic optimization of the cost and performance of charging and discharging cycles and the cost of storage.

- Charging optimization will primarily be governed by the cryogenic refrigeration system and is expected to be strongly influenced by scale, with different cost (\$/kW_{charge}) and efficiency (kg/kWh_{charge}) characteristics depending on the endpoint pressure and the type of refrigeration cycle (e.g., N₂, SMR, Cascades, etc.).
- Storage optimization is entirely determined by scale, with a cross-over between pressurized and atmospheric endpoints based on the installed cost of factory-built bullet tanks vs. field-erected tanks.
- As the charging and storage components are expected to dominate capital cost, discharge optimization will be primarily determined by cycle efficiency (kWh_{discharge}/kg) in order to maximize the useful energy produced by the expensive charging and storage systems.

Development of the cost and performance of the charging and storage systems was undertaken by Pintail's subcontractor, Wood Group. Techno-economic optimization of the discharging system was undertaken collaboratively by SwRI and Pintail Power. The development of the system specifications was undertaken by Pintail Power.

2. REQUIREMENTS

To assess the future market and competitive positioning of the LACC technology, a requirements analysis was undertaken, including:

- Examination of projections for the need for long-duration energy storage and its alternatives
- Identification of promising applications and use cases for LACC technology
- A framework for objectively evaluating LACC configurations during techno-economic optimization
- Development of the cost and performance parameters for alternative configurations of LACC technology

2.1. Energy storage terminology

There is no uniformly adopted nomenclature for energy storage applications. Rather, a range of terms are used to describe energy storage:

- Duration in terms of hours, often expressed subjectively as short, medium, long, intra-day, inter-day, weekly, and seasonal. For example, the ARPA-E DAYS program defined long-duration storage as 10 to 100 hours.
- Scale of discharge energy in the range of kWh, MWh, or GWh; the latter is sometimes referred to as large-scale. The term ‘utility-scale’ is applied without regard to size.
- Location, either in-front of the meter (IFM) or behind the meter (BTM), which sometimes reflects ownership of the storage (utility/customer), sometimes reflects tariff structure (wholesale/retail), and sometimes reflects operational control (transmission/distribution).
- Use cases, which may be:
 - Renewable time shifting
 - Transmission and distribution deferral
 - Demand charge abatement
 - Arbitrage
 - Ancillary service (frequency and voltage support)
 - Resource adequacy (capacity)
 - Reliability (backup)

Two important use cases were not considered: resiliency and hedging, which are the technical and economic aspects of dealing with low probability/high impact events such as extreme weather.

2.1.1. Chernyakhovskiy's duration framework

A useful framework was provided by Chernyakhovskiy [16] in terms of the services provided to the grid:

- Essential reliability services (ERS), sometimes referred to as ancillary services, with a response time of milliseconds to seconds and duration less than 1 hour, with potential national deployment of a few GW. Examples of ERS are:
 - Inertial response
 - Fast frequency response
 - Primary frequency response
 - Ramping reserves
 - Contingency spinning reserves
 - Replacement non-spin reserves
 - Voltage support
 - Black start capability
- Traditional peaking capacity, with a response time of minutes and a duration of 4 to 8 hours, depending on regional load profile, with potential national deployment of tens of GW.
- Capacity and diurnal energy time shifting, with a response time of minutes and a duration of 4 to 12 hours, with national potential deployment of more than 100 GW.
- Capacity and seasonal energy time shifting, with a response time minutes, duration of days to months, but with uncertain market potential.

The framework is summarized in Table 1, which correlates the need for these grid services with the grid's renewable content and suggests cost targets for ESS to provide the first three services but left seasonal storage unspecified.

Table 1. Chernyakhovskiy's energy storage framework [16]

Primary Service	Level of RE Penetration	National Potential	Duration	Response Speed	System Cost needed
ERS (Mostly Operating Reserves)	None	A few GW to maybe 10 GW	<1 hour	Milliseconds to seconds	>\$1000/kW (dominated by power cost)
Traditional Peaking Capacity	None to 50%	19s if GW (<100 GW)	4-8 hours	Minutes	\$125-350/kWh
Capacity and Diurnal Energy Time Shifting	20% to 80%	100+ GW	4-12 hours	Minutes	\$50-150/kWh
Capacity and Seasonal Energy Time Shifting	>80%?	Unknown	Days to months?	Minutes to hours	Unknown

2.1.2. SMUD's durations

The Sacramento Municipal Utility District (SMUD) published its SMUD2030 Zero Carbon Plan [17], which defined these storage durations:

- Short duration: 4 hours
- Intra-day duration: 5-12 hours
- Inter-day duration: 13 to less than 48 hours
- Multi-day duration: 48 to 168 hours (1 week)
- Long-term or seasonal: beyond 168 hours

The project used the SMUD duration terminology for all tasks.

2.1.3. Multi-day and seasonal storage

The recent paper by Sepulveda et al. [13] examined the potential for deployment of LDES in decarbonized power systems. The objective was to determine the cost and performance requirements of LDES needed to displace firm dispatchable resources. The work “used an electricity system capacity expansion model (CEM) with high temporal resolution (8,760 h) and detailed operating decisions and constraints to assess the impact of different combinations of LDES design parameters on the overall economics of decarbonized power systems.” This work examined northern and southern grid regions in North America, to determine the least cost means of delivering zero-carbon energy using a combination of:

- VRE, i.e., wind and solar
- Short duration Lithium battery storage
- LDES including:
 - PSH
 - CAES
 - Chemical (hydrogen or syngas) coupled to power cycles
 - Electrochemical flow batteries
 - Thermal storage systems of various kinds
- Firm dispatchable resources including:
 - Load-following nuclear power
 - Gas generation with carbon capture
 - Gas generation fired with blue hydrogen, i.e., steam-methane reforming of natural gas to hydrogen with capture and sequestration of carbon dioxide
 - Energy storage to support large-scale deployment of VRE with dispatchable time-shifting.

Cost and performance assumptions for energy storage alternatives were used to determine the least cost resource mix under each region's 8760 hourly load and weather conditions. First, firm dispatchable resources necessary to complement VRE without LDES were determined and set as reference cases listed in Table 2. This then was used to assess the cost targets at which LDES could displace the alternative dispatchable resources and found:

- Energy storage capacity cost and discharge efficiency are the most important performance parameters. Charge/discharge capacity cost and charge efficiency play secondary roles.
- Energy capacity costs must be $\leq$ US\$20/kWh to reduce electricity costs by $\geq$ 10%. With current electricity demand profiles, energy capacity costs must be $\leq$ US\$1/kWh to fully displace all modeled firm low-carbon generation technologies.
- Electrification of end uses in a northern latitude context makes full displacement of firm generation more challenging and requires performance combinations unlikely to be feasible with known LDES technologies.
- Finally, LDES systems with the greatest impact on electricity cost and firm generation have storage durations exceeding 100 h.

Table 2. Firm dispatchable capacity as a percentage of peak MW demand for Northern and Southern regions [13].

Dispatchable resource	Northern region	Southern region
Nuclear	48.6%	31.9%
Gas w/ CCS	51.3%	36.3%
Gas w/ Blue Hydrogen	47.5%	32.9%

Importantly, the Sepulveda et al. [13] work did *not* consider hybrid energy storage, and in particular LACC technology, as a LDES option. There is good reason to believe that LACC technology would be competitive in their techno-economic framework because:

- It is technically and economically feasible to provide storage durations in excess of 100 hours using cryogenic storage tanks such as those proven with LNG. Use of air as both a phase-change storage medium and working fluid results in exceptionally high storage density (greater than 15 MWh/m²) to reduce land-use requirements and simplify siting close to load or renewable resources.
- Incremental energy storage capacity cost is less than the critical threshold of \$1/kWh (using a 200,000 m³ cryogenic storage tank) and the self-discharge rate from evaporative losses is exceptionally low (about 0.1 wt.% per day for an 80% full tank) making long-term storage both practical and economical.
- Superior fuel efficiency of LACC, compared to conventional gas turbine-based generation, sets the stage for lower marginal cost of energy. Less fuel is consumed by LACC, which is particularly significant with expensive hydrogen or renewable natural gas, while charging can be performed with presumably lower cost renewable energy. Further efficiency improvements are available by harvesting the heat available from water of combustion, which can be productively used by the ORC bottoming cycle.

2.2. Competitive positioning

Accordingly, the team believes that the LACC technology will compete against the alternatives identified in the Sepulveda et al. study [13].

- **Energy storage.** Pumped storage hydro (PSH) has already demonstrated deployment at scale that approaches LACC, but geotechnical and siting challenges limit the potential for PSH deployments. Compressed Air Energy Storage (CAES) using large underground caverns is also a potential competitor, but also has siting limitations – the need for underground storage caverns. Due to its compactness, LACC would be the only large-scale LDES storage technology that could be deployed within a metropolitan area. The most recently available estimates from Lazard [18] for levelized cost of storage (8 hours, 100 MW discharge capacity) show a range of \$152/MWh to \$198/MWh for PSH and \$116/MWh to \$140/MWh for CAES.

- **Hydrogen fueled gas turbines.** Sepulveda et al. [13] assumed the cost of blue hydrogen at \$5.14/MMBtu plus \$10/metric ton for CO₂ sequestration cost, which is equivalent to \$2/kg [13]. Large-scale electrolysis of hydrogen is projected [15] to cost between \$1.50/kg and \$5.00/kg (\$10.55/GJ to \$35.16/GJ) in 2050, depending on the scenario (degree of optimism), production conditions, and cost of electricity used for electrolysis. The storage and pipeline transport of hydrogen to the burner tip would be an additional expense. Two potential scenarios emerge for gas turbine-based systems, depending on the need for renewable backup and the resulting capacity factor of the dispatchable resources.
 - **Simple cycle gas turbines** would likely be preferred for low capacity factor, as is the case with natural gas. However, the cost of expensive hydrogen fuel would make the LSCC fuel savings attractive, as they could be about \$50/MWh or more.
 - **Combined cycle gas turbines** typically are more cost-effective when capacity factors exceed 50%, but given the high cost of hydrogen, a modern hydrogen-fueled combined cycle could be a cost-effective alternative at lower capacity factors.
- **Natural gas fired combined cycle with CCS** would be an alternative depending on how the unit is dispatched. Startup of the CCS unit is likely to take many hours before rated capture efficiency is obtained, so cyclic duty would not be effective at GHG reduction.

Pintail Power believes that hydrogen fueled gas turbines are the chief competition for LACC systems, which would likewise be hydrogen fueled in order to provide zero-carbon dispatchable power. The primary competitive advantage for LACC would be fuel savings, the benefit of which would depend on the cost of hydrogen delivered to the burner tip. Savings could be enhanced by capturing the heat of condensation of the water of combustion, which can be fed into the low temperature ORC to improve efficiency. The condensate could be recycled to an electrolyzer, which could be advantageous if on-site hydrogen production is found to be economic compared to pipeline transportation.

2.3. Product definition

There are two aspects of scale for LDES units:

- The discharge power, P_D
- The discharge duration, t_D

For economy of scale, utilities generally favor large unit capacity. Dispatchable units for peaking are generally in the 50 to 200 MW range, and may use aero-derivative gas turbines for flexibility and fuel efficiency. Aero-derivatives are limited to about 100 MW per unit. Combined cycle plants range to 1000 MW or more and generally do not use the aero-derivative gas turbine.

The Sepulveda et al. [13] study suggests that discharge duration exceeding 100 hours will be needed for the mid-century low-carbon grid. Nevertheless, utility acquisition has been shorter duration. Most large-scale PSH units are typically discharged on a daily basis to provide energy, capacity, balancing and reserves, and an LACC unit would likely serve similar grid needs.

The requirements for LACC systems are based on anticipated market, economic, commercial, and technical needs to deploy the technology by 2030. Table 3 shows the overall requirements of the system.

Table 3. Summary of the design requirements

Metric	Requirement
Max power in discharge mode	100-1000 MW
Modularity	Ability for multiple charge or discharge trains
Max duration of liquid air discharge	13-168 hrs
Fuel source	Natural gas (with ability to utilize hydrogen)
Capacity factor	<50% with ability of daily starts

2.4. Overall plant economics

The economics of hybrid energy storage must include consideration of the fuel cost, while the long-duration nature necessitates consideration of capacity factor. Table 4 presents the key parameters used to calculate the COE. In addition, LACC plant configurations may have different numbers of charge trains, discharge trains, and storage tanks. This section derives COE in \$/MWh for an LACC plant configuration with variable numbers of charge trains, discharge trains, and storage tanks.

2.4.1. Key definitions

Table 4. Key definitions for determining cost of energy (COE)

Parameter	Unit	Description
P_C	kW	Charge electric power per train
P_D	kW	Discharge electric power per train
n_C	-	Number of charge trains
n_D	-	Number of discharge trains
n_T	-	Number of storage tanks
d_C	h	Annual charge duration. The number of hours that the system charges annually.
d_D	h	Annual discharge duration. The number of hours that the system discharges annually.

Parameter	Unit	Description
$E_C = n_C \cdot P_{C,train} \cdot d_C$	kWh	Total energy required to charge storage per year
$E_D = n_D \cdot P_{D,train} \cdot d_D$	kWh	Total energy discharged per year
M_M	kg	Total mass of the storage medium stored per year
M_L	kg	Total mass of the storage medium lost to leakage per year
$m_C = M_M/E_C$	kg/kWh	Specific mass flow rate of the storage medium into the storage tank (i.e., the amount of mass added per unit of energy to the storage)
$m_D = (M_M - M_L)/E_D$	kg/kWh	Specific discharge mass flow rate of the storage medium (Ratio of storage medium mass to the discharge energy or storage medium mass flow rate to discharge power)
$C_{D,train}$	\$	Capital cost of the discharge process equipment per train
$C_{C,train}$	\$	Capital cost of the charge process equipment per train
C_T	\$	Capital cost of a single storage tank
c_M	\$/kg	Specific capital cost of the storage medium
$C_M = M_M \cdot c_M$	\$	Capital cost of the storage medium
$C_C = C_{C,train} \cdot n_C$	\$	Capital cost of the charge subsystem
$C_D = C_{D,train} \cdot n_D$	\$	Capital cost of the discharge subsystem
$C_S = C_M + C_T \cdot n_T$	\$	Capital cost of the storage subsystem
$C_{ESS} = C_{C,train} \times n_{C,train} + C_{D,train} \times n_{D,train} + C_{tank} \times n_{tank}$	\$	Overnight capital cost of the energy storage system
$AF = \frac{i(1+i)^n}{(1+i)^n - 1}$	-	The amortization rate over n years at an annual interest of i
$CCOE = \frac{C_{ESS} \cdot AF}{E_D}$	\$/MWh	Capital cost of energy. The ratio of capital costs to the amount of energy discharged per year.
$w_C = m_C \cdot n_C \cdot P_{C,train}$	kg/h	Mass flow rate of storage medium during charge (at rated capacity)
$w_D = m_D \cdot n_D \cdot P_{D,train}$	kg/h	Mass flow rate of storage medium during discharge (at rated capacity)
w_L	kg/h	Mass flow rate of leakage out of the storage system

Parameter	Unit	Description
$P_D = n_{D,train} \times P_{D,train}$	kW	Power output
$CF_{max} = \frac{w_C - w_L}{w_C + w_D}$	-	Maximum capacity factor based on flow rates
$SER = E_C/E_D = m_D/m_C$	-	Stored energy rate (ratio of the amount of energy charged to the amount of energy discharged per cycle)
Q_D	MJ	Fuel required to discharge the system per year
$FHR = Q_D/E_D$	MJ/MWh	Fuel heat rate
c_F	\$/MJ	Specific marginal cost of fuel
c_P	\$/MWh	Specific marginal cost of purchased electricity during charge
$MCOE = FHR \cdot c_F + SER \cdot c_{P,C}$	\$/MWh	Specific marginal cost of energy discharged
OC_C	\$/year	Operational costs of a single charge train per year
OC_D	\$/year	Operational costs of a single discharge train per year
OC_T	\$/year	Operational costs of a single storage tank per year
$OCOE = \frac{OC_C \cdot n_C + OC_D \cdot n_D + OC_T \cdot n_C}{E_D}$	\$/MWh	Operational cost of energy
$COE = CCOE + MCOE + OCOE$	\$/MWh	Cost of energy

2.4.2. Levelized cost

The levelized cost of energy (LCOE) or levelized cost of storage (LCOS) metrics are useful for comparing technologies whose costs vary over the lifetime of a project for reasons such as:

- Annual performance degradation
- Capacity augmentation at specified intervals
- Accelerated depreciation
- Project financing (e.g., tax-equity flips)
- Cost escalation

The distinction between COE and LCOE is that COE is the cost for a particular year, whereas LCOE is the sum of the COE for a sequence of years, with the future COE discounted by an interest rate. LCOE accounts for time-varying performance and cash flows (performance degradation,

escalation/inflation of fuel/electricity costs and O&M expenses) as well as time varying cash flows (such as accelerated depreciation/amortization for tax purposes, tax credits and tax financing, and end-of-life salvage value). Absent the time-varying components, COE and LCOE are equivalent. To avoid potentially misleading assumptions about the time-varying components, whose future importance is discounted, this project uses COE for evaluation. Each LACC alternative is expected to have similar cost behavior over project life, so the COE is approximately equivalent to the levelized cost method.

2.4.3. Capacity factor

According to the NREL Annual Technology Baseline [19]:

Capacity factor is generally defined as the ratio of actual annual output to output at rated capacity for an entire year, using a long-term average over the lifetime of an asset, without curtailment for renewable generation. The default capacity factors listed in the ATB data spreadsheet are meant to be representative—actual plant capacity factors will depend on many factors power plant-specific conditions. The annual capacity factor ignores specific operating behavior such as ramping, startup, and shutdown that could be relevant for more-detailed evaluations of generator cost and value. Electricity generation technologies have different capabilities to provide capacity and flexibility services. These services are difficult to value and depend strongly on the system in which a new generation plant is introduced. These services are represented in electric sector models such as ReEDS and in corresponding analysis results such as the Standard Scenarios.

Analyses like the Lazard Levelized Cost of Storage [18] do not consider marginal economics, but instead assume that the ESS would operate on a daily cycle (350 days per year) between 10% and 100% depth of discharge per cycle. The storage for an LACC system would typically span multiple days, weeks, or months, rather than daily charging and discharging, but the same principal can be applied to discount the available discharge energy by 10% compared to the maximum possible energy that is stored in a year.

2.4.4. Determination of capacity factor for LACC systems

Capacity factor is limited in two ways:

- The marginal economics of the plant vs. alternative units, with the LACC unit dispatched only when its marginal cost is less than the price-setting unit.
- The availability of the plant to discharge, which requires sufficient liquid air to operate the discharge train.

The maximum capacity factor can be calculated based on the mass flow rates. Assuming that no mass is stored longer than a year, a mass balance can be used to compare the mass that is charged by the cryogenic system to the mass of liquid air that is expended through the discharge cycle and the amount that is lost through leaks, boil-off or other losses.

Combining the mass balance with the modified definition from Lazard [18], the capacity factor for a given setup was estimated to be:

$$CF = 0.90 \cdot \frac{w_C - w_L}{w_C + w_D} \quad (1)$$

The capacity factor derate factor is shown as 0.90 and is chosen in analogy with the approach used by the Lazard Levelized Cost of Storage reports [18], where it is assumed that ESS operate at 90% depth of discharge.

2.4.5. Stored energy rate

The stored energy rate (SER) is a key economic parameter, as it factors into the marginal COE (MCOE) as the ratio of charge energy to discharge energy. A smaller SER is better, which can be obtained by improving efficiency, i.e., reducing the amount of energy consumed to produce a unit of liquid air, and/or increasing the amount of energy produced using the liquid air during discharge. The LACC approach allows independent optimization of the charging and discharging processes in order to reduce SER. In addition to reducing MCOE, an optimal discharge sub-system can also reduce the size (and capital cost) of the charge and storage sub-systems by reducing the amount of liquid air required for a given annual discharge energy.

2.4.6. Marginal cost considerations

In an economic dispatch operating profile, the ESS would maximize profits from energy sales via arbitrage – charging when the price of electricity, c_P , is low and discharging when high.

- Charging is constrained by the availability of volume in the storage vessel to hold storage medium, i.e., it is not possible to charge a full vessel.
- Charging may occur at a fraction of the rated charge power P_C , for example in order to provide regulation services by increasing or decreasing the charge power.
- Both charging and discharging decisions are influenced by the time horizon at which energy markets clear. Because of system inertia, LACC would be most likely participate in day-ahead, rather than real-time markets.
- Discharging would generally occur when MCOE is less than c_P . However, since the liquid air inventory is a sunk cost once produced, discharging may also occur at negative ‘park spread’ in order to provide cash flow to cover fixed costs.
- Because BOG results in oxygen enrichment, charging and discharging may also occur to manage the oxygen fraction in the liquid air.

The specific marginal cost of discharge energy, $MCOE_D$, depends on c_P during charging. With increasing levels of VRE over-generation, electricity for charging may be available at favorable cost, so more charging capacity (trains) and storage may increase capacity factor.

During periods when VRE is not abundant, the marginal price of energy is set by the highest marginal cost resource that clears the market, such as a simple cycle gas turbine. Reducing MCOE_D makes it more likely that an ESS would clear the market, which would increase the *CF* and in turn reduce the COE.

2.4.7. Maximum capacity factor

The maximum capacity factor is fundamentally a mass balance on the liquid air and depends on the number of charge and discharge trains. The specific mass flow rate of liquid air per discharge train (m.t./MWh) m_D . The total mass flow consumed during discharge is shown in Eq. (2).

$$M_{D,annual} = E_{D,annual} \cdot m_D = d_D \cdot w_D \cdot n_{D,train} \quad (2)$$

And the total mass flow produced during charging is shown in Eq. (3):

$$M_{C,annual} = d_C \cdot w_C \cdot n_{C,train} \quad (3)$$

From a mass balance, assuming tank inventory begins and ends at the same point, the total mass discharged throughout the year is described in Eq. (4).

$$M_{D,annual} = M_{C,annual} - M_{leak,annual} \quad (4)$$

or

$$d_D \cdot w_D \cdot n_{D,train} = d_C \cdot w_C \cdot n_{C,train} - w_l \cdot n_{tank} \cdot 8760 \quad (5)$$

or

Assuming that the time frame is based on one year, the time constraint can be defined as Eq. (6).

$$d_D + d_C \leq 8760 \quad (6)$$

Solving for d_C and substituting the equality,

$$d_D = \frac{(8760 - d_D) \cdot w_C \cdot n_{C,train} - w_l \cdot n_{tank} \cdot 8760}{w_D \cdot n_{D,train}} \quad (7)$$

$$d_D = \frac{8760 \cdot w_C \cdot n_{C,train} - d_D \cdot w_C \cdot n_{C,train} - w_l \cdot n_{tank} \cdot 8760}{w_D \cdot n_{D,train}} \quad (8)$$

$$d_D \left(1 + \frac{w_C \cdot n_{C,train}}{w_D \cdot n_{D,train}} \right) = \frac{8760 \cdot w_C \cdot n_{C,train} - w_l \cdot n_{tank} \cdot 8760}{w_D \cdot n_{D,train}} \quad (9)$$

Multiplying both sides by $w_D \cdot n_{D,train}$

$$d_D(w_D \cdot n_{D,train} + w_C \cdot n_{C,train}) = 8760 \cdot w_C \cdot n_{C,train} - w_l \cdot n_{tank} \cdot 8760 \quad (10)$$

So, the maximum annual discharge hours is:

$$d_D = \frac{8760 \cdot (w_C \cdot n_{C,train} - w_l \cdot n_{tank})}{w_D \cdot n_{D,train} + w_C \cdot n_{C,train}} \quad (11)$$

Accordingly, the maximum Capacity Factor is expressed entirely in terms of system design parameters.

$$CF_{max} = \frac{d_D}{8760} = \frac{w_C \times n_{C,train} - w_l \times n_{tank}}{w_D \times n_{D,train} + w_C \times n_{C,train}} \quad (12)$$

Without locational pricing information or a study such as Sepulveda et al. [13], there isn't a basis for using marginal cost to determine capacity factor. Accordingly, in analogy with the Lazard Levelized Cost of Storage analysis [18], the economic analysis of LACC systems will use the maximum capacity factor, CF_{max} , discounted by 10% assuming that the LACC technology is able to discharge 90% of its available energy each cycle.

The key plant economic optimization question then becomes the number of charge and discharge trains, and the number of storage tanks.

2.5. Design considerations: charging and storage sub-systems

The capital cost of the cryogenic refrigeration system was expected to be the dominant factor in the Capital Cost of Energy determination, with efficiency of somewhat lesser importance to the MCOE. The project began with a benchmark specific energy consumption of 439 kWh/ton to produce liquid air [20].

Pintail Power subcontracted to Wood Group (Wood) to evaluate cost and performance trade-offs of alternative refrigeration cycles: nitrogen, mixed refrigerant, and cascade. In addition, two liquid air pressure/temperature delivery conditions were considered: atmospheric pressure (-195°C saturation temperature) and pressurized (-178°C saturation temperature). The pressurized condition would reduce energy consumption and potentially leverage LNG process designs, but requires pressure vessels which would limit the scale at which LACC could be deployed. Another consideration was the maximum size of electric motors and associated equipment for driving the compressor. Wood also considered the need for technology licensing and focused on two cryogenic processes that are openly available:

- Linde-Hampson cycle
- Claude cycle

Wood undertook process modeling to estimate efficiency. The cost estimate was calculated based on comparable equipment from both reference projects and commercially available equipment. An additional factor of 3.5 was multiplied to the equipment cost to get the total installed cost (TIC). Assuming that there are no additional civil works, including site preparation, soil decontamination, and/or soil improvement activities, this additional factor provides reasonable margin above the equipment cost to have an appropriate TIC estimate, with this level of definition. Additionally, the total O&M cost has been estimated as 2% of the capital cost.

The atmospheric-pressure Claude Cycle was used as the basis for the system cost and performance analysis. It is important to note these limitations:

- Efficiency (MWh/ton) is substantially worse than the initially assumed reference. Further investigation is warranted to understand the merits of using a more efficient process.
- The 3.5 EPC multiple of equipment cost should be amenable to value engineering, negotiation, and experience, especially when LACC projects use multiple charging trains.

Further detail is provided in Section 3.

2.6. Design considerations: discharge sub-system

The cost-effectiveness of the overall LACC system was expected to be more strongly influenced by the capital cost of the cryogenic charging and storage sub-systems than by the cost of the discharging sub-system. Accordingly, reduction of the specific mass flow during discharge m_D is expected to be a principal design consideration. However, this cannot be the optimization objective, since the trivial solution ($m_D = 0$) also results in no energy storage. Instead, a baseline liquid air flow rate, $m_D \sim 75$ kg/s, was selected for initial process screening. This value is equivalent to 50% of the Claude Cycle daily production rate, which would permit a capacity factor of up to 50% (neglecting startup delays for the charging and discharging sub-systems).

An initial process screening was undertaken by Pintail Power using Ebsilon Professional powerplant modeling software [22] to understand the impact of different design choices.

2.6.1. Gas turbine

Two gas turbines of similar power output were compared to assess whether aero-derivative or frame technology would be superior, considering gas turbine mass flow and exhaust gas temperature. Details for the two gas turbines can be found in Table 5.

Table 5. Gas turbine alternatives

Frame	Aeroderivative
SGT-800	LM9000
62.5MW	68MW
135.7 kg/s @ 595°C	186.6 kg/s @ 471°C
$\eta = 41.1\%$	$\eta = 42.1\%$

Other considerations included size limits of the aero-derivatives, versus the potential for economy of scale available with larger F-class or H-class gas turbines.

Specific mass flow rate m_D could potentially be reduced with air or steam injection into the gas turbine. Steam injection is intriguing as a potential mitigation for NO_x emissions in hydrogen

fueled systems. However, steam/air injection would not be available with an aero-derivative, so this option was discarded.

As with steam combined cycles, efficiency (fuel heat rate and specific air consumption) favors higher exhaust gas temperature, and exhaust gas heat transfer area also tended to be lower, reducing the cost of heat recovery equipment.

2.6.2. Flue gas condensation

LACC cycle conditions permit the water of combustion to be condensed and its heat of condensation to be usefully coupled into heating regasified air and/or ORC working fluid. Condensate could be recycled for steam injection, used for cooling of the liquefaction process, or used as electrolysis feedstock. Condensate temperature would be a result of air and ORC flow rate and could be arbitrarily selected although it would seem thermodynamically optimal if the flue gas temperature were close to the ambient temperature.

Recycling substantially all the latent heat of condensation for the water of combustion would require more liquid air flow to provide sufficient cooling. This would tend to reduce the fuel heat rate while increasing the SER and specific air consumption, which may not be economically optimal.

Flue gas condensation risks sulfur corrosion of the heat exchanger surfaces when consuming natural gas, so only non-condensing cycles were considered. There is nevertheless an opportunity to capture additional energy in a hydrogen-fueled LACC.

2.6.3. Fuel selection

Both natural gas and hydrogen fuels could be used with LACC, and the higher exhaust moisture content of hydrogen vs. methane must be evaluated to understand impact on the condensing heat duty. Since the aero-derivative combustions systems would not likely be modified for hydrogen (being focused on airframes), and since this project's area of interest was fossil fuel rather than hydrogen, natural gas was selected.

2.6.4. ORC working fluid

Selection of the ORC working fluid must consider technical, environmental, and safety factors including:

- Low freezing point (below the liquid air temperature)
- Low global warming potential (GWP)
- Low ozone depletion potential (ODP)
- High critical temperature for safe and economical storage when the plant is idle
- Positive condensing gage pressure to avoid air ingress at the condenser

Several hydrocarbon refrigerants were considered as working fluids, as listed in Table 6.

- R-E-140 (dimethyl ether) is attractive because of its low GWP, but must be excluded because of its high melting point, which poses a risk of freezing in the ORC condenser during startup.
- R-140 (ethane) has a good combination of performance due to its low boiling point, but there are some safety concerns with storage of ethane on hot days, when the fluid becomes supercritical. It will be important to minimize the fluid inventory and provide safe storage of the liquid in a pressure vessel when the system is not operating.
- R-290 (propane) would presumably be safer since inventory could be stored in liquid form at much higher temperature than ethane. With propane, the specific air consumption is several percent more than with ethane, even with sub-atmospheric pressure condensation, which entails larger piping diameter due to lower vapor density.

Table 6. Refrigerant alternatives

Fluid Properties	R-140 (ethane)	R-E-140 (dimethyl ether)	R-290 (propane)
Triple Point Temp [°C]	-182.78	-141.49	-187.63
Normal Boiling Point [°C]	-88.581	-24.782	-42.114
Critical Point Temp. [°C]	32.172	127.23	96.74
Critical Pressure [bar]	48.722	53.368	42.512
Critical Density [kg/m ³]	206.18	273.65	220.48
Melting Point [°C]	-182.8	-141	-187.7
ODP	0	0	0
GWP (100-yr)	5.5-10.2	1	3.3-9.5

Ethane was found to reduce specific air consumption by about 10% by permitting a lower condensing temperature while remaining above atmospheric pressure, which is desirable to avoid air ingress. In addition, ethane's higher density allowed for smaller heat exchangers.

2.6.5. Rating condition

The availability of gas turbine inlet air cooling is advantageous for maintaining system power output on hot days. When operating at cooler ambient conditions, it is expected that liquid air flow would be reduced or that low-grade heat could be supplied from a process, service cooling system, or the ambient. Rather than the customary ISO rating condition of 15°C, a rating condition of 40°C was selected.

2.6.6. Coupling of inlet air cooling

The cooling duty for inlet air chilling was evaluated with two coupling locations:

- Cold regasified air between heat exchangers 4 and 5
- Cold ORC working fluid following the recuperator

It is anticipated that a single-phase liquid coolant would circulate between the inlet air chiller (8) and the ORC preheater (13). Although the ORC working fluid could be used for heat transfer in inlet air chiller (8), a propylene-glycol antifreeze solution permits customary inlet air heat transfer equipment to be used and avoids the risk of introducing a flammable agent at the gas turbine inlet.

Coupling to the ORC working fluid (depicted in Figure 2) was preferable from an exergetic perspective (smaller temperature difference) and reduced the impact of the cooling load on the air cycle to simplify off-design operation and control.

2.7. Reference process design

Based on the review of the design alternatives, the reference process design shown in Figure 2 was identified and selected for in-depth analysis.

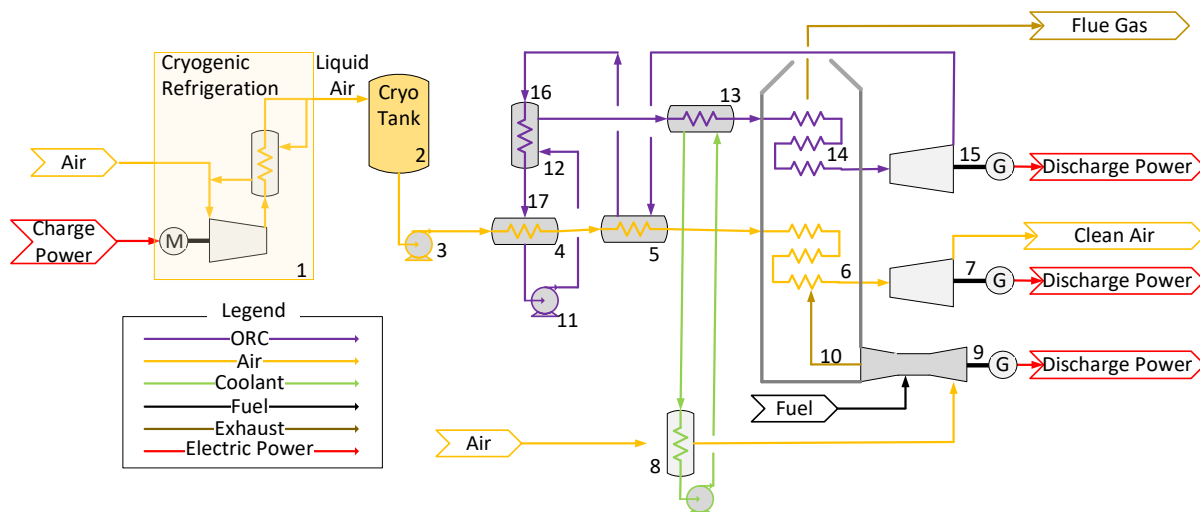


Figure 2. Reference process design

The reference process design included the following components cross-referenced to the diagram:

1. **Cryogenic refrigerator.** During charging, a cryogenic refrigeration system uses low-cost or excess renewable energy to cool air to the liquid state. Various processes have been developed and perfected, but all rely on compressors, heat exchangers, and expanders to cool gas and reject heat to the environment. In an LACC application, the compressors are motor driven to consume electricity.

2. **Storage tank.** Liquid air is stored at near-atmospheric pressure in an insulated tank permitting long-term storage of weeks or months. For example, evaporative loss (BOG) of LNG ranges from 0.012 wt.% per day for an 80% full tank to 0.12 wt.% per day for a 10% full tank [15]. Losses would be only slightly higher at the lower bubble point of liquid air (-198°C) compared to LNG (-162°C).
3. **Cryopump.** A conventional cryopump, such as used for LNG send-out, pressurizes liquid air from the storage tank.
4. **Regasification.** The pressurized liquid air is converted back to vapor by absorbing heat rejected during condensation of the ORC working fluid.
5. **Preheating.** The pressurized regasified air is further heated by cooling the ORC working fluid prior to its condensation.
6. **Exhaust gas heating.** The pressurized air is then heated to high temperature using exhaust gas from a gas turbine.
7. **Air turbine generator.** Pressurized hot air flows through a turbine generator, similar to a steam turbine, to produce power. The clean, dry air is exhausted to atmosphere.
8. **Inlet air cooling.** Ambient air is cooled, increasing the air density to allow the gas turbine to operate at its maximum power point and maximize the exhaust heat available for extracting stored energy. Heat absorbed from the inlet air is transferred via a cooling loop, filled with anti-freeze coolant, to pre-heat the regasified air.
9. **Gas turbine generator.** Any suitable gas turbine can provide the exhaust heat used to drive the discharge cycle. By controlling the inlet air temperature via the cooling loop, this turbine can always operate at optimum conditions to maximize the cycle efficiency and power output.
10. **Exhaust heat recovery.** Exhaust heat from the gas turbine is extracted by two heat exchanger coils. The first coil heats pressurized air from the cooling loop pre-heater; the second coil heats the ORC working fluid.
11. **Pump.** Liquid refrigerant is pumped to pressure.
12. **Recuperative heating.** The pressurized liquid refrigerant is preheated in a recuperator by low pressure refrigerant exhausted from the ORC turbine.
13. **Preheating.** Further heating is obtained by heat transfer from the coolant loop that provides gas turbine inlet air chilling.
14. **Exhaust gas heating.** The refrigerant is evaporated and superheated by gas turbine exhaust.
15. **ORC turbine generator.** The pressurized hot refrigerant produces power in a turbine and exhausts as low-pressure refrigerant.

16. **Recuperative cooling.** The low-pressure refrigerant vapor is cooled in the recuperator by the condensed refrigerant liquid.
17. **Condenser.** Finally, the ORC loop is closed in the condenser, where refrigerant transfers its latent heat of cooling to the liquid air, which absorbs latent heat of evaporation.

2.7.1. Performance of discharge system

Performance of turbomachinery, heat transfer equipment, vessels, and piping components were modeled with efficiencies, pressure drops, and heat losses representative of combined cycle equipment. Table 7 shows the key assumptions and performance metrics for the discharge system. The ORC operates outside customary conditions, so similar performance assumptions were made for its turbine and heat exchangers. The Ebsilon Professional gas turbine library's vendor performance correction factors were used to adjust for inlet and exit conditions.

Table 7. Discharge system performance

Parameter	Performance
Gas turbine	62 MW Siemens SGT-800
Ambient Temperature	40°C
Fuel	Natural gas:
Net Power	116,979 kW-discharge
Net Fuel Heat Rate (FHR)	5138 kJ/kWh-discharge HHV 4631 kJ/kWh-discharge LHV
Net Stored Energy Rate (SER) @ 0.439 MWh/ton	3672 kJ/kWh 1.02 kWh/kWh-discharge
Net Specific Air Consumption	2.32 kg/kWh-discharge
Liquid Air Flow Rate	271 metric tons/hour (309 cubic meters/hour)

2.7.2. Cost of discharge system

A rough estimate of the cost of the discharge subsystem is \$1,600/kW-discharge, a 50% premium over a comparably sized steam combined cycle [33]. The cost savings from the simplicity of the air cycle, without a steam condenser, are offset by the complexity of the ORC, additional heat exchangers, and the cryogenic conditions. Cost estimates were refined as the project progressed.

2.8. Summary

The team performed a requirements analysis to understand trade-offs and define how the project would be evaluated going forward. Factors such as the competitive landscape, the LACC operating environment and design requirements were considered. From the technical perspective, the specific liquid air consumption m_D was considered to be a key design parameter, as it drove the cost of the charging system and marginal cost of energy.

3. CRYOGENIC CHARGING AND STORAGE SYSTEM

This section describes trade studies undertaken to evaluate the cost and performance of the charging and storage systems for use with LACC. Because cryogenic refrigeration is both capital and energy intensive, it is vital to estimate the cost and performance of alternative processes, and to assess the validity of assumptions from the literature, notably the benchmark specific energy consumption of 439 kWh/tonne for production of liquid air [20].

3.1. Decoupling of charge and discharge

LACC is distinguished from alternative cryogenic energy storage technologies in two principal ways:

- The use of fuel in a gas turbine to provide power and heat, i.e., diabatic vs. adiabatic process
- De-coupling of the charging and discharging systems.

Adiabatic LAES recovers, stores, and recycles cold temperatures, as shown in Figure 3, in order to reduce the energy consumption of the charging process, whereas LACC uses the cold energy to increase power production during discharge. As a result:

- Adiabatic LAES has lower specific energy consumption during charging – about 0.297 MWh/ton for atmospheric pressure storage [23].
- Adiabatic LAES has higher specific air consumption during discharging – 333 to 552 kJ/kg (10.8 to 6.5 kg/kWh) [23].

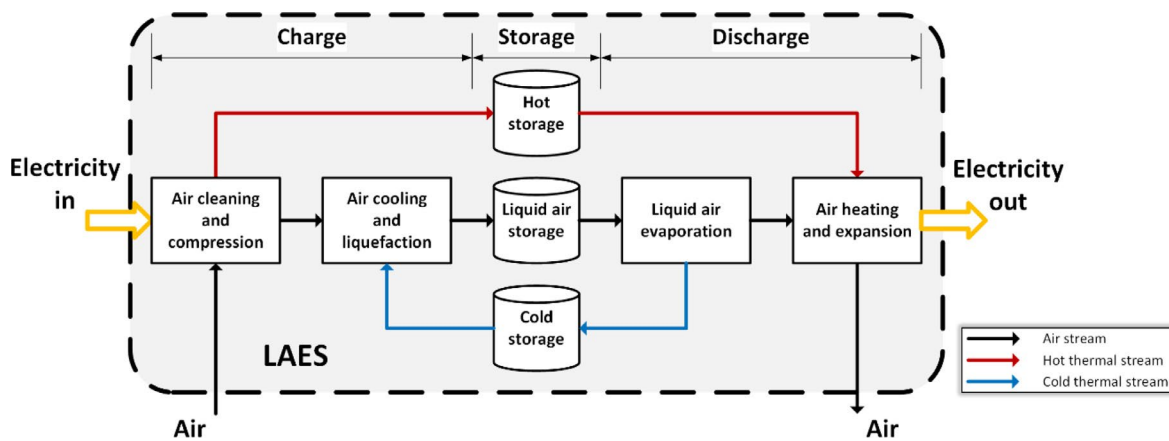


Figure 3. Adiabatic LAES process. Figure taken without permission from Vecchi et al. [23].

The LACC approach:

- Uses one fifth to one third as much liquid air per unit of electrical output (albeit with about half of the electrical power resulting from the gas turbine), so the charging energy per unit of discharge energy is lower than the adiabatic system.

- Eliminates the capital cost and complexity of the cold recycle and storage system.
- Allows a conventional liquefaction system to be employed without the integration challenges of the cold recycle, e.g., charging when cold storage is depleted, such as during system startup, after an extended layup, or as a result of off-design operation.

3.2. Charge cycle trade studies

To quantify the potential benefit of the LACC approach Wood was contracted to perform the charging and storage system trade study aimed at selection of a suitable cryogenic process and a cost estimated for producing liquid air [24]. The study considered the practical size limits of electric motor driven compressor trains and the expectation that minimizing capital cost would be essential for the overall economics of LACC. Wood was charged with evaluating these options consistent with the requirements analysis:

- Liquid production rates of 50% and 100% of the nominal discharge flow rate of 75 kg/s
- Atmospheric and pressurized liquid air end points for storage

During this cryogenic cycle analysis, Wood considered two different processes to generate liquid air – the Claude Cycle and the Linde-Hampson Cycle.

A pressurized sphere could be used for a smaller scale pilot plant when the required discharge rate is low enough to allow for the smaller storage volume. Lower power requirements, as well as smaller and cheaper equipment, are clear benefits to the charging cycle when a pressurized storage vessel is applicable. With the larger liquid air requirements of utility-scale LACC systems, atmospheric tanks have been considered in order to provide larger storage volumes of liquid air. However, due to the limitations of commercially available equipment, the production rate of liquid air is limited to 6,480 tons/day per train. Any discharge rates for the LACC cycle over that amount will need to be addressed with the addition of duplicate trains of air liquefaction.

Due to its lower cost and greater efficiencies, as can be seen in Table 8 and Table 9 below, the Claude Cycle shows it could be very advantageous for liquid air production upstream of the LACC.

Table 8. Power and efficiency study for charge system

Cryogenic Cycle	Power [MWe]	Liquid Air Flow Rate [mtpd]	Efficiency [MWh/ton]	Comments
Claude Cycle	185	6,480	0.685	Liquid air stored at atmospheric pressure
Claude Cycle	82.5	3,240	0.611	Liquid air stored at 7 bar
Linde-Hampson Cycle	245	6,480	0.907	Liquid air stored at atmospheric pressure
Linde-Hampson Cycle	97.5	3,240	0.722	Liquid air stored at 7 bar

Table 9. Air liquefaction cost summary

Cryogenic Cycle	Total Installed Cost	O&M Cost	Comments
Claude Cycle	\$526.0MM	\$10.52 MM/year	200k m ³ atmospheric storage tank, 6480 mtpd flow
Claude Cycle	\$251.7MM	\$5.03 MM/year	3k m ³ pressured storage tank, 3240 mtpd flow
Linde-Hampson Cycle	\$612.1MM	\$12.24 MM/year	200k m ³ atmospheric storage tank, 6480 mtpd flow
Linde-Hampson Cycle	\$269.1MM	\$5.38 MM/year	3k m ³ pressured storage tank, 3240 mtpd flow

One benefit of either cryogenic cycle is that they are both considered open art, so licensing should not be required for the use of the cycle itself. However, it should be noted that the cost of the mixed refrigerant in the Linde-Hampson cycle and the potential licensed equipment in both cycles will need to be factored into the budget as additional costs associated with these air liquefaction cycles.

3.2.1. Claude cycle

Figure 4 shows the cycle diagram for the Claude cycle with atmospheric storage conditions. The Claude cycle was anticipated to be the most desirable for the supply of liquid air to the LACC due to its lower equipment count and the ability to cool with a single fluid compressed at a single pressure level. The cooling can be done without an additional refrigerant because of the cooling achieved from expansion. About 75-80% of the compressed air is sent to the expander, and it is then recycled and used to cool the remaining air. Despite the large amount of recycle used for the cooling of the air, the Claude cycle still ends up as the more efficient of the two air liquefaction cycles.

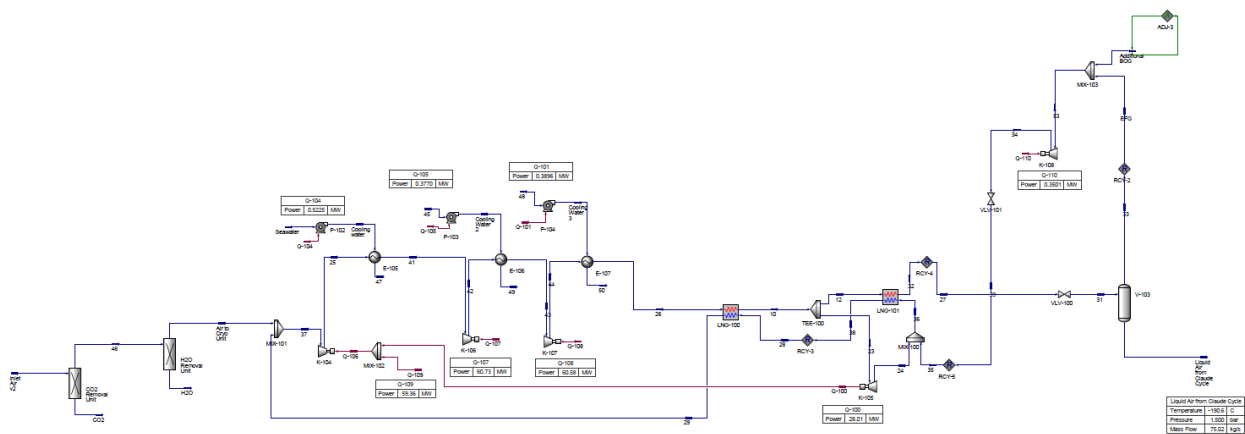


Figure 4. Claude cycle diagram

3.2.2. Linde-Hampson cycle

The Linde-Hampson cycle provides another viable option to charge the storage unit with liquid air. This is a proven option that has been used throughout the world in different applications of cryogenics. Figure 5 shows the cycle diagram for the Linde-Hampson cycle with atmospheric storage conditions. In order to achieve cold enough temperatures in this cryogenic cycle, the addition of a mixed refrigerant (MR) cycle or propane is required. While the mixed refrigerant is an additional expense compared to the Claude cycle, use of a MR is common and does not cause any operational issues that may hamper the feasibility of the Linde-Hampson cycle. However, similar to the Claude cycle described above, there will be some additional licensing costs associated with use of the cryogenic heat exchangers provided by the licensor.

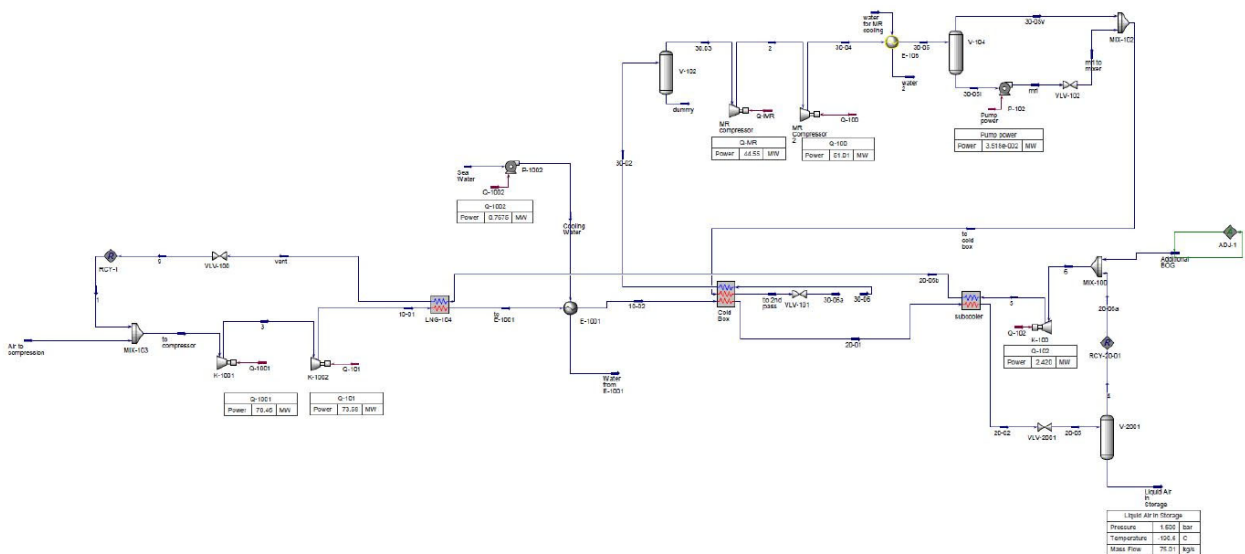


Figure 5. Linde-Hampson cycle diagram

The Linde-Hampson cycle begins with a similar look to the Claude cycle in that it starts with multiple stages of compression at the inlet of the air cycle. However, the Linde-Hampson cycle does require a higher compression ratio due to its design. Also, because there is no expander in this cycle, that larger expansion work is lost. Downstream of the compressors, the air is sent through four separate heat exchangers, with one of them being the cryogenic heat exchanger that

uses MR to cool the air to low temperatures. In this study, only mixed refrigerant has been used as the refrigerant for this cycle as the use of propane would lead to lower efficiencies and higher costs. After cooling the air, the fluid is sent through an expansion valve and into the storage unit. A recycle line to an end-flash gas (EFG)/BOG compressor is taken from the top of the storage unit and is used for cooling air in two of the heat exchangers.

3.2.3. Charge system cost estimate

Wood's equipment cost estimate was calculated based on comparable equipment from both reference projects and commercially available equipment. An additional factor of 3.5 was multiplied to the equipment cost to obtain the total installed cost. Assuming that there are no additional civil works, including site preparation, soil decontamination and/or soil improvement activities, this additional factor provides reasonable margin above the equipment cost to have an appropriate TIC estimate with this level of definition. Additionally, the total O&M cost has been estimated as 2% of the capital cost, as can be seen below in Table 10.

Table 10. Charge system cost breakdown

Parameters	Claude Cycle	Linde-Hampson Cycle
Capital Costs	\$526.0MM	\$612.1MM
Equipment Costs	\$129.7MM	\$154.3MM
Construction Costs	\$324.0MM	\$385.5MM
200k m ³ storage	\$72.3MM	
100k m ³ storage	\$46.1MM	
O&M Costs (per year)	\$10.52MM	\$12.24MM
O&M Cost (per day)	\$14,800	\$19,600

Storage tanks of 200,000 m³ and 100,000 m³ have both been priced (there is about a \$26MM difference between the price of each tank). However, the size and number of the storage tanks will be determined by both the discharge rate, the charging time, and the time between charging, including the startup time.

The scale of the contemplated storage systems is noteworthy, as the 116,979 kW baseline plant consumes 308.6 m³/h. Accordingly, the 200,000 m³ storage tank would be capable of discharging 75.8 million kWh, so the capital cost of storage would be approximately \$1/kWh of discharge energy.

Start-up times are variable depending on if it is a warm start-up or cold start-up. During a cold start-up, with liquid air in circulation from the storage tanks to the cold part of the cryogenic cycle, the start-up period can be as low as 2.5 hours. However, during a warm start-up, the time can almost double, up to 4.5 hours. Reduction of warm start-up time is limited by the time required to

cool off the brazed aluminum heat exchangers, as they are very sensitive to thermal stress. Potential to use coil wound exchangers could improve the cool down time. However, this would be dependent on manufacturer analysis and would likely lead to higher capital costs with marginal improvements in start-up times. This operational design should be further analyzed in the future to avoid additional start-up costs during liquid air charging.

The estimate for the start-up costs seen in Figure 6 below is based on 60% of the power requirement over the duration of the cold or warm start. As can be assumed, because the power requirement is larger for the Linde-Hampson cycle, the start-up costs are also larger for the cycle. Similarly, since the operating expenses are based on the power requirement per train, the Claude cycle shows lower operating expenses.

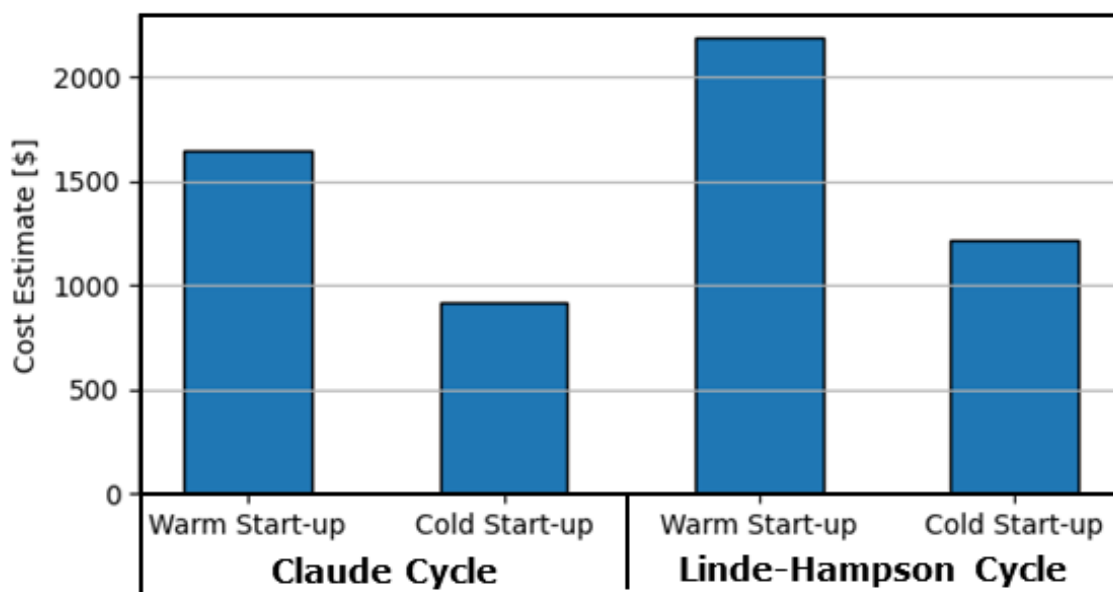


Figure 6. Charge system start-up costs

4. ORC COMPONENT SIZING

As discussed previously, the LACC approach is largely composed of commercially available equipment, with the exception being the novel ORC, whose size and operating conditions are outside the range of common industrial practice. Accordingly, the technical feasibility of the ORC equipment for the LACC service conditions was investigated.

4.1. Turbines

4.1.1. Axial turbine sizing

Conceptual design of the axial turbine was performed with aerodynamic and rotordynamic performance considerations for the ORC turbine. Aerodynamic design was based on correlations by Balje to define the physical size of the stages (tip diameter and blade height) yielding the optimal efficiency for a given specific speed [25]. For a given aerodynamic design, the rotor layout is generated based on length and diameter ratios relative to the stage sizing. The rotordynamic behavior is predicted by an undamped critical speed analysis and a screening criterion for stability. Finally, other general screening criteria are also applied to assess the feasibility of the design.

Aerodynamic sizing

The boundary conditions for the aerodynamic design are taken from the baseline cycle design. The shaft rotational speed and number of stages are varied to identify the impact on turbine efficiency and other parameters that would limit the physical design (e.g., rotordynamics, etc., as discussed in the subsequent sections). For a given shaft speed and number of stages, the loading of each stage is taken to have identical ideal adiabatic enthalpy change across each stage. This allows the definition of the specific speed N_s , given by Eq. (14), where N is the shaft rotational speed, $\dot{V}_3$ is the volumetric flow rate at the exit of the stage, and h_{ad} is the adiabatic enthalpy change for the stage. The optimal specific diameter D_s , defined by Eq. (15) in terms of the stage tip diameter D , is determined for the peak efficiency point as given in the Balje correlations. Similarly, the optimal blade height ratio is determined by the correlation. This process is carried out stage-by-stage to define all of the turbine stage diameters and efficiencies. The various lengths are estimated by scaling relative to the blade height for each stage and the stage 1 hub diameter, as discussed in the following section.

$$N_s = \frac{N \dot{V}_3^{0.5}}{h_{ad}^{0.75}} \quad (13)$$

$$D_s = \frac{D h_{ad}^{0.25}}{\dot{V}_3^{0.5}} \quad (14)$$

Rotor layout

The rotor layout comprises the following elements: the bladed section, a balance piston seal, shaft end seals, journal bearings, a thrust bearing, and a coupling. Refer to the layout shown in Figure 7. The bladed section includes space for the stator nozzle vanes and rotor blades. The balance

piston seal, adjacent to the turbine inlet, sees inlet pressure on the inboard side and exit pressure on the outboard side. The balance piston diameter is chosen to produce a desired thrust load to balance the thrust from the bladed section. The shaft end seals restrict the flow of the process fluid (at exit pressure) leaking to atmosphere. For the conceptual design, space provisions for dry gas seals (DGSs) are considered. The journal bearings are outboard of the shaft end seals. The thrust bearing and coupling are chosen to be on the same end of the shaft. Table 11 summarizes the dimensional scaling used for the various elements of the rotor design. Table 12 summarizes the inertia models used for the components whose inertias are not fully captured by the beam finite element model of the rotor described in the subsequent section.

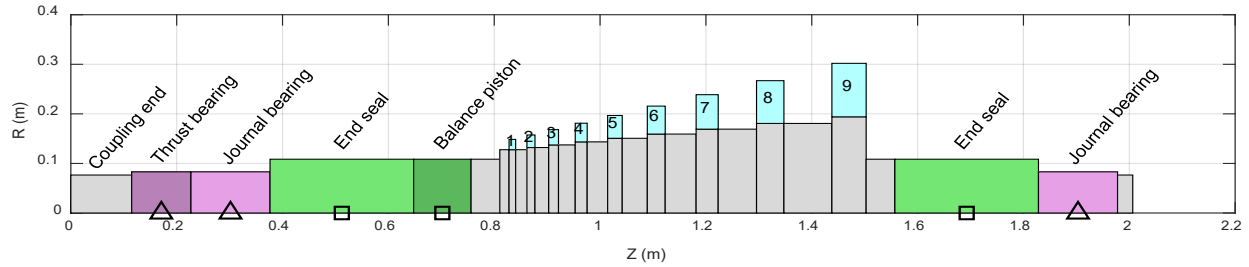


Figure 7. Rotor layout for a 9-stage 10 krpm axial turbine

Table 11. Rotor layout scaled dimensions.

Rotor layout dimension (ratio definition)	Value
Journal bearing diameter to stage 1 hub diameter	0.65
Journal bearing length to diameter ratio	0.90
Thrust bearing length to journal bearing length	0.75
End seal shaft diameter to stage 1 hub diameter	0.85
End seal length to diameter ratio	1.25
Balance piston length to end seal diameter ratio	0.50
Coupling shaft diameter to stage 1 hub diameter	0.60
Coupling shaft length to diameter ratio	0.75
Rotor blade length to blade height	0.60
Stator nozzle length to rotor blade length	1.40
Axial length between bladed section and seals relative to seal diameter	0.25

Table 12. Component inertia properties

Component	Inertia modeling
Balance piston seal	Properties of hollow cylinder with outer diameter 1.5 times shaft diameter for the full length
Thrust collar	Properties of hollow cylinder with outer diameter 1.75 times journal bearing diameter and 0.25 times the length
Coupling	Properties of hollow cylinder with outer diameter 1.25 times shaft end diameter and 0.75 times the length
Dry gas seal (DGS)	Representative mass (m), polar inertia (I_p), and transverse inertia (I_t) properties scaled by seal size: $m/(LD) = 460 \text{ kg/m}^2$ $I_p/(LD) = 0.85 \text{ kg}$ $I_t/(LD) = 0.70 \text{ kg}$

The rotor layout is converted into a finite element beam model. The shaft is discretized into cylindrical beam elements according to the lengths and diameters produced by the rotor layout process described above. In addition to nodes at diameter changes, nodes are also defined at the locations of journal bearings and discrete inertias. The beam elements are modeled as Euler-Bernoulli beams with stiffness, mass, and gyroscopic properties, which are assembled into the system mass, stiffness, and damping matrices. The discrete inertia properties are defined in Table 12. Component inertia properties are applied to the respective shaft nodes, modifying the system mass and gyroscopic matrices, and bearing stiffness and damping properties are applied to the bearing node locations, modifying the system stiffness and damping matrices.

A standard rotordynamics evaluation typically includes undamped critical speed (UCS) map analysis, imbalance response analysis, and stability analysis. The UCS map is a plot of the system undamped natural frequencies (typically coincides with resonance at critical speeds) as a function of bearing stiffness and assesses the bearing and shaft relative flexibility. The imbalance response predicts the vibration of the shaft subjected to imbalance forces, and it results in predicted Bode plots that show the location and relative damping of critical speeds. Stability analysis evaluates the damped eigenvalues of the system to predict the stability characteristics corresponding to each natural frequency. For the purposes of the conceptual design, rotordynamic feasibility will be assessed primarily by invoking the Level 1 stability screening criteria defined by API 617. This is a plot of the shaft critical speed ratio (CSR) – the ratio of running speed to the first critical speed on rigid supports – versus the average gas density of the machine. The Level 1 criteria define a “Region A” with a maximum CSR of 2.5 and a maximum gas density of 60 kg/m³ where experience has validated previous designs for lower risk of instability. In general, machines with higher gas density have higher destabilizing forces from stages and seals, and higher CSR increases the possibility of exciting the sub-synchronous natural frequency. If a machine lies outside of Region A, it simply means that higher-fidelity analysis of shaft seals, etc., is required to quantify the machine stability. That level of detail is beyond the scope of the current conceptual design,

which will only consider the effect of bearings. Moreover, the CSR will be estimated from the UCS map, where predicted natural frequencies intersect with reasonable estimates for bearing stiffness. Finally, it is assumed that an acceptable rotordynamic design could ultimately result, through selection and design of the appropriate bearings and seals, simply by ensuring that the design passes the Level 1 screening criteria.

As previously mentioned, the conceptual design approach relies on estimates for bearing stiffness. The state of the art for oil-film bearings for high performance turbomachinery is the tilting pad journal bearing. There are numerous parameters that govern tilting pad journal bearing performance, but the bearings in this study are described as follows: 5-pad, load-between-pivot, 60° pad length and 50% pivot offset, dimensionless preload of 0.5, and length-to-diameter ratio of 0.6. The pad clearance-to-journal diameter ratio typically ranges between 0.0015 to 0.0020, so the value of 0.00175 was used. Dimensionless performance of this bearing is obtained using Calculation 41 from Someya with the specified scaling for bearing length-to-diameter ratio [26]. Dimensional performance parameters are obtained by considering the shaft size and load determined from the rotor layout and lubricant dynamic viscosity of 0.0146 Pa-s, which corresponds to ISO-VG32 oil at 60°C.

Additional design considerations

The preceding sections described the methodology and conceptual design calculations for aerodynamic design, rotor layout, and rotordynamics modeling. It was explained that rotordynamic feasibility is assessed based on the API 617 Level 1 stability screening criteria. However, there are additional considerations that should be evaluated for the conceptual design to be acceptable. An aerodynamic design rule of thumb employed on designs in other applications has been to limit the adiabatic stage loading to 74.7 kJ/kg per stage – this sets the minimum number of stages required by the design. In addition, maximum hub surface speeds should be limited to 300 m/s, though it is recognized that higher speeds are possible.

Shaft size dictates the torque able to be transmitted through the coupling as well as bearing and seal velocities. Assuming an AISI 4140 steel with 655 MPa tensile yield strength, the shear yield strength would be 378 MPa based on von Mises theory. Ensuring that the shaft can withstand up to 15 times the nominal torque, which is possible while traversing torsional natural frequencies during startup, means that the allowable shear stress should be up to 25 MPa. In the case of surface speeds, it is recommended that journal bearings be less than 91.4 m/s, while DGS speed limits can be up to 152 m/s. Maximum bearing unit load is 1.38 MPa.

Conceptual design sweeps

As described previously, the shaft rotating speed and number of stages are variables to be chosen by the designer. This section provides results for 6-10 stages and design speeds from 5-20 krpm. Figure 8 shows the resulting isentropic efficiency trends, displaying increasing magnitudes with speed and number of stages. Figure 9 shows the minimum hub diameter (first stage) and maximum tip diameter (last stage), which indicates that shaft diameter is largely dependent on speed, though hub diameters tend to be lower with more stages. The minimum and maximum blade heights tend to increase with number of stages and speed (not shown for brevity).

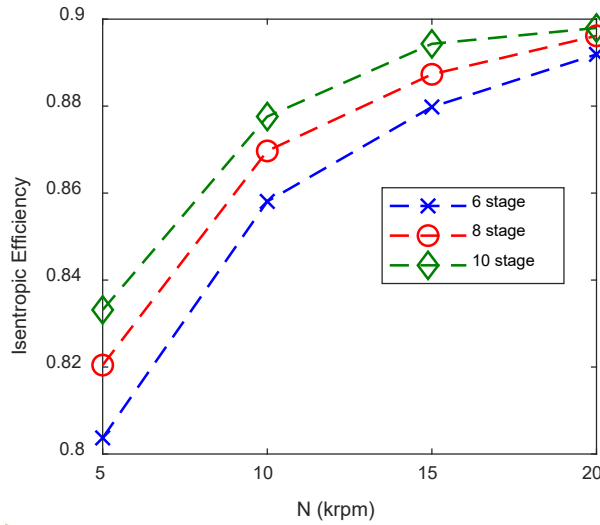


Figure 8. Isentropic efficiency

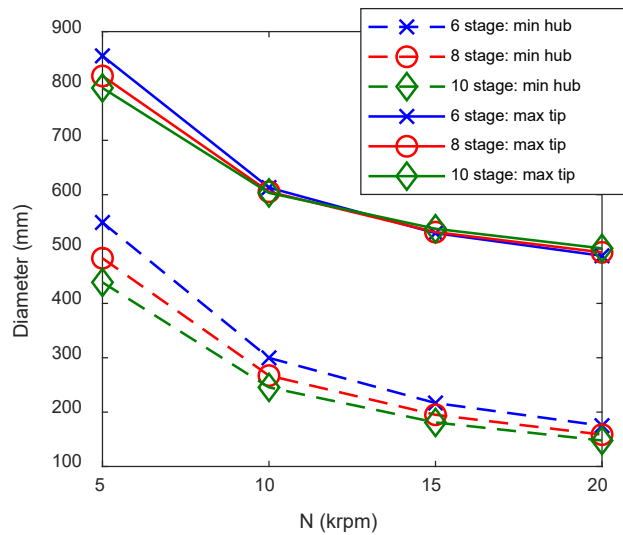


Figure 9. Minimum hub diameter and maximum blade tip diameter

Bearing span tends to decrease with speed, driven by the reduction in shaft diameter, and the expected increase in bearing span with number of stages is apparent at higher speed – see Figure 10. However, at low design speed, a greater proportion of the shaft length is due to the seals, driven by shaft diameter, which is why the 6-stage machine has a greater bearing span than the 10-stage machine at 5 krpm (see Figure 11).

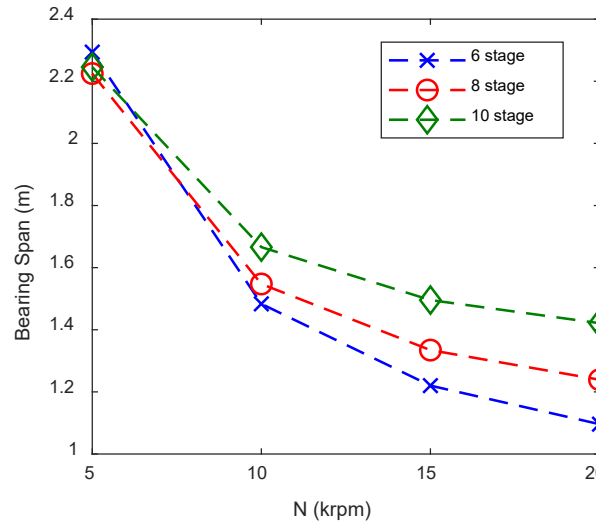


Figure 10. Bearing span

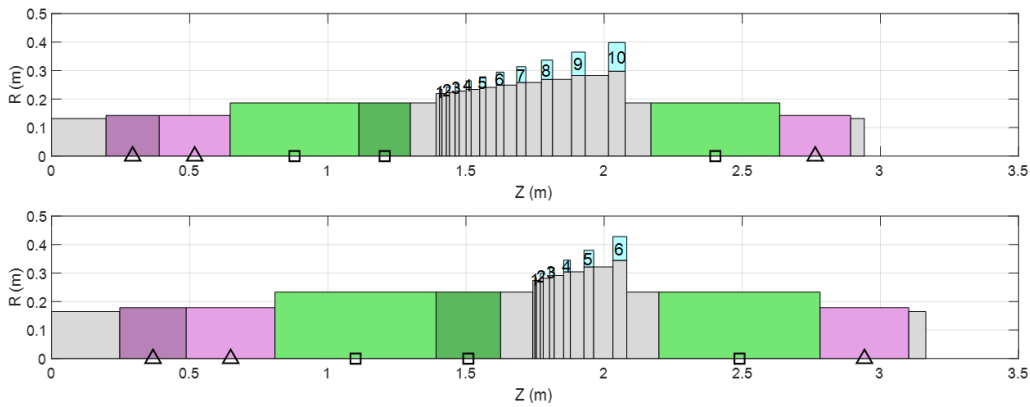


Figure 11. Comparison of 6- and 10-stage 5 krpm turbine layouts

Figure 12 shows the CSR calculations. Increasing the number of stages and the speed results in lower natural frequencies and higher CSR. The average gas density of the machine is 41.0 kg/m^3 , which corresponds to a maximum CSR of 2.2 required for Region A. This corresponds to upper limits of $\sim 17 \text{ krpm}$ for an 8-stage machine and $\sim 14 \text{ krpm}$ for a 10-stage machine. However, it is noted that there is margin available in the bearing design to further reduce CSR by increasing bearing stiffness, if necessary, so these limits are not strict.

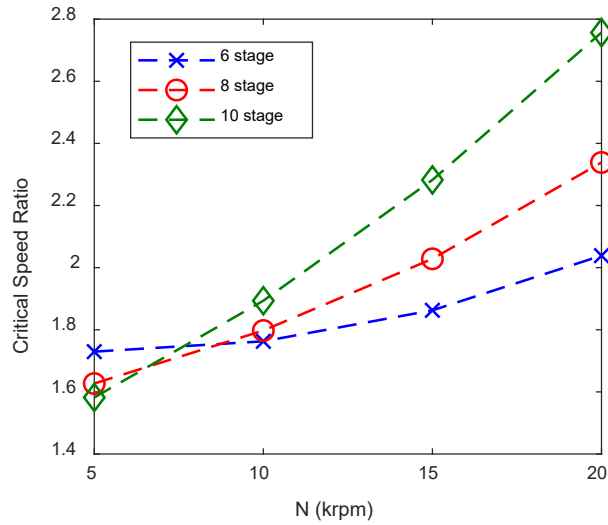


Figure 12. Critical speed ratio

Figure 13 shows the ratio of maximum shaft shear stress, at the coupling end of the turbine, to the allowable shear stress based on a 15 times factor of safety to the nominal torque load. Therefore, an acceptable design requires a value less than unity. A 6-stage design must have a speed less than 15 krpm, and a 10-stage design must have a speed less than 10 krpm. The journal bearing surface speed relative to the maximum allowable speed is shown in Figure 14. Values less than unity are acceptable, which eliminates the 6-stage design entirely. Since bearing unit loads are low (< 20% of the maximum), it is possible to reduce the journal bearing diameter ratio to reduce the surface speed. However, that would subsequently increase the shaft shear stress, since the shaft end must be smaller than the bearing journal diameter. Not shown for brevity, it is merely noted that the seal speed ratios are all lower than the bearing speed ratios, and maximum hub velocities are less than the 300 m/s limit for all configurations.

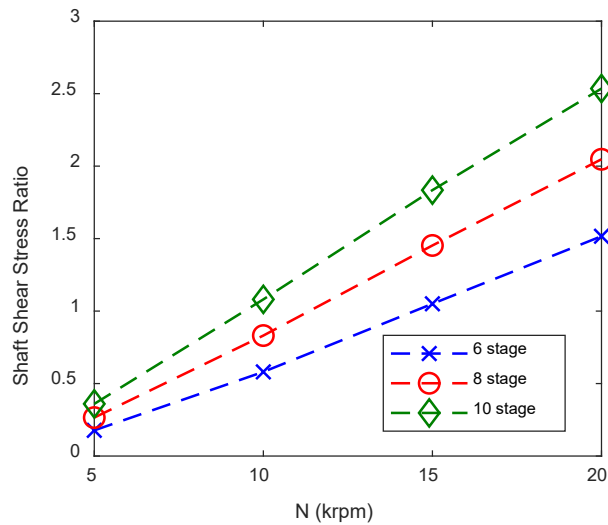


Figure 13. Shaft shear stress to allowable shear stress ratio

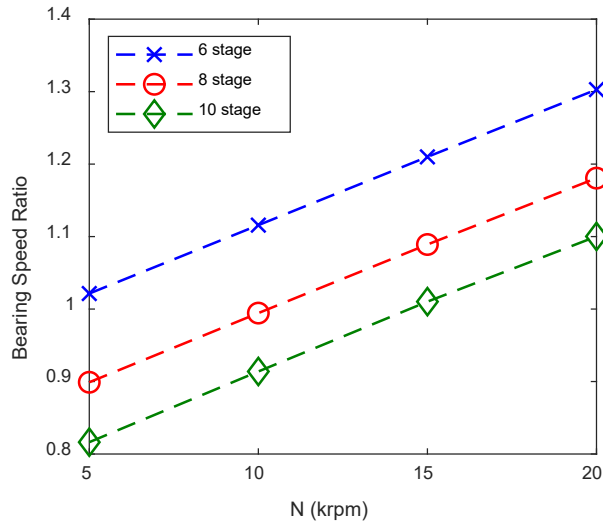


Figure 14. Bearing surface speed to maximum surface speed

Preliminary conclusions from the above study are as follows: It is desirable to have as many stages with as high of a design speed as possible to maximize efficiency. Increasing the number of stages and speed has a negative effect on rotordynamics in terms of the predicted CSR, but there is margin in the bearing design, so this does not appear to be a limiting factor in this application. There is a clear trade between requiring as large of a shaft size as possible to handle torque transmission and smaller bearing size to limit surface speeds. With the goal of maximizing shaft speed to maximize efficiency, it appears that an optimal configuration has between eight and nine stages with a design speed around 10 krpm.

Based on these observations, a more-detailed sweep was performed spanning 7-10 stages and 7-12 krpm. The key trade of shaft size and efficiency is presented as a single plot in Figure 15. The shaded region represented by shaft shear stress ratio < 1 and bearing speed ratio < 1 indicates the acceptable design space for shaft size. The efficiency contours indicate that peak efficiency exists for the combination of maximum bearing speed ratio and shaft shear stress ratio. Correspondingly, the optimal configuration based on the current constraints can be defined as point A or B. Point A is an 8-stage turbine operating at 10 krpm and has 87.0% isentropic efficiency. Point B is a 9-stage turbine operating at 10 krpm and has 87.4% isentropic efficiency (shown in Figure 7). Further optimization could be performed by modifying the scaling parameters of Table 11 to shift one of the constant-stage curves to intersect the corner of the acceptance region, but potential gains would be marginal.

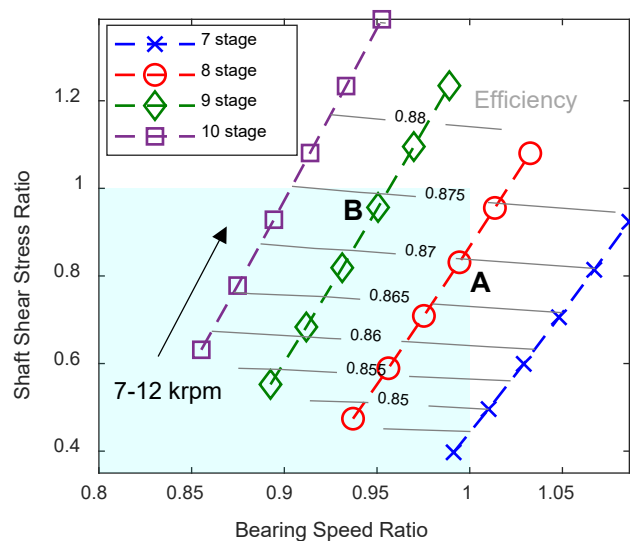


Figure 15. Comparison of shaft shear stress ratio, bearing speed ratio, rotational speed, and efficiency

4.1.2. Radial turbine sizing

The radial turbines were analyzed using the RTD code created by NASA [27]. This code requires inputs for thermodynamic properties at the inlet and outlet. Additionally, the code requires geometric considerations such as the stator inlet angle, the stator exit angle, the ratio of the exit hub-to-tip diameter, rotor blade exit angle, etc. Assumptions were made for these properties based on several sources [27, 28]. Other assumptions for the code included leakage rates and pressure losses in the volutes (0.1%) and return channel (0%). The code was then able to sweep through the stages and rotational speed to calculate the thermal efficiency of each stage and the size of each stage.

It was determined that due to the high-pressure ratio of the machine (28:1), a single-shaft system was unable to be used for a radial turbine. This was done by sweeping the speed of each stage to determine the possible operation range of each stage. The overlap was iterated on until a two-shaft, four-stage (total) system was able to be designed. An additional number of stages were also evaluated, but the high-pressure ratio made high speeds problematic for the low-pressure stages and vice versa.

The first shaft had a maximum operational range between 26-27 krpm. This shaft was able to reduce the pressure down to 2.7 MPa and generate 9.6 MW after accounting for leakage. The second shaft further reduced the pressure down to 0.54 MPa and was able to generate an additional 8.4 MW of power. Table 13 shows the estimated values for all four stages and shafts.

Table 13. Conceptual design of two-shaft, four stage radial turbine.

Parameters	Stage 1 (Shaft 1)	Stage 2 (Shaft 1)	Stage 3 (Shaft 2)	Stage 4 (Shaft 2)
Rotational speed [krpm]	26.5	26.5	11.5	11.5
Number of blades [-]	14	15	14	15
Total-to-total pressure ratio [-]	2.141	2.333	2.098	2.265
Total stage efficiency (w/leakage) [-]	93.0	95.3	93.0	95.2
Inlet total pressure [MPa]	12.74	6.03	2.71	1.31
Inlet static pressure [MPa]	8.87	4.09	1.90	0.90
Outlet total pressure [MPa]	6.03	2.71	1.31	0.604
Outlet static pressure [MPa]	5.99	2.62	1.30	0.59
Inlet total temperature [K]	608	550	508	466
Inlet static temperature [K]	579	530	487	447
Outlet total temperature [K]	550	510	466	429
Outlet static temperature [K]	549	508	465	427
Power output (w/leakage) [MW]	4.81	4.83	3.99	3.99

After the aerodynamics were calculated, the size of the overall turbine could be estimated. The high-speed shaft was calculated to be 120 mm in diameter with each blade having a tip diameter of around 500 mm. Figure 16 shows the shaft, blade sizing, and flow path for the high-speed shaft. The low-speed shaft had a diameter of around 300 mm with the blades extending to over 1000 mm in each case. Figure 17 shows the shaft, blade sizing and flow path for the high-speed shaft. This stage arrangement could lend itself to an integrally geared machine architecture, but this option was not investigated further at this time.

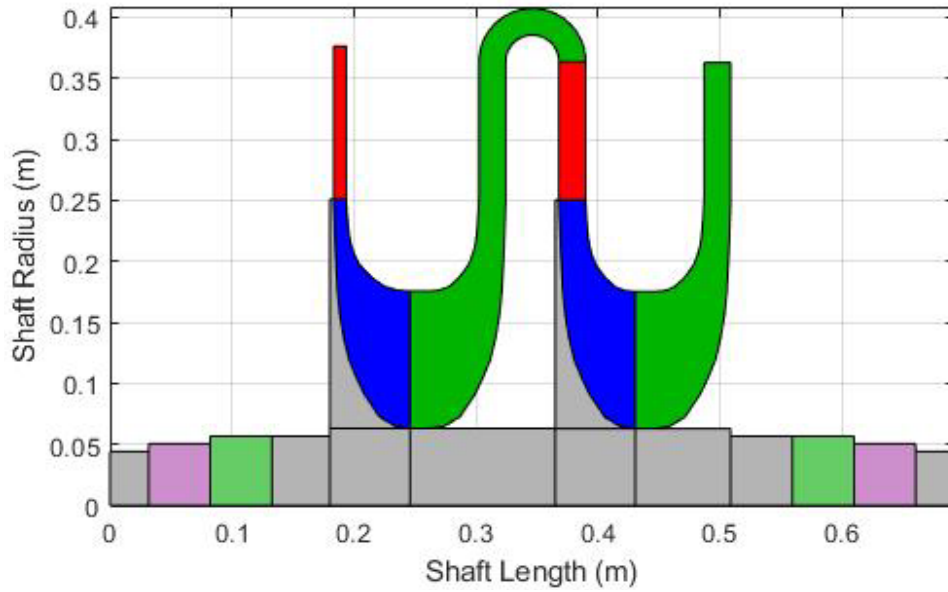


Figure 16. Conceptual design of the high-speed shaft.

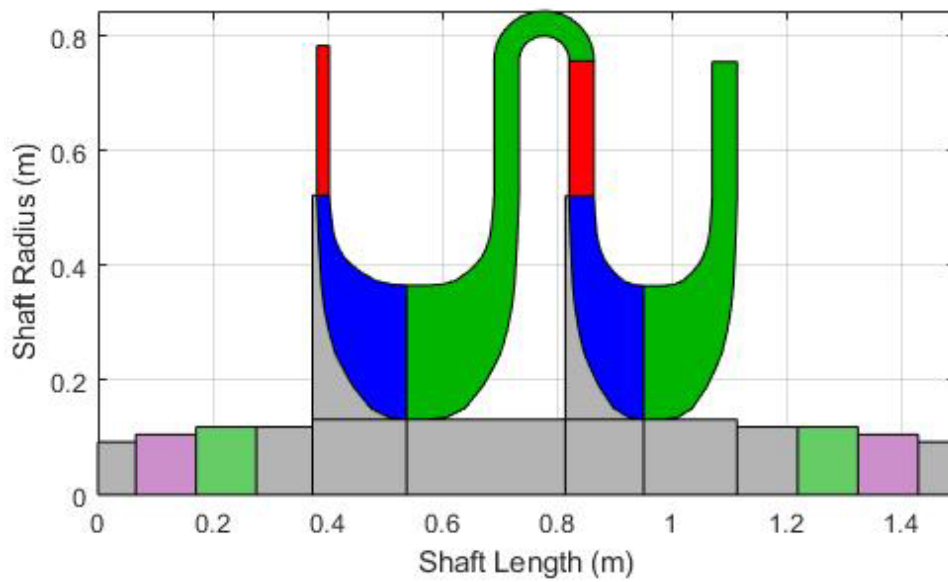


Figure 17. Conceptual design of the low-speed shaft.

4.2. Heat exchangers

The ORC in the LACC requires multiple heat exchangers to transfer heat between the air stream and ORC. In addition, a recuperator is also located in the ORC. Three different heat exchangers have been evaluated to estimate the sizing and performance in order to improve the cycle modeling of the LACC.

- The air preheater (unit #5 in Figure 2) is designed to add heat from the ORC turbine exhaust back into the regasified air stream. Both of the fluids in this heat exchanger are in the gaseous phase.
- Air Evaporator/ORC Condenser (#4 in Figure 2) is the primary method for converting the high-pressure liquid air back into the gas phase as well as liquifying the ORC fluid in preparation for pumping. This heat exchanger is designed to transfer 21 MW from the ORC to the air. Both fluids are required to undergo phase change in this cycle.

ORC recuperator (#15 in Figure 2) transfers the remaining ORC heat after the air preheater to the liquid ORC stream coming from the pump. This heat exchanger transfers 1 MW of heat. It is expected that the high temperature outlet will start liquifying at the exit of this heat exchanger (quality around 0.95).

The three different types of heat exchangers were evaluated for each using the Aspen EDR (v11.0). This program is able to optimize the size and performance using built-in design rules for shell-and-tube, plate-fin, and gasketed plate heat exchangers. The program was able to estimate the size, pressure drop, and cost (for shell-and-tube and gasketed plate heat exchangers) to provide important details for the design of the overall system. Additional parameters such as headers for shell-and-tube and the material were based on recommendations from Shah and Sekulic [30].

4.2.1. ORC-to-air preheater

The ORC preheater is designed to add heat back to the air before it recovers the waste heat from the exhaust flow. The preheater has the highest temperatures but the lowest pressures of all three heat exchangers. As a result, the team decided to use 304 stainless steel as the material. Table 14 shows the conditions for the air preheater. The heat required to transfer was determined to be 18.9 MW of heat from the ORC fluid to the air.

Table 14. Conditions for the air preheater

Parameters	Value
Stream 1 Composition	Air
Stream 1 Inlet Pressure [MPa]	7.2
Stream 1 Inlet Temperature [K]	204
Stream 1 Outlet Temperature [K]	423

Parameters	Value
Stream 1 Mass Flow Rate [kg/s]	75.3
Stream 2 Composition	Ethane
Stream 2 Inlet Pressure [MPa]	0.45
Stream 2 Inlet Temperature [K]	438
Stream 2 Outlet Temperature [K]	220
Stream 2 Mass Flow Rate [kg/s]	44.7

The shell-and-tube heat exchanger was optimized for the ORC fluid on the shell-side of the heat exchanger (high temperature, low-pressure) and the gaseous air passing through the tubes. The effective surface area for one unit was determined to be 242.1 m^2 over a total of 911 tubes. The inside diameter of the shell was 737 mm. In order to accommodate all of the energy needed to be exchanged, the software calculated that the optimal design would require a total of 30 units, three series of 10 units in parallel. To utilize shell-and tube heat exchangers, the total cost would be \$3.4MM as each unit costs \$114k. Due to the excessive number of units, it has been determined that the shell-and-tube heat exchanger is not ideal for this application.

The second type of heat exchanger examined for the preheater was a stainless steel gasketed plate. The gasketed plate combines alternating parallel plates to transfer heat between the fluid. Figure 18 shows the dimensions of the preheater as a gasketed plate heat exchanger. As such, the surface area of a single heat exchanger is $4,374 \text{ m}^2$, almost 20 times the calculated surface area for a shell-and-tube. This resulted only requiring two units in parallel for a cost of \$362k (\$181k/unit).

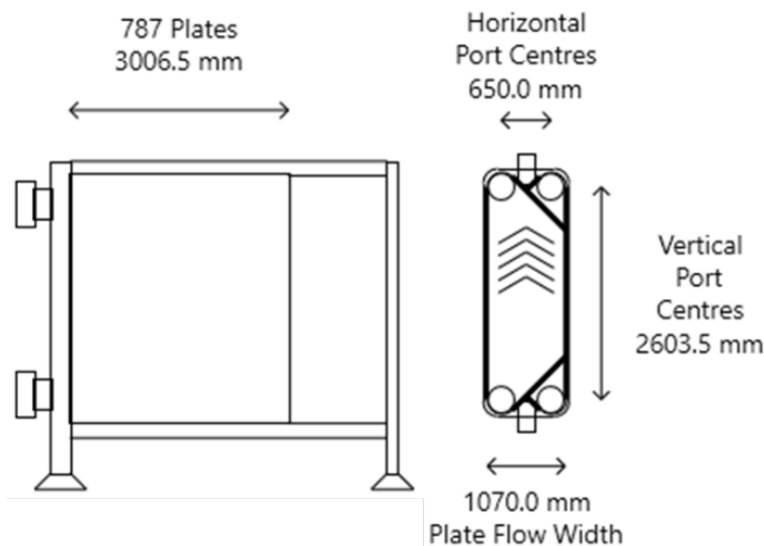


Figure 18. Schematic of gasketed plate heat exchanger for the air preheater.

A plate-fin heat exchanger was also examined for the preheater conditions. Plate-fin heat exchangers use fins in addition to metal plates to add surface area to either side and enhance the heat transfer process. The plate fin heat exchanger was 4,714 m² on the air side of the heat exchanger and 9,428 m² on the ORC-side for each unit. A total of five units were required in parallel for this design. The EDR program is not able to estimate the cost for plate-fin heat exchangers, so a cost comparison was not possible. Figure 19 shows the diagram for the plate fin heat exchanger.

The best design for the air preheater appears to be a gasketed plate based on the current design conditions. The cost of the gasketed plate is about one-tenth the cost of a shell in tube and similar in size when the number of parallel units are considered. The gasketed plate has a total volume of 16.7 m³ compared to 11.1 m³ for the plate fin heat exchangers.

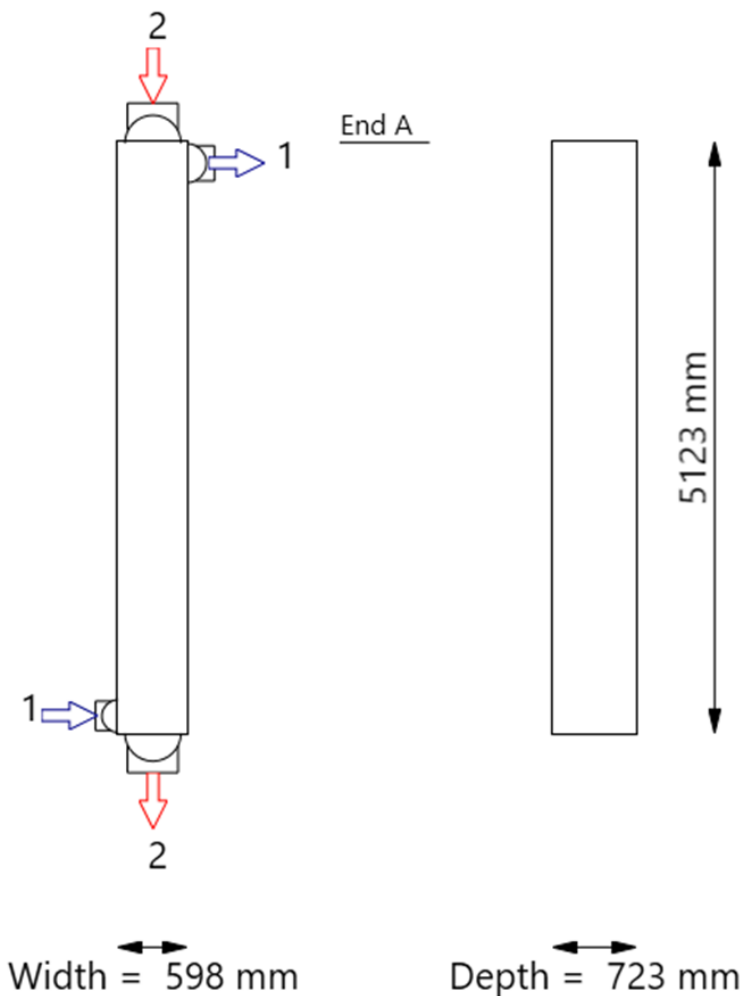


Figure 19. Plate fin heat exchanger schematic for the air preheater. Stream 1 is air and stream 2 is ethane

4.2.2. ORC condenser / air gasifier

The ORC condenser or air regasifier is the driver for phase change of both fluids in the system. Table 15 shows the key parameters of the system. The system requires that it transfers 21.5 MW of heat from the ORC into the liquid air stream to vaporize it. As the air is entering in cryogenic temperatures, two different materials were considered based on the recommendations from Shah and Sekulic. Aluminum is typically the material of choice for cryogenic fluids, but the high pressure of the air would increase the cost prohibitively on both the shell-and-tube designs and the gasketed plates as well as run into other issues with the high pressures required. Therefore, these heat exchangers were also tested with stainless steel and displayed significantly improved performance. Table 15 Shows the conditions for the ORC condenser / air gasifier.

Table 15. Conditions for the ORC condenser / air gasifier

Parameters	Value
Stream 1 Composition	Air
Stream 1 Inlet Pressure [MPa]	7.2
Stream 1 Inlet Temperature [K]	81
Stream 1 Outlet Temperature [K]	204
Stream 1 Mass Flow Rate [kg/s]	75.3
Stream 2 Composition	Ethane
Stream 2 Inlet Pressure [MPa]	0.29
Stream 2 Inlet Temperature [K]	207
Stream 2 Outlet Temperature [K]	194
Stream 2 Mass Flow Rate [kg/s]	44.7

The results for the shell-and-tube heat exchanger are shown below. The fluid was designed with the air in the tubes due to the high pressures, and the ethane was on the shell-side. The effective surface area for a single unit was estimated to require 4,205 m² over a total of 786 tubes. Overall, it was estimated that for shell-and-tube it would require a series of nine units in parallel twice to transfer all of the heat (18 total units). The diameter of the shell is 686 mm all units. The overall cost of the system would be \$99k per unit for a total of \$1.8MM. Using aluminum, the cost was estimated as \$1.5M with a similar number of heat exchangers required.

The gasketed plate was again considered a better option for the air gasifier due to the reduced capital cost estimate. The heat exchangers required 837 m² of surface area to condense the ethane

and evaporate the air in the same heat exchanger. Only a single unit was required for this heat exchanger totaling \$70k as compared to three heat units for the plate-fin design. It also required slightly more volume than the plate-fin heat exchanger at 4.7 m³ but was less complex as a single unit. The schematic of the heat exchanger can be seen in Figure 20.

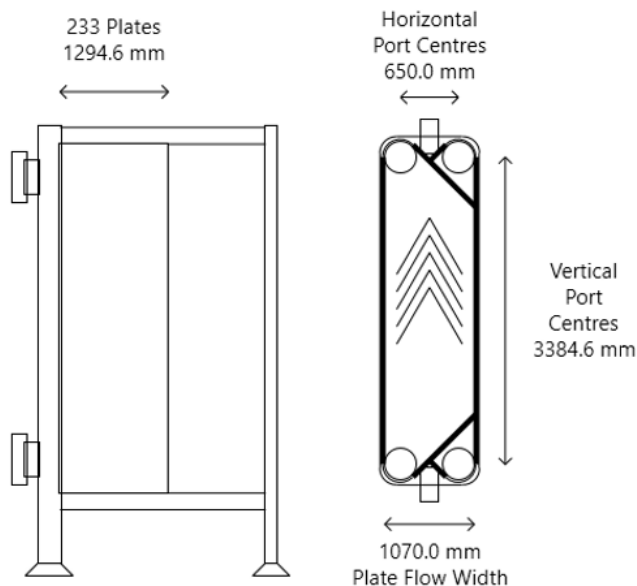


Figure 20. Schematic for gasketed plate heat exchanger for the ORC condenser / air gasifier

4.2.3. ORC recuperator

The final heat exchanger examined was an ORC recuperator. This heat exchanger aims to add heat to the high-pressure liquid stream and reduce the heat load required to liquify the ethane in the gasifier. The total heat transferred in this heat exchanger was estimated to be 1.1 MW, significantly smaller than the previous two. The temperatures are all around 210 to 220 K for inlet and outlet streams. Stainless steel was once again used as the material for all three heat exchanger types. Table 16 shows the conditions for the heat exchanger.

Table 16. Conditions for the ORC recuperator

Parameters	Value
Stream 1 Composition	Ethane
Stream 1 Inlet Pressure [MPa]	13.0
Stream 1 Inlet Temperature [K]	201
Stream 1 Outlet Temperature [K]	211
Stream 1 Mass Flow Rate [kg/s]	44.7

Parameters	Value
Stream 2 Composition	Ethane
Stream 2 Inlet Pressure [MPa]	0.37
Stream 2 Inlet Temperature [K]	220
Stream 2 Outlet Temperature [K]	207
Stream 2 Mass Flow Rate [kg/s]	44.7

The shell-and-tube design only required two units in parallel for the ORC recuperator. The design has an effective surface area of 186.8 m^2 and was estimated to cost \$144k for each unit. A total of 762 tubes were placed in each shell. The overall length of the unit was determined to be 4 m and has a diameter of 813 mm.

The gasketed-plate heat exchanger only required a single unit for the recuperator. The design had a surface area of 827.4 m^2 for each side of the heat exchanger and a total of 299 plates. The cost estimate was \$70k. Figure 21 shows the dimensions of the heat exchanger.

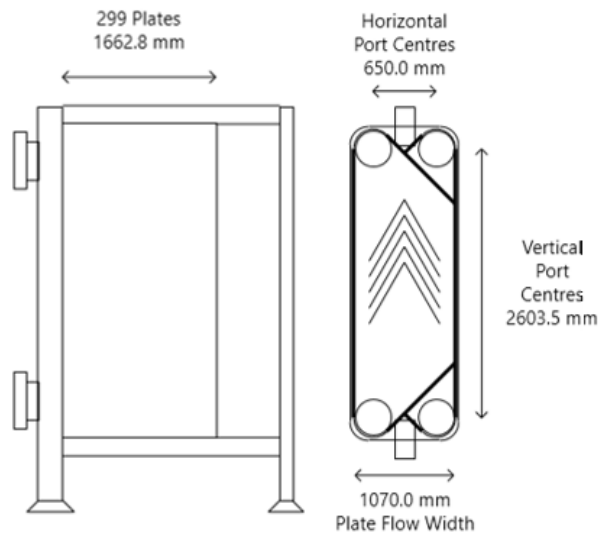


Figure 21. Schematic of gasketed plate heat exchanger for the ORC recuperator.

The plate fin heat exchanger also required two separate units in parallel, similar to the shell-and-tube design. These heat exchangers were estimated to require 976 m^2 on the hot-side and 495 m^2 on the cold-side. Each heat exchanger was estimated to have a footprint of 1.3 m^3 .

Additional research on the recuperator is required to get a cost estimate for the plate fin heat exchanger as the plate fin requires additional units. The gasketed plate is also three times the size of the heat exchanger.

5. COMMERCIAL-SCALE LACC DISCHARGE CYCLE DESIGN

This section describes the evolution of the discharge cycle design from the reference process described in Section 2 to a commercial scale design. The evolution included process model enhancement to support parametric studies for development of an optimal Pareto front, cost estimation, and then selection of the final design using engineering judgement.

5.1. Process model enhancements

Optimization was based on the reference process design shown in Figure 2, and repeated below for convenience.

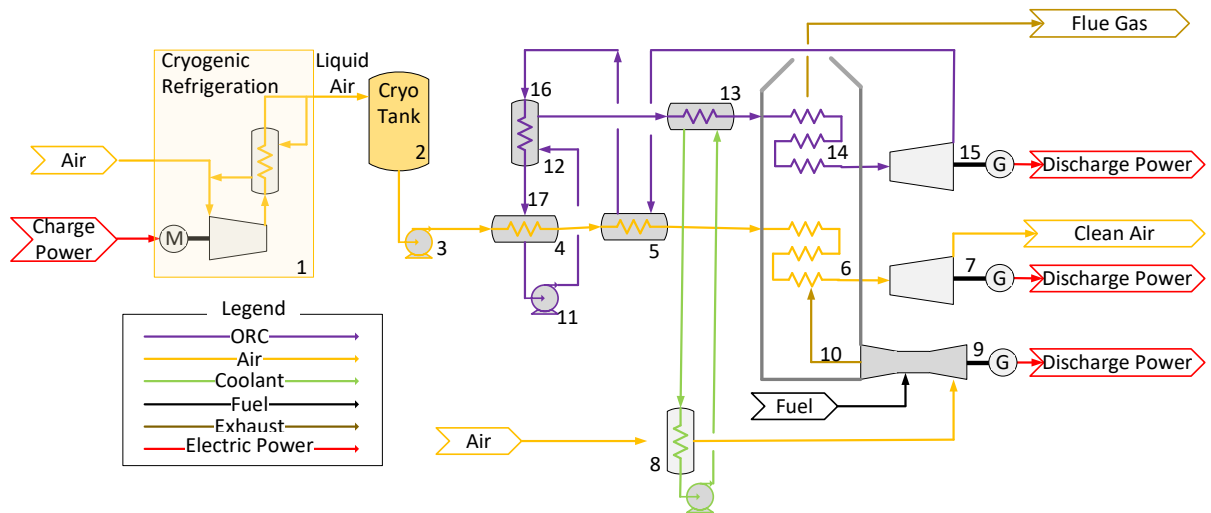


Figure 22. Reference process design

The discharge cycle was modeled using Ebsilon Professional power plant software. Equipment specifications for the unit processes (e.g., pumps, turbines, heat exchangers) included efficiency, pressure drops, and heat transfer behavior. Process conditions (turbine inlet temperature and pressure and exhaust pressure, and ORC condensing pressure) were also specified. A typical graphical user interface is shown in Figure 23.

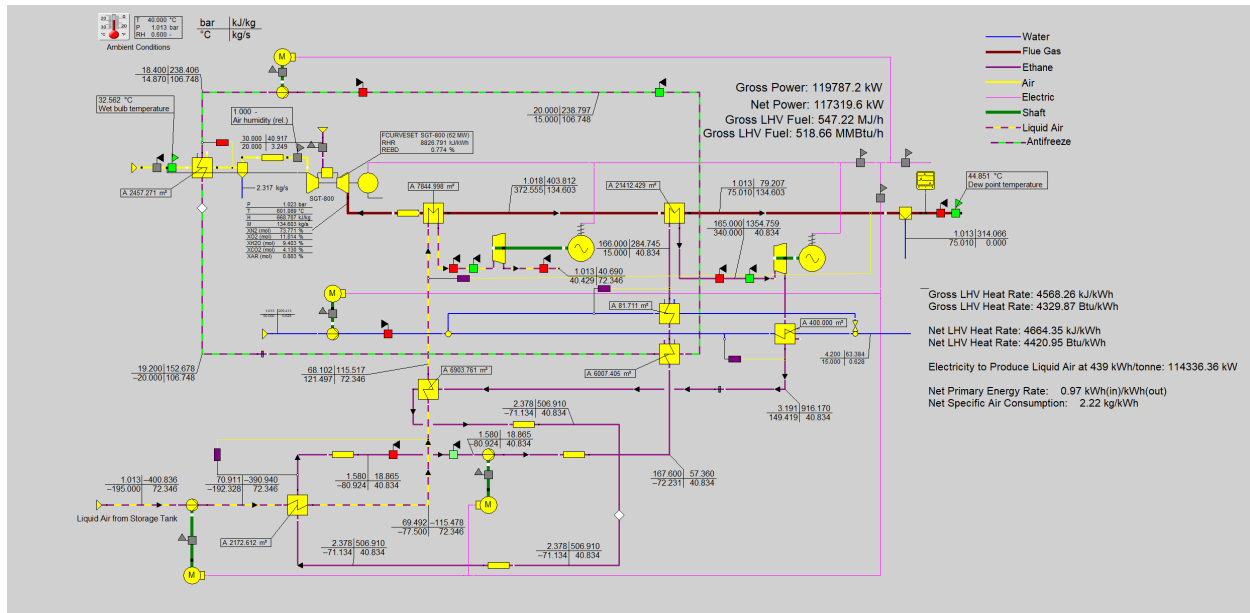


Figure 23. Graphical interface for discharge cycle process model

The model solves for thermodynamic conditions (temperatures, pressures, flow rates of each fluid stream) satisfying the physical and process parameters by adjusting the flow rates of liquid air and ORC fluid. Convergence will only occur for physically realizable combinations of parameters and process conditions.

Both equipment specifications and process conditions could be specified by an Excel programming interface to permit automation of parametric studies. The resulting process conditions and summary performance (power output, fuel heat rate, stored energy rate, specific air consumption, etc.) are then available in Excel for further analysis.

The reference model used during the requirements analysis was updated to expand the range of process parameters, and improve stability. The updates included:

- Changing the liquid air flow control to target flue gas exhaust temperature at the stack, rather than downstream of the air heater to permit the heating duty to be optimized between the regasified air and ORC subsystems.
- Adding service water heat exchangers to maintain consistent ORC fluid temperature entering the exhaust gas heater and exiting the ORC turbine. This change is expected to improve off-design performance, but has little impact on the design point performance used for optimization (other than introducing some additional pressure loss).
- Calibration of the gas turbine performance against confidential performance estimates provided by Siemens Energy for the SGT-800 gas turbine.

A series of sensitivity runs were undertaken using the Excel interface, to further understand cycle model sensitivity to input parameters and make necessary tuning adjustments, for a total of 119 cases to prepare for the cost/performance optimization study.

5.2. Cost modeling

In order to perform the techno-economic analysis, the capital costs of a specific design must be calculated. The capital costs for the commercial scale application have been estimated for most major equipment based on several sources as shown in Table 17. The major equipment has been divided into nine different types of equipment, and various methods were used to calculate the costs for each piece of major equipment. Some of the equipment was based on cost correlations for sCO₂ equipment as the densities, pressures and temperatures for the equipment were similar. For the ORC turbines, the equations from Weiland et al. have been used. These were determined to be the best approximation for ethane as it has a similar density as the sCO₂ quotes used to generate the curve. Vendor quotations were used for the gas turbine, air turbine, and exhaust heat recovery equipment, which is essentially unchanged across the parameter range of interest. These equations also include a temperature dependent factor, denoted as f_t , that is fully defined in the reference.

All of the major equipment can then be summed to determine the total equipment costs to which a turnkey factor of 3.5 is applied to calculate the total overnight cost (TOC) summary of the system. The turnkey factor was chosen based on the experience of Wood. The current estimate for the TOC is \$709 MM for a baseline system, at 117 MW discharge power, with 75 GWh (641 hours) of discharge and a charging system capable of producing liquid air at the discharge flow rate. Actual sizing of storage and charging capacity will depend on the application.

Table 17. Specific costs of major equipment

Type of equipment	Specific cost / Equation	Source
Charge System	\$443.7MM	Wood Group ¹
Generators	$C[\$] = 108,900P_e[MW]^{0.5463}$	Weiland et al. ²
Heat Exchangers	Various	Aspen EDR ³ , Quotations ⁴ , Weiland et al. ²
Motors	$C[\$] = 131,400P_e[MW]^{0.5611}$	Weiland et al. ²
ORC Pump	\$283.40/mtpd	Equipment specification being developed for vendor quote
Liquid Air Pump	\$92.60/mtpd	Equipment specification being developed for vendor quote
Air Turbine	\$325/kW	Equipment specification being developed for vendor quote
SGT-800 Gas Turbine	\$18.4 MM	Vendor Quotation ⁵
Exhaust Heat Recovery	\$5 MM	Estimate scaled from equivalent size Heat Recovery Steam Generator

Type of equipment	Specific cost / Equation	Source
Storage Tank	$C[\$] = 262 \cdot V[m^3] + 19.9 \cdot 10^6$	Wood Group ¹
ORC Turbine (Radial)	$C[\$] = 182,600P_{sh}[MW]^{0.5561} \cdot f_t$	Weiland et al. ²
ORC Turbine (Axial)	$C[\$] = 406,200P_{sh}[MW]^{0.8} \cdot f_t$	Weiland et al. ²

1. Bulte, A., Salazar, T., 2021. Liquid Air Combined Cycle Air Liquefaction Study Report. Internal Report for FE-0032002.
2. Weiland, N. T., Lance, B. W., & Pidaparti, S. R. 2019. SC02 power cycle component cost correlations from DOE data spanning multiple scales and applications. In Turbo Expo: Power for Land, Sea, and Air (Vol. 58721, p. V009T38A008). American Society of Mechanical Engineers.
3. Aspen Technology, Inc. 2021. Exchanger Design and Rating, V11.0. Estimates the cost of heat exchangers based on the material and design conditions. Uses internal database from Aspen that includes multiple types of heat exchangers.
4. Quote from Custom Cooling. Includes equipment costs. February 2022.
5. Turnkey package, Siemens confidential indicative quote dated March 2021.

5.3. Discharge cycle trade studies

Seventeen parameters in the Ebsilon model were selected to investigate the cost and performance trade-offs of the discharge system. Table 18 shows the parameters that were studied and the ranges that were investigated for individual design sweeps. For each of these sweeps, a single variable was chosen to manipulate while holding the remaining variables constant. For the first two variables focusing on the flow rate, the flow rates were not modified directly, but each had a reference temperature associated with the flow rate. The liquid air flow rate was referenced to the exhaust gas exit temperature (after component 14 in Figure 2). The ORC flow rate was referenced to the temperature of the air stream after exiting the liquid air vaporizer (after component 4 in Figure 2).

Table 18. Independent variables for the discharge trade studies

Variable	Design point	Min. value	Max. value
Liquid Air Flow Rate Reference Temp. [°C]	50	50	100
ORC Flow Rate Reference Temperature [°C]	-70.1	-90.1	-70.1
Air Turbine Inlet Temperature [°C]	527	427	627
Air Turbine Inlet Pressure [bar]	67.1	52.1	82.1
ORC Turbine Inlet Temperature [°C]	219	219	419
ORC Turbine Inlet Pressure [bar]	156	126	186
ORC Condensing Pressure [bar]	1.8	1.4	4.4
Air Turbine Isentropic Efficiency [%]	88	80	94
ORC Turbine Isentropic Efficiency [%]	88	80	94

Variable	Design point	Min. value	Max. value
Liquid Air Pump Isentropic Efficiency [%]	80	65	85
ORC Pump Isentropic Efficiency [%]	80	65	85
Cooling Loop Pump Isentropic Efficiency	80	65	85
Liquid Air Vaporizer / ORC Condenser Pressure Drop [%]	2.0	0.5	4.0
Air Preheater Pressure Drop [%]	2.0	0.5	4.0
ORC Recuperator Pressure Drop [%]	2.0	0.5	4.0
Air Preheater Effectiveness [%]	80	50	95
ORC Recuperator Effectiveness [%]	80	50	95

Three metrics were chosen to compare the discharge system trade-offs. The capital costs of the discharge system were calculated based on cost models previously presented. The discharge energy is an estimate for the total energy that is discharged over a given year, and the stored energy rate (SER) is the ratio between charged and discharged energy. The main take-away of the analysis was that most of the variables had a relatively minor effect on either the performance or economics. The five variables that had the largest impact were the liquid air flow rate, the ORC flow rate, the air turbine inlet temperature, the ORC turbine inlet temperature, and the air turbine isentropic efficiency. The liquid air flow rate was varied between 260-290 tonne/hr and the effects on the three performance metrics are shown in Figure 24. Based on the model, the capital costs of the discharge system (Figure 24a) had a minimum around 268 tonne/hr, while both the energy discharged (Figure 24b) and the stored energy rate (Figure 24c) continued to rise as the flow rate increased.

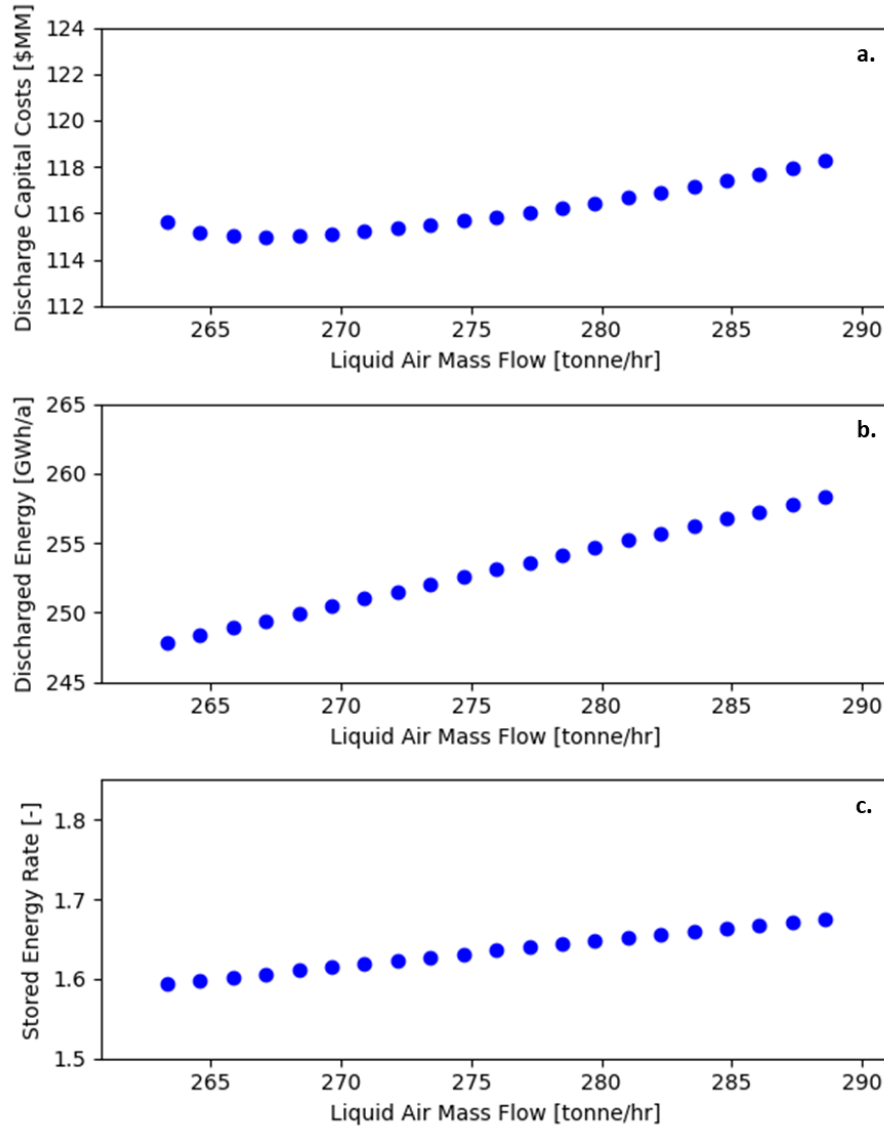


Figure 24. Comparison between the liquid air mass flow rate and three dependent metrics.
a) Discharge capital costs, b) discharged energy, c) stored energy rate.

Additionally, the three metrics were compared to each other to understand the potential trade-offs as the liquid air flow rate was varied, as shown in Figure 25. The plots show that decreasing energy output results in reduced CAPEX and a lower SER, although the change in CAPEX was only about 3%. Examining Figure 25c, it can be noted that a reduction in the SER (an improvement) was inversely correlated with output power. This is because SER is heavily influenced by the liquid air production and consumption, so an increase in the flow rate would result in an increase in the charge system power requirements and charge system CAPEX.

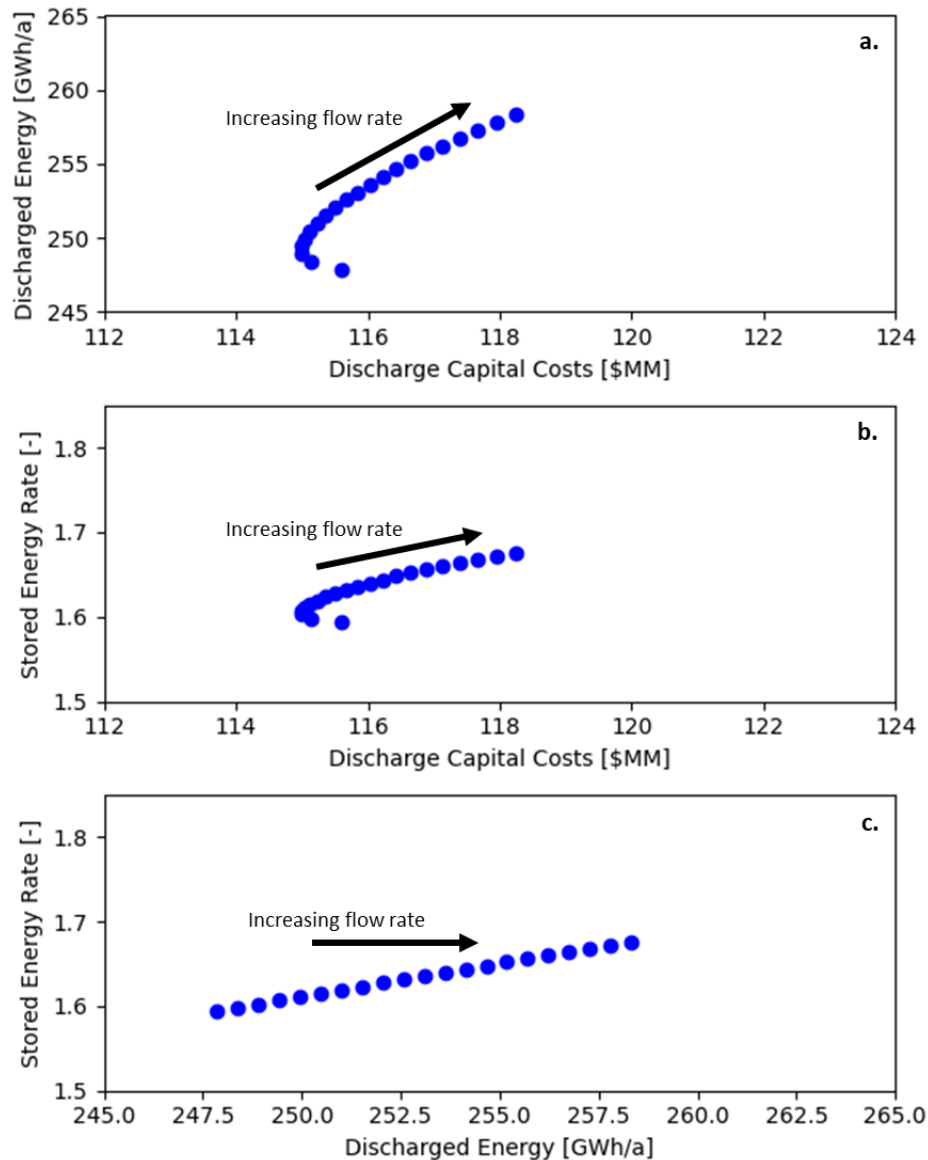
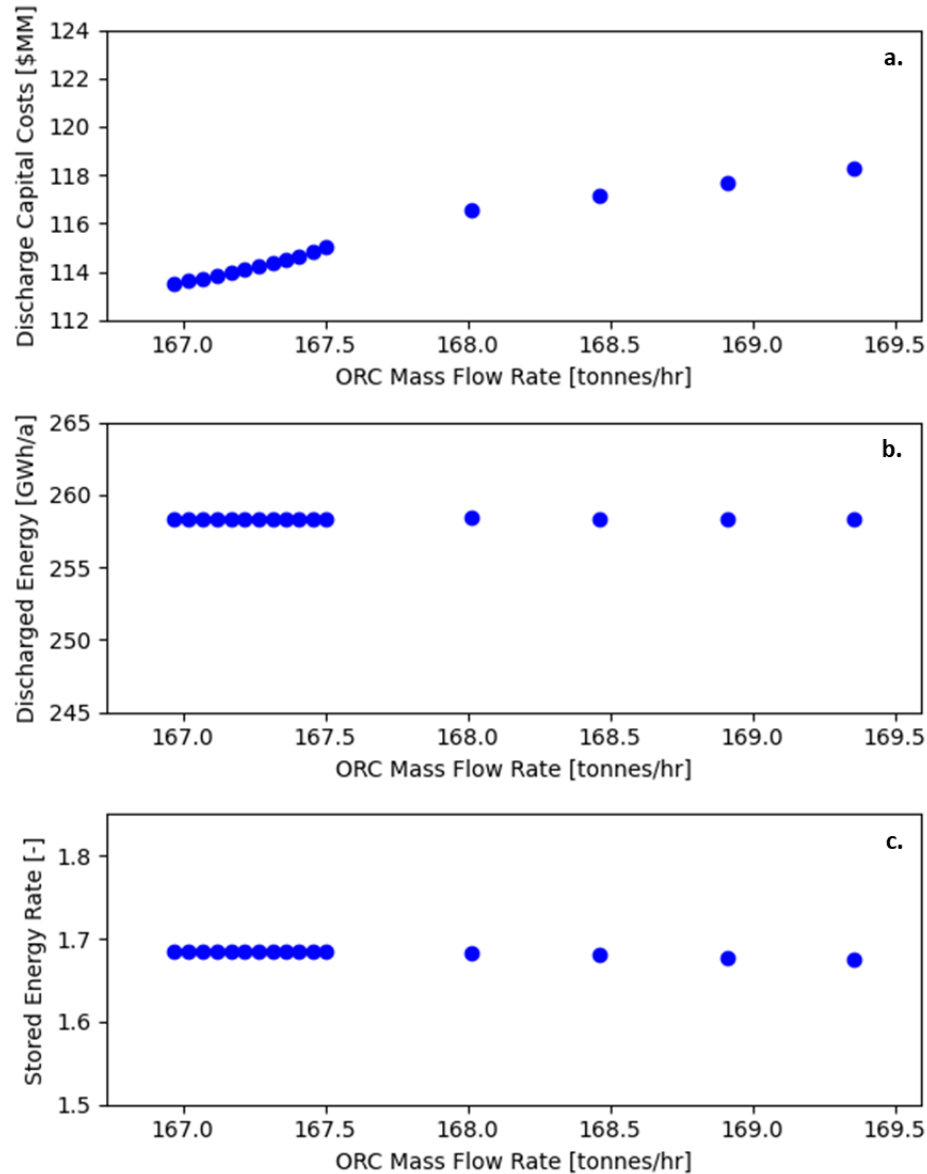


Figure 25. Comparison of the amount of energy discharged, SER, and discharge capital costs as the liquid air flow rate was swept between 260-290 tonne/hr. a) Discharge capital costs v. discharged energy, b) discharge capital costs vs. stored energy rate, c) discharged energy vs. stored energy rate.

The ORC flow rate was swept by altering the temperature control for the liquid air vaporizer (temperature after Component 4 in Figure 2). Through this method, the ORC flow rate was swept between 165-170 tonne/hr. Figure 26 shows the sweep of the ORC flow rate vs. the three dependent parameters. It can be noted that most of the points were bunched between 167-167.5 tonnes/hr. This result is due to the liquid air temperature being too high (above -80°C) and the ORC fluid being unable to completely undergo phase change. At the condensing pressure of 1.83 bar, the saturation temperature is -76.8°C .



*Figure 26. Comparison between the ORC mass flow rate and three dependent metrics.
a) Discharge capital costs, b) discharged energy, c) stored energy rate.*

Comparison of the dependent variables together is shown in Figure 27. Only one variable, the discharge capital costs, had a significant change with varying the ORC flow rate despite the flow rate only changing by 3 tonne/hr.

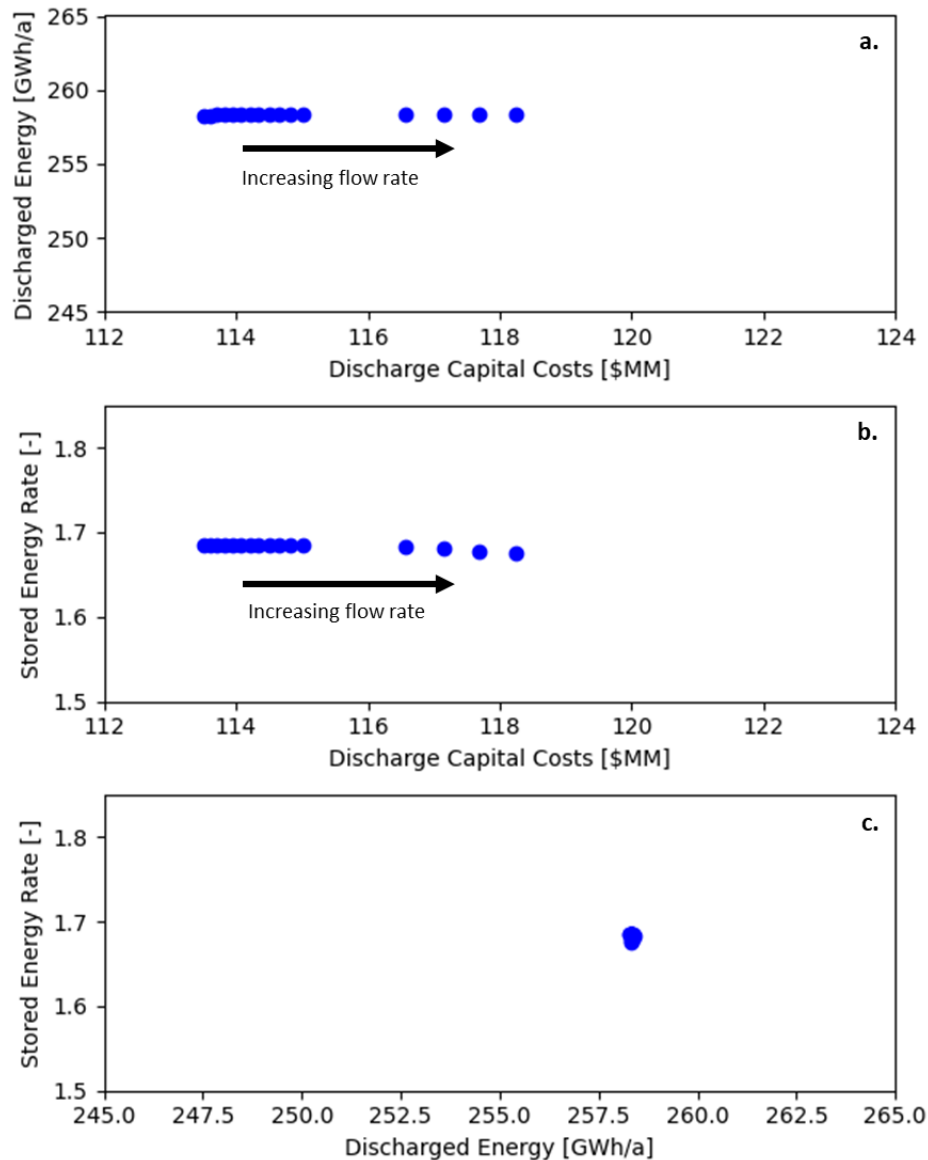


Figure 27. Comparison of the amount of energy discharged, SER and discharge capital costs as the liquid air flow rate was swept between 165-170 tonne/hr. a) Discharge capital costs vs. discharged energy, b) discharge capital costs vs. stored energy rate, c) discharged energy vs. stored energy rate.

The air turbine inlet temperature (TIT) was swept between 400-600°C as shown in Figure 28. The TIT had an interesting effect on the system design; increasing the temperature resulted in less power output from the ORC system due to less heat being available to it above a 475°C air turbine TIT (Figure 28b). Up to 475°C, the ORC system had similar performance, which resulted in a maximum power output increasing with each step. Once the temperature went above 475°C, the reduced power output from the ORC had a more significant effect, resulting in less power output from the system overall. In addition, to reach the higher temperatures, the liquid air flow rate had to be reduced, which decreased the SER at a faster rate as shown in Figure 28c.

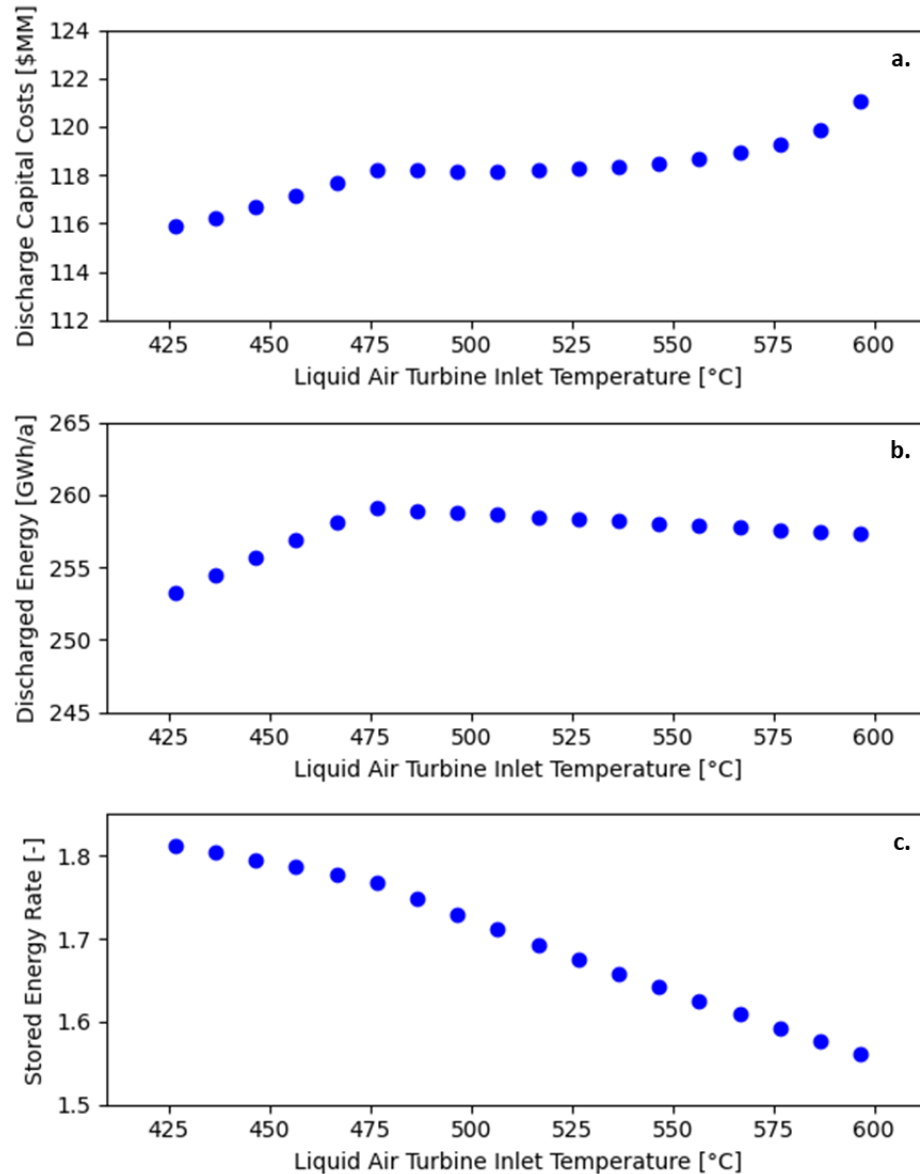


Figure 28. Comparison between the liquid air turbine inlet temperature and three dependent metrics. a) Discharge capital costs, b) discharged energy, c) stored energy rate.

The three dependent variables were compared to each other in Figure 29 to show the trade-offs with the regasified air TIT. The low temperature points show that there is a trade-off between an additional 3 GWh/a for each increase of \$1MM, but above the temperature inflection point, the energy drop-off decreased while the costs increased (shown in Figure 29a). The reduction in liquid air flow rate from the high temperatures showed significant improvement in the SER with a minimal change to the energy produced. Overall, the SER dropped by 0.13 for each GWh/a decrease in discharge energy (shown in Figure 29c).

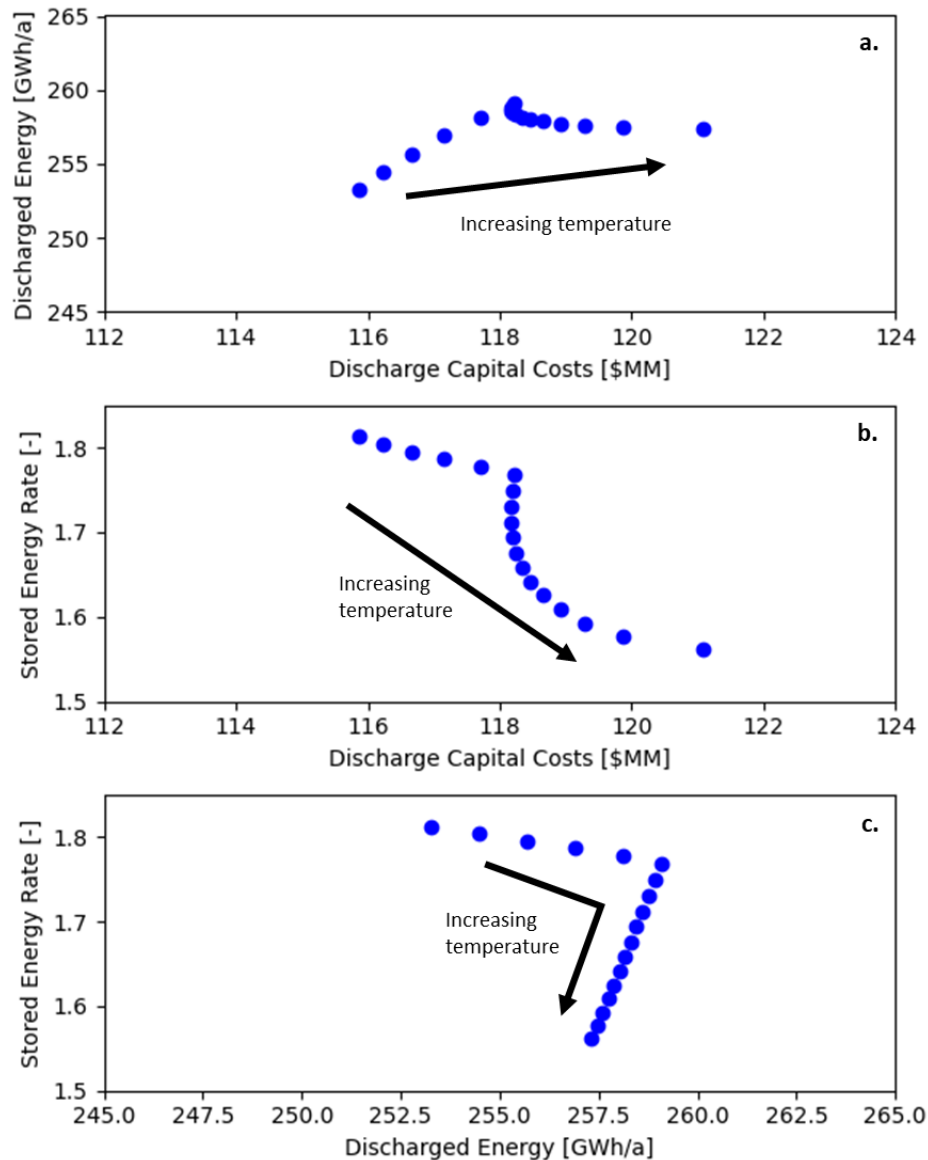


Figure 29. Comparison of the amount of energy discharged SER and discharge capital costs as the liquid air turbine inlet temperature was swept between 400-600°C. a) Discharge capital costs vs. discharged energy, b) discharge capital costs vs. stored energy rate, c) discharged energy vs. stored energy rate.

ORC TIT was swept between 200-350°C as shown in Figure 30. The results show that the capital costs and power output increased while the SER decreased with increasing inlet temperature. Both flow rates for the two fluids decreased as the temperature was increased. This led to the minor increases to the overall power output of the overall system. As the increase in power from the ORC turbine was higher (4.5 MW) than the decrease in the liquid air turbine (-3.0 MW), the overall power input minimally increased with increasing ORC TIT, but there was a significant decrease in the SER due to the reduced liquid air flow rate.

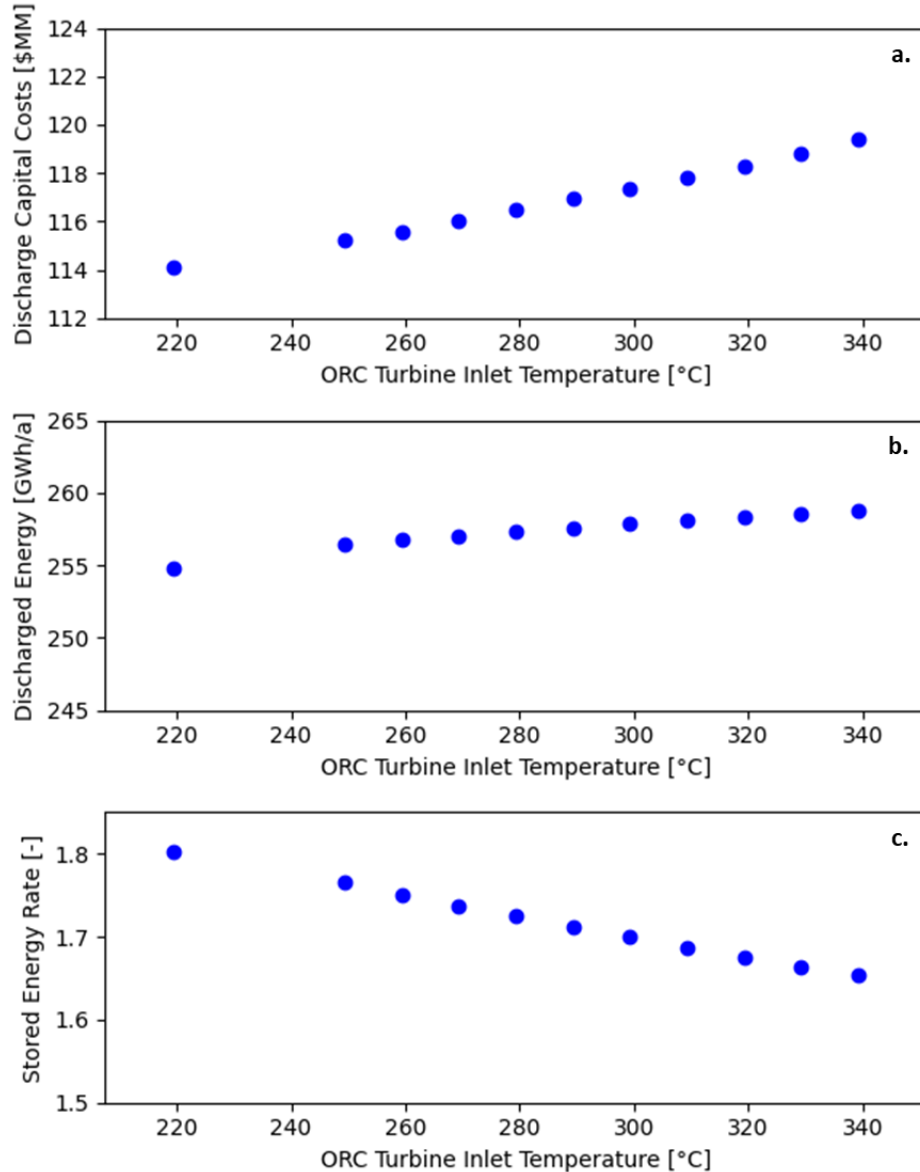


Figure 30. Comparison between the ORC turbine inlet temperature and three dependent metrics.
a) Discharge capital costs, b) discharged energy, c) stored energy rate.

Increasing the ORC TIT had obvious trade-offs for each dependent variable as shown in Figure 31. Comparing the discharge costs vs. the amount of energy discharged showed relatively little change in the amount of power being produced with each additional \$1MM in capital, only increasing the amount of energy discharged by around 0.8 GWh/a (shown in Figure 31a). The SER decreased by around 0.03 for a similar amount. This shows that the increase in ORC TIT resulted in simultaneous improvements to the SER (decreased) and discharged energy (increased) (shown in Figure 31c). This is due to the improvements to the ORC system, which had an increase in the power output for a reduction of flow rate while the reduction in air flow rate brought down the amount of energy used to charge the storage tanks.

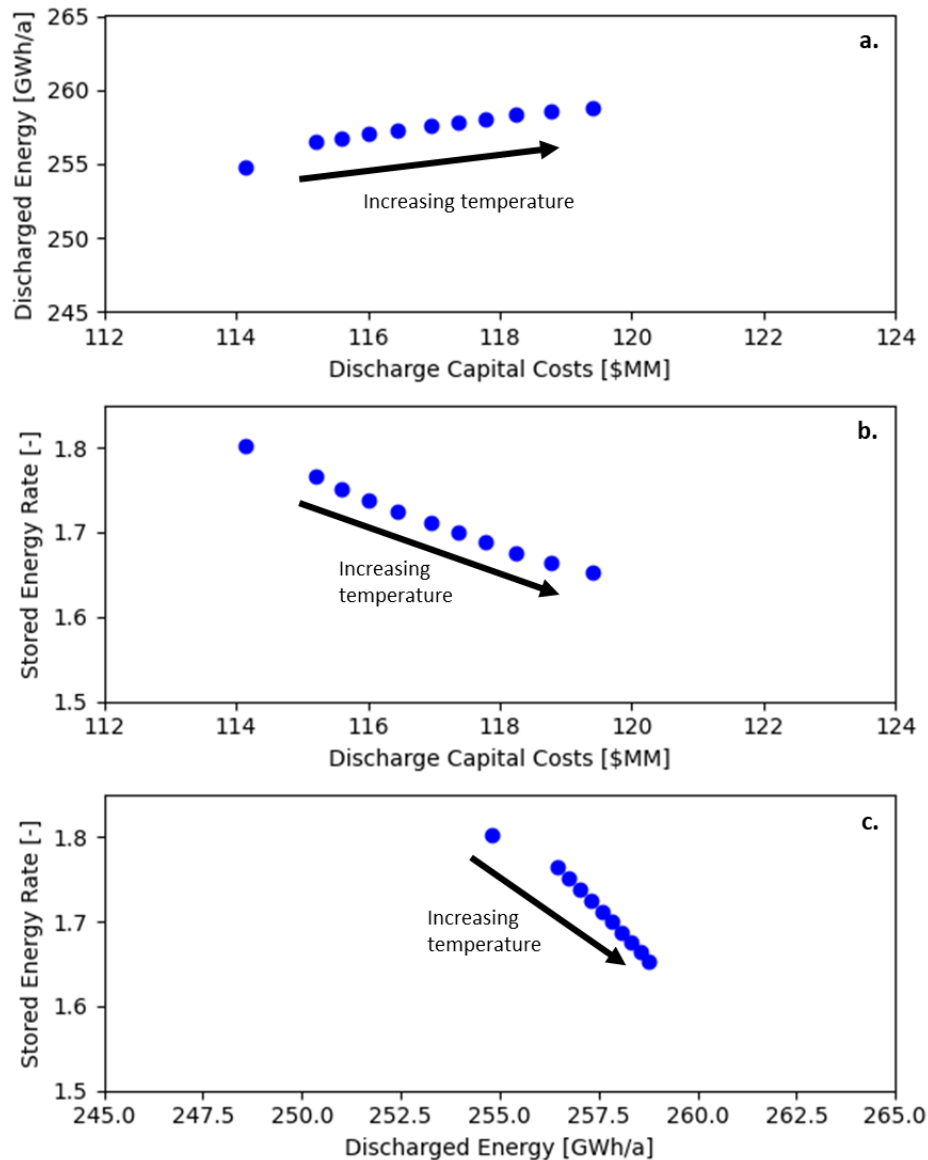


Figure 31. Comparison of the amount of energy discharged, SER and discharge capital costs as the ORC turbine inlet temperature was swept between 200–350°C. a) Discharge capital costs vs. discharged energy, b) discharge capital costs vs. stored energy rate, c) discharged energy vs. stored energy rate.

The last variable that had significant effect on the results was the liquid air turbine efficiency as shown in Figure 32. Turbine efficiency has two main effects on a cycle. The first is the increase in power out of the turbine, and the second is the reduced outlet temperature from the turbine. For this cycle, only the first effect is important. The increasing efficiency of the machinery did increase the costs of the turbine, but it had more significant effects on the other two variables. As a result, an increase of 15 GWh/a was observed over the range of efficiencies compared to an increase of less than \$1MM over the same range.

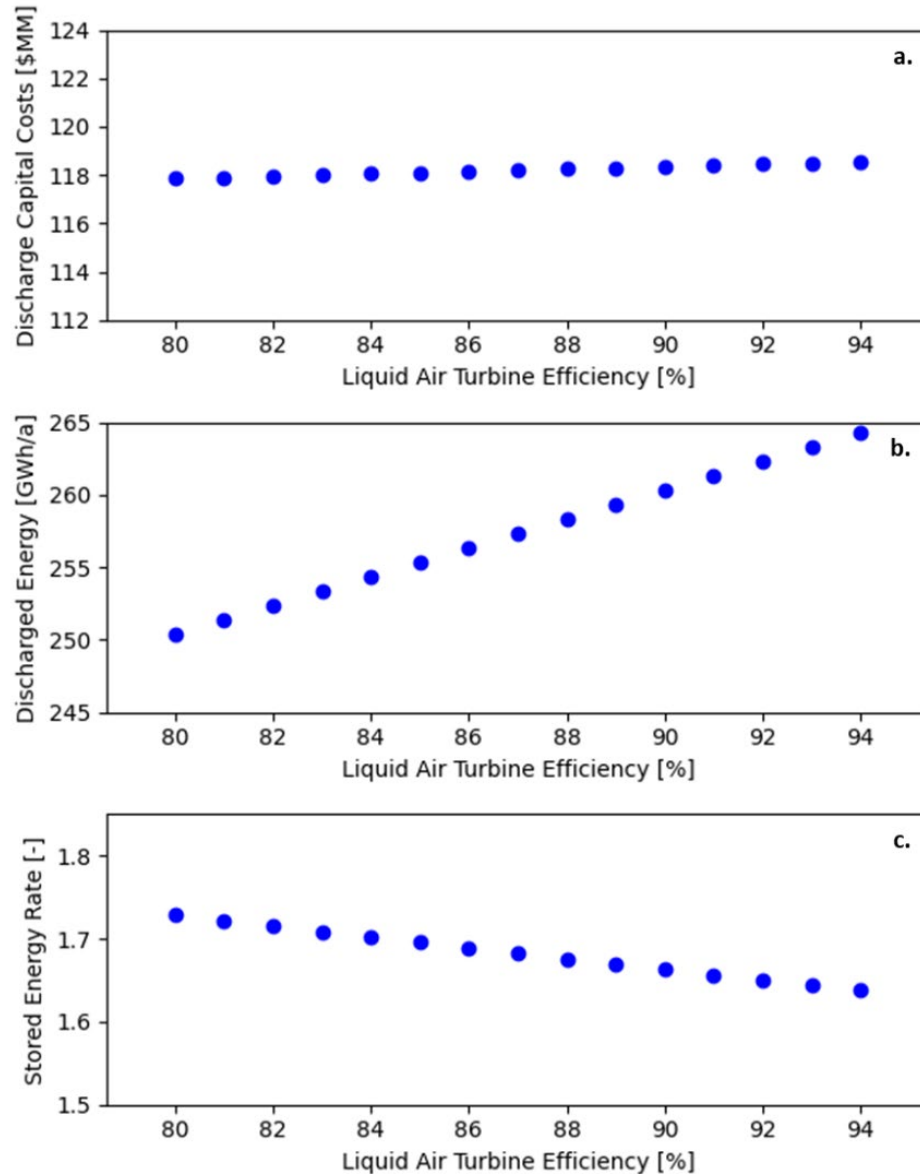


Figure 32. Comparison between the liquid air turbine efficiency and three dependent metrics. a) Discharge capital costs, b) discharged energy, c) stored energy rate.

The efficiency trends show that the increase in costs for developing a more efficient turbine has a significant improvement for minor costs as shown in Figure 32. Overall, the turbine costs are relatively minimal compared to the other components but have a significant impact on the performance. The energy discharged increased approximately 6% over the range of efficiencies, while the capital costs increased by less than 1% (shown in Figure 32a).

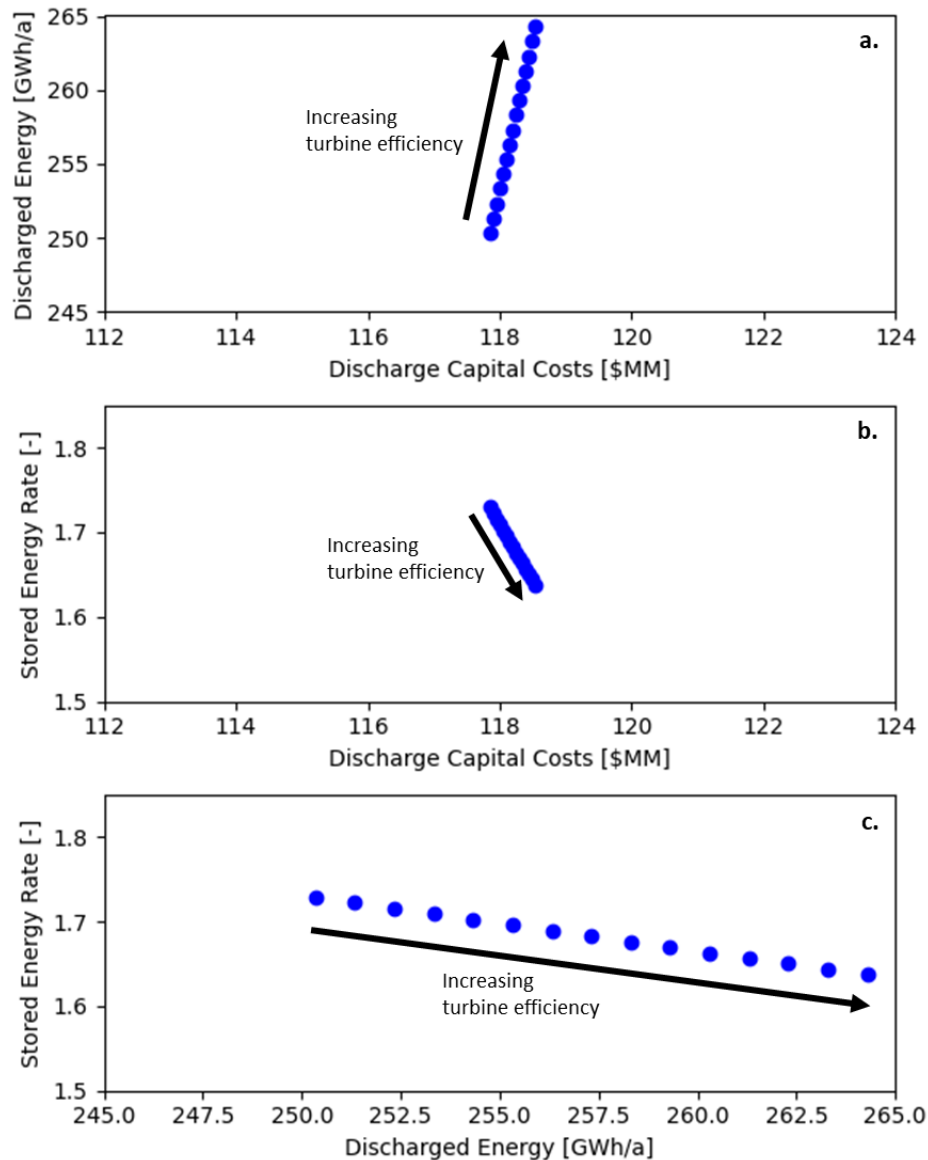


Figure 33. Comparison of the amount of energy discharged, SER and discharge capital costs as the liquid air efficiency was swept between 80-95%. a) Discharge capital costs vs. discharged energy, b) discharge capital costs vs. stored energy rate, c) discharged energy vs. stored energy rate.

5.4. Discharge cycle design optimization

After the seventeen variables were individually swept, the five variables with the highest impact were examined with a more robust method. Each of these variables were given new ranges, chosen to ensure that the results were both feasible (temperatures in heat exchangers did not cross) and numerically stable (model converged to a feasible solution in this range previously). The variables were then swept systematically to test all possible combinations of the five variables with five points each for a total of 3,125 different simulations. The new results are compared using the previous metrics in Figure 34.

The factorial analysis showed significant improvements for the design. By examining the turbine inlet temperatures for the air and ORC streams simultaneously, it was shown that maximizing both temperatures resulted in the highest energy output and reduced liquid air consumption. This also allowed the most energy to be extracted from the exhaust gas stream by optimizing the ratio of heat extracted by the liquid air subsystem and the ORC subsystem. The previous baseline results had the regasified air extracting 35.4 MWth from the exhaust stream to 48.2 MWth of energy with the ORC stream. Examining one of the new optimized points (see Selection C in Figure 34) showed an increase in the amount of energy extracted by the regasified air stream to 36.3 MWth and decreased the amount of energy removed by the ORC to 44.6 MWth, resulting in the air turbine increasing its power output to 41.9 MWe (up from 40.6 MWe) while the ORC turbine decreased to 17.7 MWe (down from 19.0 MWe). Overall, the system had a minor increase in net power output to 118.3 MWe (from 117.9 MWe) while the SER decreased from 1.68 to 1.51.

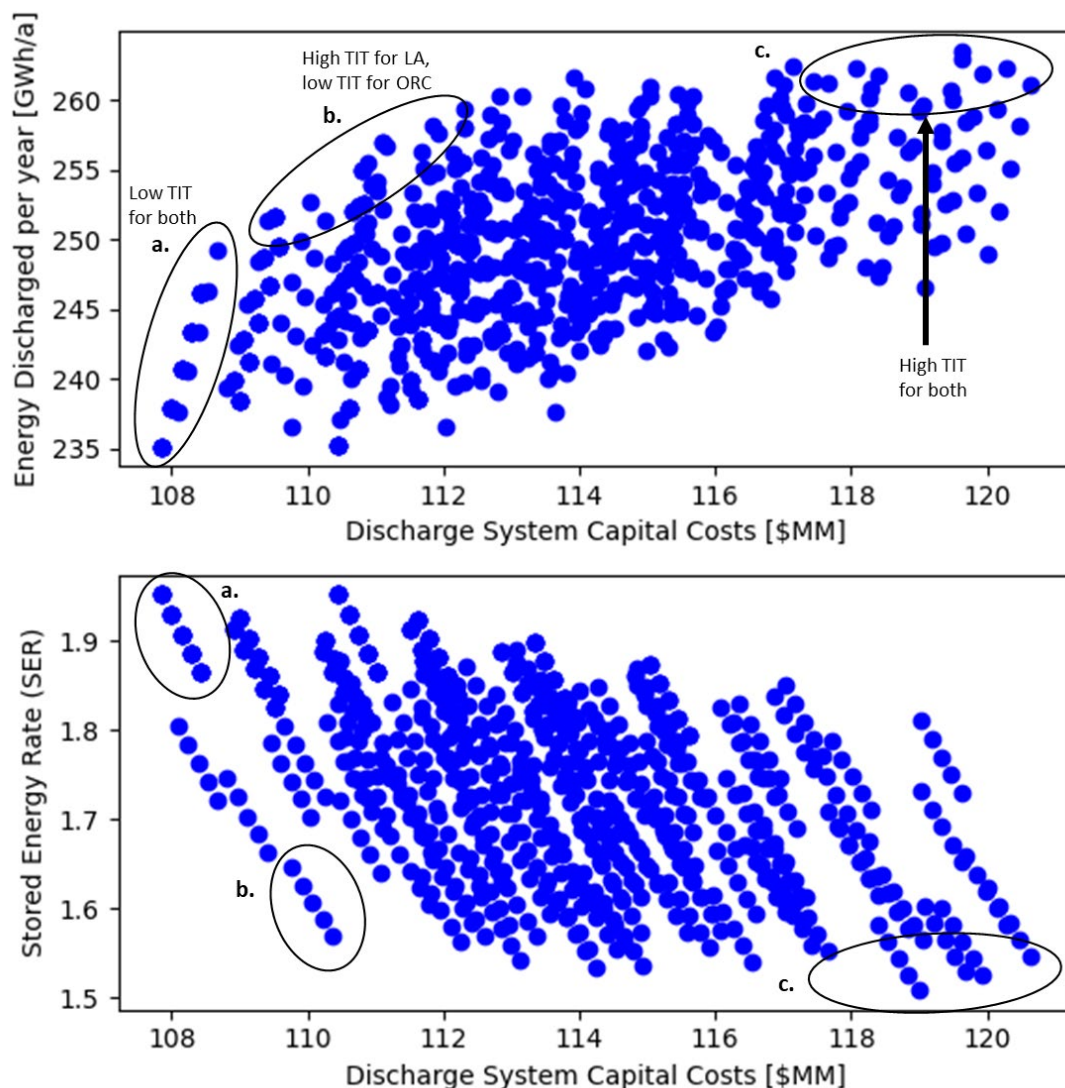


Figure 34. Factorial analysis of the cycle modeling. a) Selection of points with low TIT for both the liquid air and ORC subsystems. b) Selection of points with high TIT for the liquid air subsystem and low TIT for the ORC subsystem. c) Selection of points with high TIT for both subsystems.

5.5. Final design selection

Based on the factorial analysis, it was confirmed that the relatively high CAPEX of the charging system would favor low SER, or equivalently low specific air consumption (kg/kWh). The Ebsilon process model was modified to adjust process conditions accordingly, and some engineering judgement was used to make the following changes:

- The ORC recuperator was found to have very small heating duty, so it was removed. The elimination of pressure drop through the recuperator allowed the ORC condensing pressure to be reduced, which offset the thermal performance lost by removing the recuperator, while also reducing CAPEX of the discharge train.
- Although high regasified air TIT was found to be beneficial, very close approach temperatures resulted in excessive heat transfer area. Increasing the approach temperature reduced the size and cost of the exhaust gas heat exchanger.
- With a larger approach temperature, the target flue gas temperature was increased above the acid dew point temperature, which is expected to reduce cost and improve reliability.

This revised performance model is used for the marginal cost components (fuel and air consumption), and the discharge train capital cost estimate was refined accordingly based on vendor equipment quotes as available, with scale factors for EPC, process contingency, and fees for Pintail Power's technology.

In addition, a design review suggested that an intermediate heat transfer loop should be interposed between the exhaust gas heat transfer coil and the hot ORC refrigerant destined for the ORC turbine. This change greatly reduces the volume and mass of ORC refrigerant, allows the ORC system to be more compact, and eliminates leakage of flammable ethane refrigerant into the flue gas, reducing the area subject to Class I Div 2 hazard classification. The final process design, with the additional heat transfer loop is shown in Figure 35.

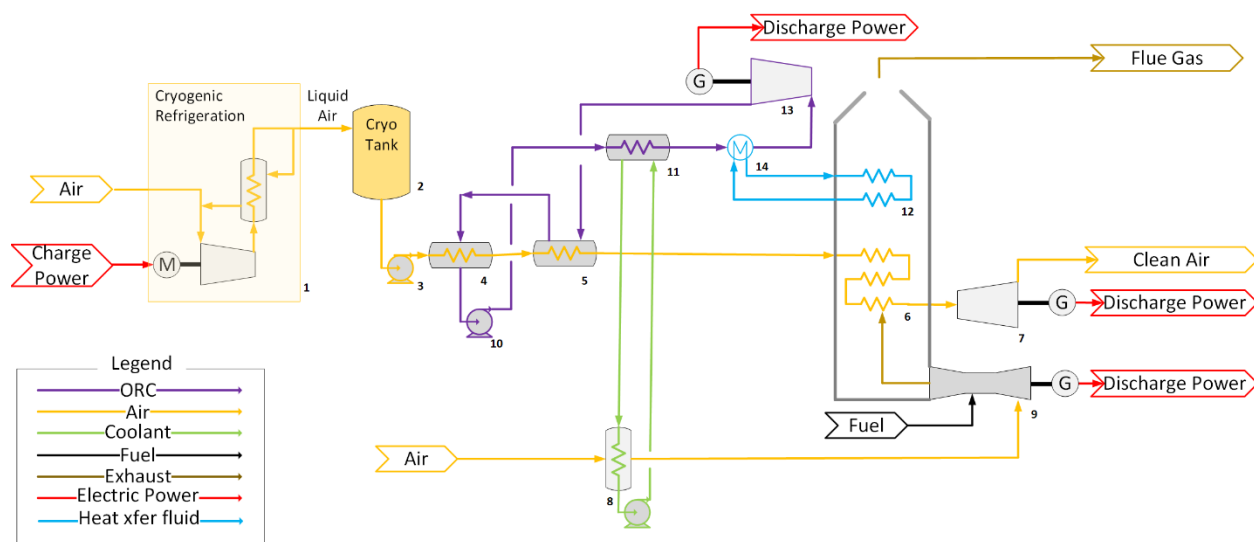
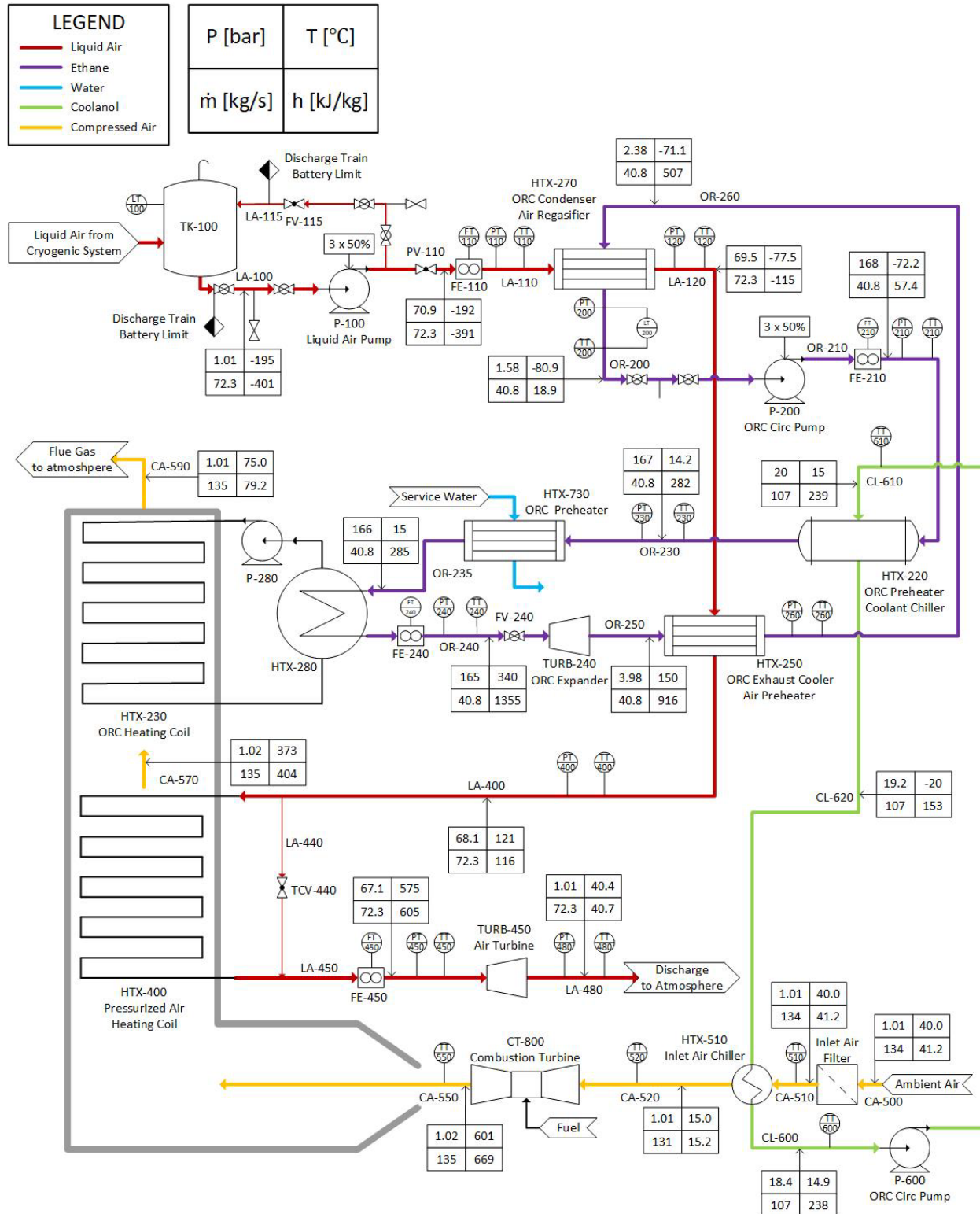


Figure 35. Final design

Process conditions for the final design are shown in Figure 36.



5.5.1. Performance

Performance of the final design is shown in Table 19.

Table 19. Discharge train performance – final design

Parameter	Performance
Net Power	117,319 kW
Net Fuel Heat Rate (FHR)	5068 kJ/kWh HHV 4664 kJ/kWh LHV
Liquid Air Flow Rate	260 metric tons/hour 2.25 kg/kWh

5.5.2. Cost estimate

The equipment cost estimate for the major equipment in the final design was developed from vendor quotes and estimates as itemized in Table 20.

Table 20. Discharge train equipment cost estimate – final design

Equipment	Cost	Source
SGT-800 genset	\$23,000,000	Vendor quote from October 2022 for a packaged genset
Air-turbine generator	\$4,850,000	Vendor quote for a condensing steam turbine generator, air turbine will be similar
ORC axial turbine generator	\$3,262,590	Weiland formula, multiplied times 10 for conservatism
Exhaust heat recovery	\$6,850,000	Vendor quote from March 2021 for a heat recovery system for an SGT-800, including unneeded equipment and therefore conservative
Cryo pump	\$866,625	Vendor quote from Oct 2022
Cryo heat exchangers	\$1,969,200	Vendor quote from August 2022
Inlet air chilling system	\$1,000,000	Estimate partly supported by quote, scope includes both HTX and circulating pump
Equipment Total	\$41,798,415	

A turnkey factor (3.5 multiple) is applied to the equipment total to arrive at the TOC, which results in a specific cost of \$1247/kW. For comparison the cost of a 120 MW steam combined cycle is about \$958/kW for a 1x1 (1 AE64.3A gas turbine and 1 steam turbine) [31]. Accordingly, the discharge train cost estimate is approximately a 30% premium over the steam combined cycle,

which represents cost savings from the simplicity of the air cycle (without a steam condenser), offset by the complexity of the ORC, additional heat exchangers, and the cryogenic conditions, as well as technology fees and risk premium.

6. LACC ECONOMICS

6.1. Final design economic assumptions

The techno-economic analysis uses the cost and performance data developed during the project, which is summarized in Table 21 and the COE equations from section 2.5. For all of the below results, it is assumed that the system will begin discharging energy once the storage volume is completely full with a fixed average value for the cost of energy during charging. The 10% capacity derate factor includes down-time for mode switching (2-hours), scheduled maintenance and other factors that would cause the actual capacity factor to be less than the theoretical maximum described in Section 2.4.7.

Table 21. List of assumptions for techno-economic analysis

Assumption	Baseline Value	Source
Liquid air tank volume [m ³]	100,000	Assumed value
Charge train capacity [mtpd]	6,480	[24]
Number of charge trains	1	Variable
Number of storage tanks	1	Variable
Number of discharge trains	1	Variable
Capacity factor derate [-]	90% of CF _{max}	Assumed value
Specific energy to produce liquid air [MWh/ton]	0.685	[24]
Discharge train performance - Net power (kW) - Net fuel heat rate (kJ/kWh) HHV - Liquid air flow rate (mtpd)	117,319 5068 6250	Calculation
Equipment cost - Charge train - Discharge train - Storage tank	\$129,700,000 \$41,798,415 \$46,100,000	[24] Estimate (See Table 20) [24]
Weighted Average Cost of Capital [-]	0.062	EIA AEO 2021 [1]
Number of years for capital interest [-]	30	EIA AEO 2021 [1]
Specific cost of fuel [\$ / GJ]	3.98	EIA AEO 2021 [1]

Assumption	Baseline Value	Source
Specific cost of charge electricity [\$/MWh]	35	Lazard LCOS v7.0
Storage tank boil-off rate (%/day)	0.066	[15]
Fixed O&M rate [per cent of overnight cost]	0.02	[24]
Charge turnkey factor [-]	3.5	[24]
Discharge turnkey factor [-]	3.5	[24]
Storage tank turnkey factor	3.5	[24]

6.2. Effect of plant configuration

The economics of the LACC plant depend on its configuration:

- Number of charge trains,
- Number of discharge trains, and
- Number of storage tanks.

Total overnight cost. The total plant cost is a linear function of the number of charge and discharge trains for various plant configurations with a single 100,000 m³ tank.

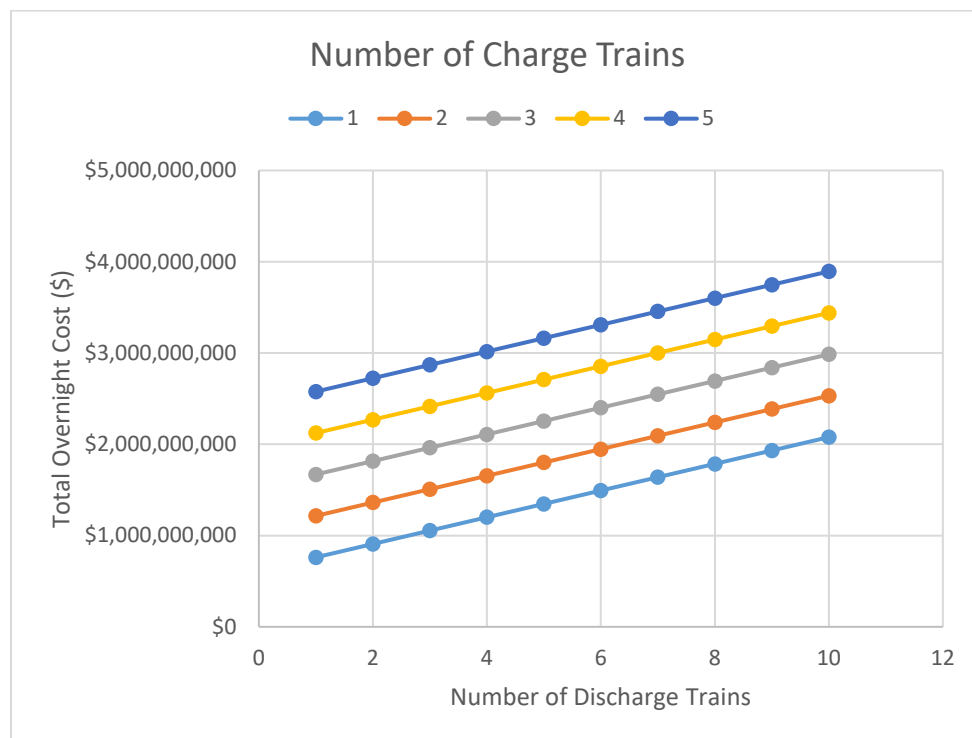


Figure 37. Total overnight cost for a single tank system

Specific cost of energy capacity. A figure of merit for comparison of alternative storage technologies is the specific cost of stored energy capacity (\$/kWh). Figure 38 shows the cost to be \$94.71/kWh for a configuration with a single 100,000 m³ storage tank and 5 charge/9 discharge trains. As noted in Table 10 the cost of a 200,000 m³ tank is a modest fraction of the overall plant capital and would reduce the specific cost to \$48.52/kWh and double the duration.

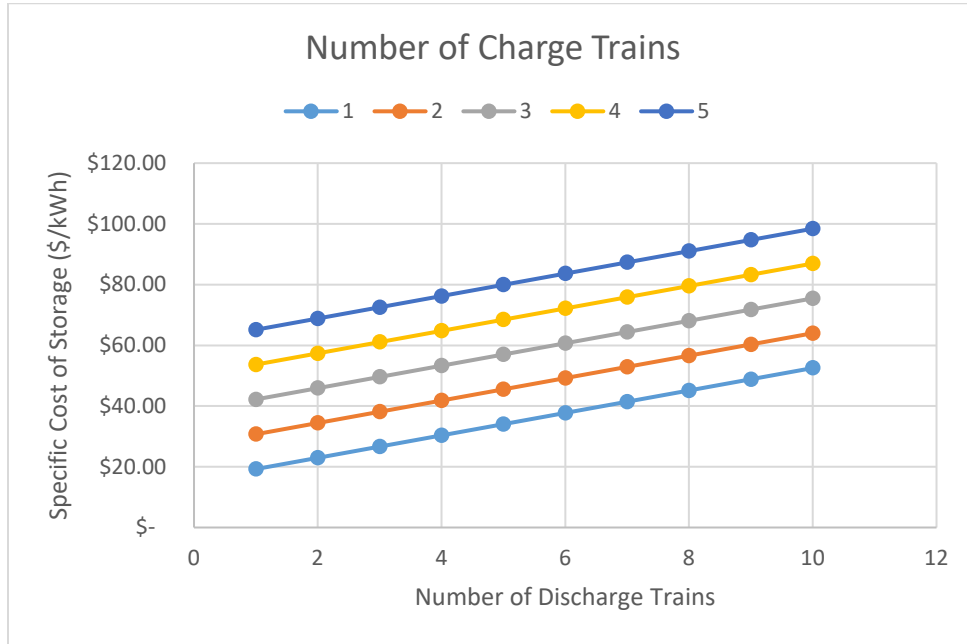


Figure 38. Specific cost of storage for a single tank system

Capacity factor. The capacity factor is a function of the balance between charging and discharging capacity, increasing in proportion to the number of charge trains, but decreasing as the number of discharge trains increases.

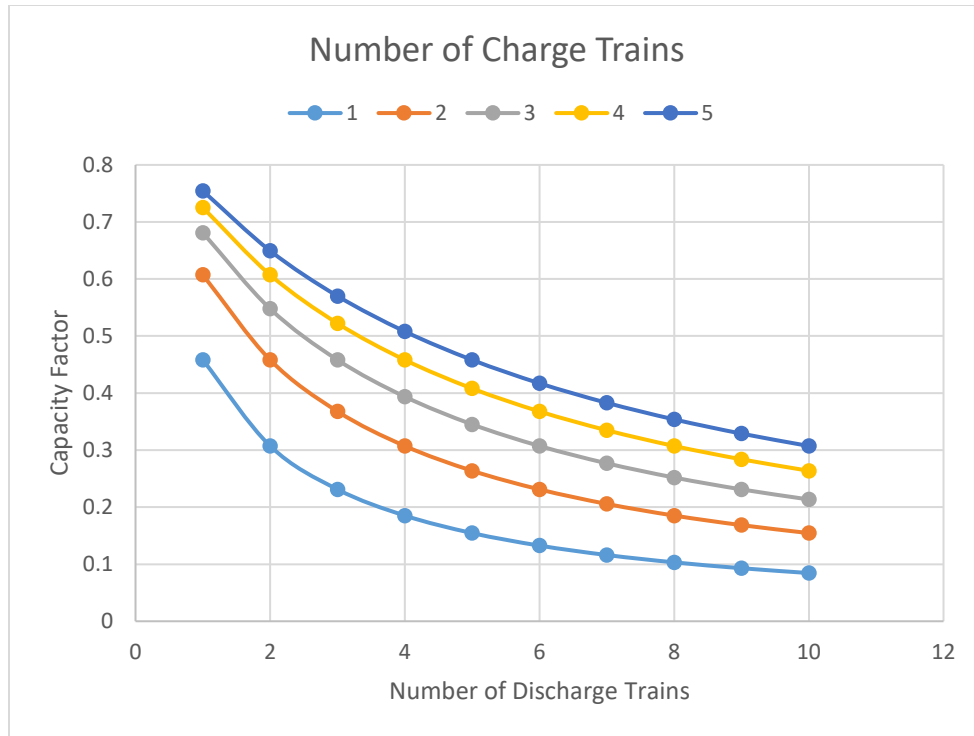


Figure 39. Capacity factor for different plant configurations

For a given number of charge trains, the annual discharge energy is proportional to the number of charge and discharge trains as limited by the capacity factor.

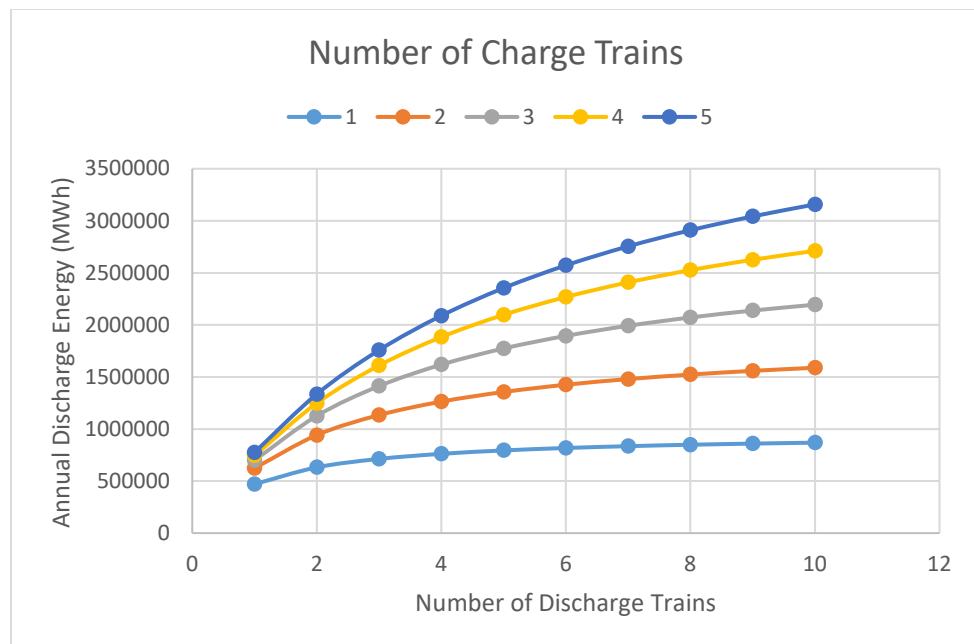


Figure 40. Annual discharge energy for various plant configurations

For many plant configurations, the cost of energy is relatively constant, although the case with a single charge train illustrates that there is a minimum cost configuration associated with the charge capacity; i.e., if the system runs out of liquid air for discharge, it is less able to cover its costs. The

configuration with the lowest cost of energy (\$189.62/MWh) is for 5 charge trains and 9 discharge trains – 1056 MW, 39.6 GWh, or 37.5 hours of storage.

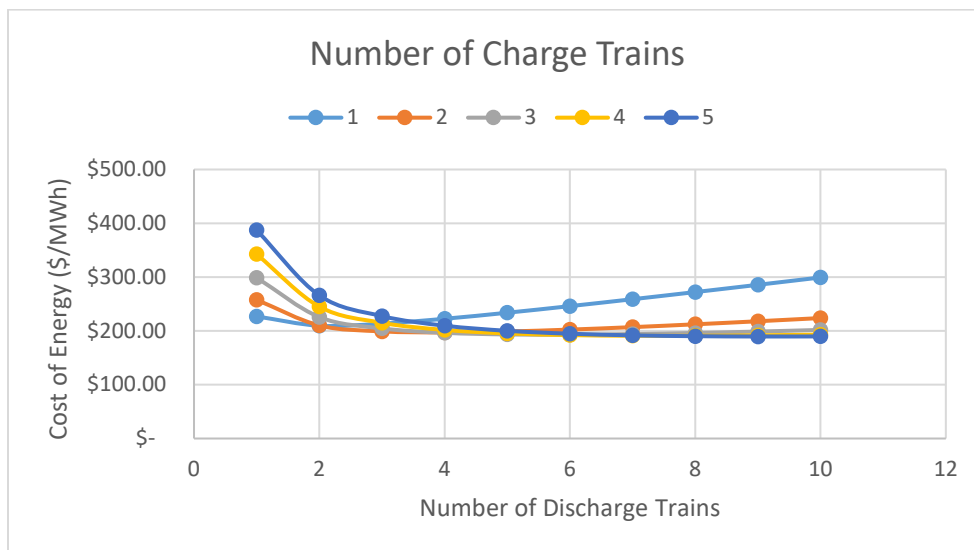


Figure 41. Cost of energy for various plant configurations

The assumptions for this section were based on assuming that the system has minimal idle time and that the system is either charging or discharging. Future analyses for the techno-economics would include real-world scenarios of marginal energy prices. This would allow for more realistic analyses for operation and impact when the system would be able to buy or sell energy.

6.3. Cost sensitivity

The cost assumptions in Table 21 were varied to assess their impact on COE. The configurations with the lowest COE were 5 charge and 9 discharge trains, with the sensitivity results summarized in Table 22.

- Turnkey factor sensitivity reflects capital cost reduction opportunities identified below.
- Specific fuel cost sensitivity assumes availability of hydrogen fuel at the DOE Hydrogen Shot target and adjusts the fuel heat rate to 5.343 GJ/MWh to reflect the difference in heating value between hydrogen and natural gas [20].
- Specific electricity cost assumes lower market prices for charging due to renewable curtailment.
- Specific charge energy cost assumes liquefaction efficiency identified in Akhurst et al. [20].
- Weighted average cost of capital is an arbitrary change for evaluating sensitivity.
- The all of the above sensitivity varies the sensitivity factors simultaneously.
- The best case simultaneously varies the negative (cost reducing) sensitivity parameters.

- The worst case simultaneously varies the positive (cost increasing) sensitivity parameters.

The capital COE has the largest impact as shown by the turnkey factor. The marginal cost components (specific charge energy and electricity price) are the next most important cost reduction components. Fuel cost and the weighted average cost of capital (WACC) were the least sensitive parameters.

Table 22. Minimum cost and configurations for different assumptions.

Parameter	Value	Cost of Energy [\$/MWh]	Relative change [%]	Sensitivity [\$/MWh per %]
Baseline	Table 21	\$188.81	--	--
Turnkey factor	3.0	\$172.23	-14.9%	-1.88
Specific Fuel Cost (H ₂ @\$1/kg)	\$7.05/GJ	\$207.12	+235%	+0.08
Specific Electricity Cost	\$15/MWh	\$154.14	-50%	-0.69
Specific Charge Energy	0.439 MWh/m.t.	\$160.59	-36%	-0.78
WACC	7.5%	\$189.91	+21%	+0.15
All of the above parameters		\$162.86	-26%	-1.00
Best case		\$129.35	-31%	-1.92
Worst case		\$212.38	+12.5%	+1.89

6.3.1. Cost reduction opportunities

Capital cost. As the most sensitive component of the COE, capital cost reductions should be targeted.

- Future FEED studies will refine the engineering design and permit better estimation of material quantities and construction labor hours.
- Multiple parallel trains are amenable to reduction of engineering hours, construction experience curves, and quantity discounts.
- The large scale of the project should improve the opportunity for vendor negotiation.

Energy inputs. There are opportunities to reduce both the fuel heat rate and the stored energy rate by condensing the water of combustion to provide additional heat into the ORC. In the commercial design, the ORC absorbs about 52,886 MW of thermal energy (9.193 MW from gas turbine inlet air cooling and 43.7 MW from gas turbine exhaust gas cooling) to produce 17.6 MW for a gross cycle efficiency of 33.3%. Fuel input to the gas turbine is 152 MW LHV (169 MW HHV).

Assuming that the water of combustion was condensed, and the heat was converted to power at 25% efficiency, an additional 4.25 MWe could be obtained, a power increase of 3.5%. The power increase reduces both the fuel heat rate and stored energy rate by this amount, which would reduce the marginal cost of energy by \$2.13/MWh.

Water of combustion represents about 15.7% of the higher heating value of hydrogen vs. about 9.7% for natural gas, which is reflected in the fuel thermal input of 180.3 MW-th, or 23.3 MW-th heat of condensation. At 25% conversion efficiency about 7MW-e of additional power would be available, an increase of 6%, with the potential to reduce marginal cost of energy by \$4.25/MWh.

Economy of scale. Experience with steam combined cycles shows the advantage of turbo-machinery scale-up, which can also be applied to LACC.

- Moving from the 1x88MW SGT-800 steam combined cycle to the 176MW 1xSGT6-2000E reduces the specific cost 23% from \$1080/kW to \$833/kW [32].
- Steam turbine size is commonly increased by feeding steam from two or three heat recovery steam generators, and a similar arrangement could be used with LACC to achieve economy of scale with the air turbine. Changing from the 1xSGT-800 steam combined cycle to the 2xSGT-800 reduces specific cost 15% from \$1080/kW to \$917/kW.

6.4. Comparison to alternatives

6.4.1. LDES

As part of the DOE Energy Storage Grand Challenge, Pacific Northwest National Laboratory (PNNL) has developed cost estimates and LCOS for a range of LDES technologies. PNNL published installed cost (\$/kWh) and LCOS in its 2022 Energy Storage Cost and Performance Database [33], summarized in Table 23 for 100 MW systems with 24-hour duration (24,000 MWh). This is somewhat smaller than the 37 GWh LACC commercial plant.

Table 23. Costs of LDES technologies in 2030

Technology	Specific Cost of Energy Storage Capacity [\$/kWh]		Levelized Cost of Energy [\$/MWh]	
	Low	High	Low	High
LACC	\$19.25	\$98.41	\$129	\$212
CAES	\$42.18	\$65.02	\$110	\$110
PSH	\$159.07	\$242.27	\$110	\$140
Hydrogen (bidirectional)	\$37.44	\$60.16	\$170	\$210
Vanadium Redox Flow	\$166.35	\$292.88	\$230	\$350
Lithium-ion	\$213.38	\$270.45	\$240	\$330
Lead Acid	\$294.58	\$334.19	\$390	\$440
Zinc Air (1000 MWh)	\$337.46	\$408.43	\$940	\$1680

With larger and/or multiple storage tanks the specific cost of energy capacity of LACC technology is competitive with all alternative LDES technologies evaluated by PNNL. Except in the best case scenario, the COE of LACC technology is somewhat more than CAES and PSH, but its siting flexibility permits LACC to avoid the cost of new transmission lines from remote CAES and PSH sites.

6.4.2. Gas turbines

As noted in section 2.3, rather than LDES, hydrogen-fueled gas turbine-based power plants are expected to be the major competition for LACC technology. The COE of gas turbine-based systems as a function of capacity factor is shown in Figure 42, based on costs (capital, O&M) and performance from the Annual Energy Outlook [1], WACC from Table 21, and assuming \$1/kg hydrogen.

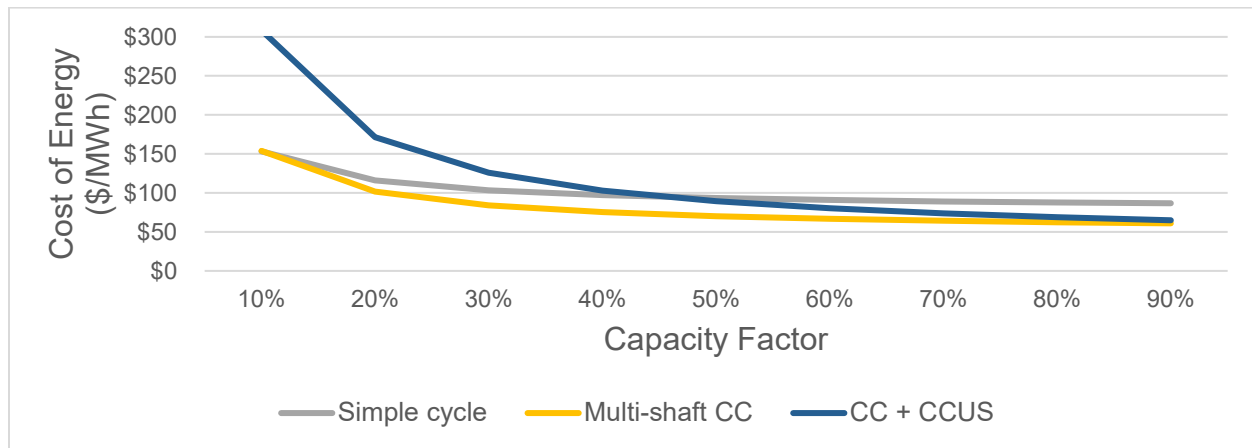


Figure 42. COE for gas turbine power plants fueled with hydrogen or equipped with carbon capture

To compete with gas turbines, the LACC technology will require cost reduction and performance improvements that have already been identified, including:

- Economy of scale with larger gas and air turbines
- Experience (learning)
- EPC value engineering (turnkey factor)
- Higher efficiency by condensing the water of combustion
- Recycling of condensate to hydrogen electrolyzers to reduce water consumption

7. TECHNOLOGY MATURATION PLAN

This section is taken from the technology maturation plan (TMP) that was submitted as one of the deliverables for the project.

7.1. Technology readiness level

The Liquid Air Combined Cycle (LACC) is a hybrid waste-heat recovery and energy storage system. While many elements of the LACC are based on technically mature subsystems, the complete LACC system is now at Technology Readiness Level (TRL) 3 based on detailed simulations, techno-economic trade studies, and the optimization of a commercial-scale system performed in this program. This is up from TRL 2 at the start of the project. Table 24 summarizes the TRL of the overall system, subsystems, and critical components. As described in this report, the LACC can be advanced to TRL 4/5 through the development of the demonstration-scale system focused on validating low-TRL components like the ORC turbine.

Table 24. Technology readiness levels for LACC

System or subsystem	TRL	Description
Liquid Air Combined Cycle (LACC) full system	3	Detailed simulations and trade studies completed. A commercial scale system has been finalized. Key technical risks have been identified.
Cryogenic liquefaction	9	Commercially available in several different approaches (nitrogen cycle, mixed refrigerant cycle, absorption chilling, or cascades of different cycles).
Liquid air storage	9	Commercially available equipment exists for pressurized (e.g., bullet tanks up to $\sim 1\text{e}3\text{ m}^3$) and near-atmospheric pressure (e.g., steel-lined concrete tanks over $\sim 2\text{e}5\text{ m}^3$) storage.
Cryogenic pump	9	Commercially available for liquid nitrogen, liquefied natural gas (LNG), etc.
Combustion turbine	9	Commercially available; virtually any type of combustion turbine, existing assets or new installations, could be used for LACC.
Inlet air cooler	9	Commercially available.
Air-exhaust gas heater	8	Commercially available for similar applications, e.g., air heaters or heat recovery steam generators.
ORC-exhaust gas heater	7	Commercially available in lower pressure applications; customary design.
Air turbine	8	Commercially available for similar applications such as compressed air energy storage (CAES); essentially a modified steam turbine.

System or subsystem	TRL	Description
Liquid air regasifier/ ORC condenser	3-4	Custom design for application, with each side of the heat exchanger based on proven technology: cryogenic condensation on the cooled side and cryogenic regasification on the heated side. Unique application requirements include startup, load-following, off-design, and trip/sudden load rejection. Commercial providers are prepared to offer heat exchangers to meet these requirements, suggesting a higher TRL may be appropriate.
ORC turbine	4	Custom design for the application due to the high-pressure ratio.
ORC Pump	4	Custom design for the application due to the pressure ratio and max pressure. The high-pressure ratio means that the LACC may require multiple stages of pumping to reach the maximum pressure required.

A key objective was to identify a commercial-scale design for the LACC. Figure 1 shows the process flow diagram (PFD) for this system. The determination of the requirements was based on the market need, cost-effectiveness compared to alternatives, and utility acceptance of cryogenic storage energy storage technology. The key components enumerated in the figure, are described in Section 2.8.

Market need. LACC is positioned to meet the need for Long-Duration Energy Storage (LDES) for important emerging applications:

- Inter-day discharge duration (24 to 48 hours)
- Multi-day discharge duration (48 to 168 hours)
- Seasonal time shifting of several months.

LACC's characteristics that make it well-suited to serve these applications include:

- Exceptionally low cost of storage due to the use of a free and ubiquitous storage medium (air) and proven large-scale cryogenic storage tanks, the combination of which reduce storage costs to less than \$1/kWh.
- Exceptionally high storage density due to the use of air as both a phase-change storage medium and working fluid, resulting in discharge energy density of the storage system greater than 15,000 kWh/m³.
- Exceptionally low self-discharge rate to permit long-term storage with minimal losses, as evaporative losses are typically about 0.02% per day (for an 80% full tank).
- Decoupled and independent sizing of the charging and discharging systems to provide design and operational flexibility.
- Substantially reduced fuel consumption, about half of a simple cycle gas turbine and typically about 20% less than a conventional combined cycle using the same gas turbine.

Accordingly, Pintail Power anticipates that LACC systems will be deployed at GW-scale for LDES applications to store renewable resources and provide essential grid services. Storage and time-shifting of renewable energy provides capacity and reliability to the grid when renewable energy is not available.

Competition. The future need for renewable backup spanning many consecutive days can be met by these competitive approaches:

- **Energy storage.** Of energy storage systems, only Pumped Storage Hydro (PSH) and Compressed Air Energy Storage (CAES) have the potential to be deployed at a scale that approaches LACC. Geotechnical and siting challenges limit the potential for PSH and CAES deployments. Due to its compactness, LACC would be the only large-scale LDES storage technology that could be safely deployed in a metropolitan area.
- **Hydrogen fueled gas turbines.** Large-scale electrolysis of hydrogen is projected to cost between \$1.50/kg and \$5.00/kg (\$10.55/GJ to \$35.16/GJ) in 2050 [15], depending on the scenario (degree of optimism), production conditions, and cost of electricity used for electrolysis. The storage and pipeline transport of hydrogen to the burner tip would be an additional expense. Two potential scenarios emerge for gas turbine-based systems, depending on the need for renewable backup and the resulting capacity factor of the dispatchable resources.
 - **Simple cycle gas turbines** would likely be preferred for low capacity factor, as is the case with natural gas. However, the cost of expensive hydrogen fuel would make the LACC fuel savings attractive, as they could be about \$50/MWh or more.
 - **Combined cycle gas turbines** typically are more cost-effective when capacity factors exceed 50%, but given the high cost of hydrogen, a modern hydrogen-fueled combined cycle could be a cost-effective alternative at lower capacity factors.
- **Natural gas fired combined cycle with carbon capture and storage (CCS)** would be an alternative depending on how the unit is dispatched. Startup of the CCS unit is likely to take many hours before rated capture efficiency is obtained, so cyclic duty would not be effective at GHG reduction.

Pintail Power believes that hydrogen-fueled gas turbines are the chief competition for LACC systems, which would likewise be hydrogen-fueled in order to provide zero-carbon dispatchable power. The primary competitive advantage for LACC would be fuel savings, the benefit of which would depend on the cost of hydrogen delivered to the burner tip. Savings could be enhanced by capturing the heat of condensation of the water of combustion, which can be fed into the low temperature ORC to improve efficiency. The condensate could be recycled to an electrolyzer, which could be advantageous if on-site hydrogen production is found to be economic compared to pipeline transportation.

Commercial Applications. Competitive advantages of LACC include:

- A storage medium that is both freely available without supply chain constraints and zero-cost.
- Compact, high density storage, which can also be relatively inexpensive when using customary large-scale storage tanks such as are used for Liquefied Natural Gas.
- The low loss rate, less than 0.1%/day, which enables seasonal storage applications.

These cost and scale advantages may be particularly appealing in applications requiring storage at the scale of a PSH System (tens of GWh). LACC could be safely sited near or within a metropolitan area to provide long duration storage closer to load and more cost-effectively than Pumped Storage.

The major challenge to LACC commercialization is the very hybrid nature of the technology – its requirement of a fuel source to power the gas turbine. However, the fuel requirement can be turned to provide reliable fuel supply for polar vortex events such have caused challenges for gas fired power plants. Using liquid air at -195°C pipeline gas can be liquefied at -178°C using relatively inexpensive heat exchangers. During periods of low gas prices, some liquid air could be circulated through a heat exchanger and the cold gas returned to a boil-off gas (BOG) compressor for re-liquefaction. The LNG would provide on-site fuel storage, without risk of pipeline delivery disruptions due to well-head freeze-offs, such as what occurred in ERCOT in 2021, or diversion of pipeline gas from power to heating loads as occurs in the Northeast.

LACC can provide synergistic benefits in the use of hydrogen as an alternative to carbon-based fuels. Using a heat exchanger and BOG compressor as above, the density of hydrogen can be increased by a factor of 3 or more to reduce the footprint of on-site hydrogen storage. The discharge cycle can be configured to condense the water of combustion from the gas turbine exhaust, making about 18% more thermal energy available for use in the ORC bottoming cycle. The condensate can be recycled to the on-site hydrogen electrolyzer to minimize water requirements.

Pintail Power believes that the additional fuel synergies will enable smaller LACC systems to be competitive.

7.2. Post project plans

The post-project plans focus identified areas of risk in order to advance the technology to TRL 6 and beyond. This research is not being performed under this award due to budgetary, schedule, and technical constraints. Three major broad activities are anticipated.

7.2.1. Engineering optimization

Toward the end of the project, Pintail Power identified a process modification to improve safety and operability of the discharge train. Figure 43 shows a heat exchanger (14) and heat transfer loop interposed between the exhaust gas heat transfer coil and the hot ORC refrigerant destined for the turbine. This optimization greatly reduces the volume and mass of ORC refrigerant, simplifying inventory management, allows the ORC system to be more compact, and separates flammable ethane refrigerant from the flue gas, reducing the area subject to Class I Div. 2 hazardous area

classification. Preliminary assessment of performance indicates the impact on critical parameters (heat rate, stored energy rate, specific air consumption) will be minimal. The post project effort is primarily related to component sizing, costing, and performance modeling.

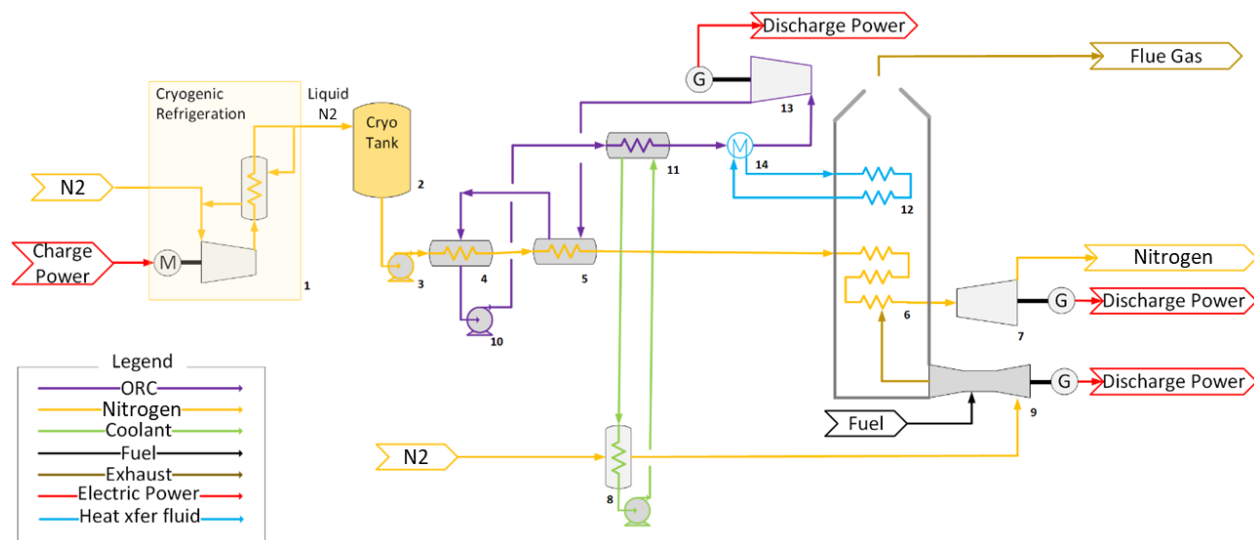


Figure 43. Process flow diagram of engineering optimized LACC

7.2.2. Demonstration-scale system

It is anticipated that a pilot-scale demonstration system will be constructed to validate the performance and operability of the LACC technology. A pilot-scale system at about one-tenth of the commercial reference plant would be based on an industrial gas turbine, such as the 6500 kW Solar Turbines Taurus 65. Exhaust gas temperature is somewhat less than the SGT-800 used as the commercial-scale reference, but operating conditions would otherwise be similar. Commercialization could proceed after demonstration of the pilot-scale discharge train.

Rather than constructing a cryogenic liquefaction system, cryogenic liquid nitrogen would be purchased from commercial suppliers for use during testing.

7.2.3. ORC component validation

The major components of the LACC that are at high TRL with significant technical risks are associated with the ORC loop. Accordingly, a component validation loop will be used to demonstrate the safe and reliable operation of the ORC and bring each component and the ORC system to TRL 6. The loop shown in Figure 44 will enable verification of the ORC components at the operational conditions of the demonstration-scale pilot plant.

Instead of using gas turbine inlet air and exhaust as the heat sources, electric heaters will control the temperature of a thermal fluid in order to provide the requisite ORC temperatures at the exits of units 11 and 14.

Liquid nitrogen would be purchased from an external source for storage in a bullet tank adjacent to the loop. A cryopump would pressurize the air for feeding the two heat exchangers 4 and 5, with regasified nitrogen discharged to atmosphere.

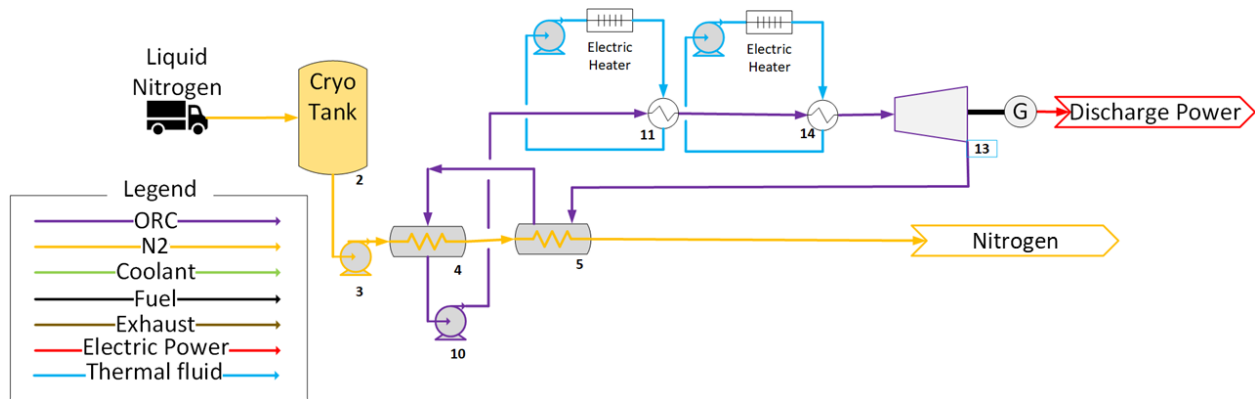


Figure 44. ORC component validation loop

7.2.4. Commercialization

Pintail Power anticipates that LACC technology will be licensed to interested utilities, developers, and EPCs.

8. DEMONSTRATION-SCALE LACC DESIGN

The conceptual design specification for the reference process design has been adapted to a reduced-scale LACC specification, including conceptual sizing of key components. This reduced-scale LACC is about 10% of the power output of the reference system with a nominal net power output of 13530 kW. The specification of the demonstration-scale LACC includes identification of a gas turbine, conceptual sizing of key components, and performance predictions.

8.1. Performance

The performance of the demonstration scale LACC system was estimated using the same Epsilon Professional model as for the commercial-scale system by replacing the 62MW Siemens SGT-800 gas turbine with a 6292 kW (ISO) Solar Turbines Taurus 65 unit (Figure 45). In order to achieve a thermal balance (i.e., a converged result) with the changes in gas turbine exhaust conditions, several adjustments were made to:

- Air turbine inlet temperature was reduced from 575°C to 525°C
- Flue gas temperature was increased from 75°C to 100°C
- Ambient air temperature was reduced from 40°C to 30°C

None of these adjustments would be expected to materially impact the ability of the system to demonstrate operability. The estimated performance of the demonstration scale design is shown in Table 25.

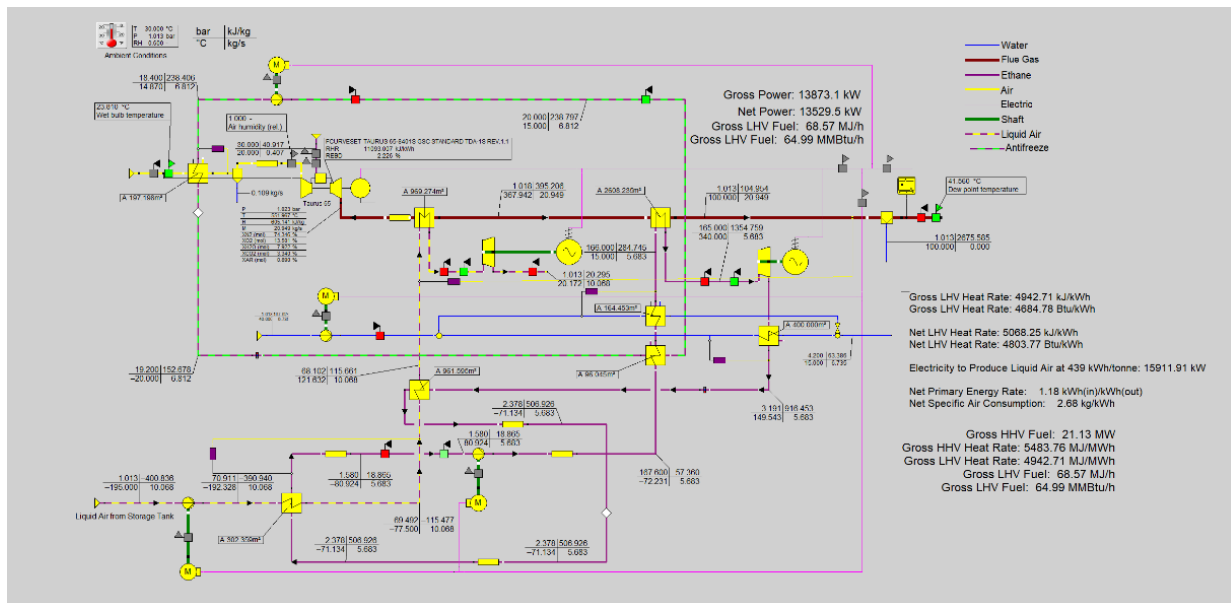


Figure 45. Demonstration-scale performance model

Table 25. Demonstration scale performance

Parameter	Performance
Net Power	13530 kW
Net Fuel Heat Rate (FHR)	5484 kJ/kWh-discharge HHV
Liquid Air Flow Rate	2.68 kg/kWh 363 metric tons/hour

8.2. Plant arrangement

It is expected that the demonstration plant would be focused on the discharge operation and performance, and accordingly would purchase liquid air or liquid nitrogen to fill a storage tank, similar to the ORC component validation rig described in section 7.2.3.

The layout would resemble a steam based combined cycle:

- Gas turbine generator set
- Heat recovery unit, similar to a once-through HRSG (without steam drums), with heat transfer loops for regasified air and for heat transfer fluid
- Air turbine generator set
- ORC subsystem
- Electrical and mechanical auxiliary systems

8.3. Cost

Cost estimates have not been developed, but for a first-of-a-kind demonstration, it is reasonable to apply some scale factors to the specific cost of the commercial scale discharge train (\$1247/kW) to reflect the effects of smaller scale. (The 3.5x EPC multiple for a First-Of-A-Kind (FOAK) commercial unit already includes contingency). For the demonstration unit, a specific cost multiplier of 2x to 3x is appropriate pending FEED studies and preliminary engineering. Accordingly, the cost of a demonstration unit cost is in the range of \$33 to \$50 million.

9. SUMMARY AND CONCLUSIONS

LACC integrates cryogenic energy storage with gas turbine exhaust heat to provide large-scale, LDES using largely proven equipment. LACC eliminates the geotechnical, hydrological, and land-use constraints of PSH, or the underground caverns required for CAES.

The LACC technology is a unique approach to LAES that focuses on reducing the capital and operating costs of large-scale energy storage by combining proven equipment and processes adapted from the power and process industries. An important design principle of LACC is systems engineering, with clear interfaces between sub-systems to minimize interaction, reduce technical and financial risk, and permit the use of best-in-class equipment provided by top-tier vendors.

The project team of Southwest Research Institute (SwRI) and Pintail Power (Pintail) analyzed and optimized Pintail's patented LACC to develop the conceptual design and specification for commercial scale systems. Key accomplishments included:

- Analysis of the technical market requirements for LDES in a decarbonized grid, including assessment of competitive positioning.
- Trade studies of cryogenic charging and storage systems led to baseline cost and performance estimates which were key to the economic analysis.
- An ORC, used to increase power output and provide a thermal bridge between the gas turbine exhaust heat and the cryogenic liquid air operates outside the customary range of cryogenic refrigeration systems, and was a major focus of the design effort.
- Trade studies of the discharge system, including cost and performance modeling with thousands of cases, were used to identify Pareto fronts near system optimal points and guide selection of a final design.
- An economic analysis examined plant configurations comprised of multiple charge and discharge trains to develop cost curves, understand sensitivity to assumptions, and compare results to the technical market requirements. A dispatch model was not used and an assumption that the system is discharging upon completely charging the system. More detailed analysis including site/region-specific marginal price scenarios would be important future work.
- A TMP identified the key steps needed to advance LACC technology to commercial scale, including the validation of ORC components followed by discharge train demonstration at pilot scale.
- A demonstration scale system was proposed using a small gas turbine, in order to gain operating experience prior to commercialization.

The results support that LACC technology is well suited to LDES applications due to its relative compactness and independence from the geological features required for alternative LDES technologies (PSH and CAES), and its potential for reduced cost than other alternatives. The

results also suggest that hydrogen-fueled gas turbines represent a major competition to LDES technologies and that further cost and performance improvements are needed.

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APPENDIX: PUBLICATIONS

The following is a list of publications produced on this project (DE-FE0032002).

Completed:

1. Conlon. W., Venetos, M., and Rimpel, A., 2022, “Liquid Air Combined Cycle,” GT2022-80426, ASME Turbo Expo 2022, Rotterdam, NLD, June 13-17.
2. Pryor, O., Rimpel, A., and Conlon, B., 2022, “Organic Rankine Cycle Turbine and Heat Exchanger Sizing for Liquid Air Combined Cycle,” GT2022-82263, ASME Turbo Expo 2022, Rotterdam, NLD, June 13-17.

In-process at the time of this report:

3. Pryor, O., Conlon. W., Rimpel, A., and Venetos, M., 2023, “ORC2023, “Design and Modeling of a Demonstration-scale ORC Cycle for the Liquid Air Combined Cycle,” 7th International Seminar on ORC Power Systems, Seville, Spain, September 4-6.